SNJB's

Late Sau. Kantabai Bhavarlalji Jain

College of Engineering

(An Autonomous Institute Affiliated to Savitribai Phule Pune University, Pune)

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ESTD - 1928

SNJB

Curriculum and Evaluation Scheme for Third Year B. Tech. in

Computer Engineering with Multidisciplinary Minor

To be implemented for 2024-28 Batch

(With Effect from Academic Year 2026-27)

CHAIRMAN
BOARD OF STUDIES COMPUTER ENGINEERING
SNJB's
LSKBJ COLLEGE OF ENGINEERING
Chandwad Dist.Nashik

SNJB's COLLEGE OF ENGINEERING Chandwad (Nashik) Pin-423101

CHAIRMAN
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SNJB's
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Chandwad Dist.Nashik

Vision of the Institute

Transform young aspirant learners towards creativity and professionalism for societal growth through quality technical education.

Mission of the Institute

1. To transfer the suitable technology, particularly for rural development.
2. To enhance diverse career opportunities among students for building a nation.
3. To acquire the environment of learning to bridge the gap between industry and academics.
4. To share values, ideas, and beliefs by encouraging faculty and students for the welfare of society.

Vision of the Computer Engineering Department

To empower young generations for significant contributions in the field of computer engineering through excellence in knowledge, technical education, and innovation to cater the industrial demands and societal needs.

Mission of the Computer Engineering Department

1. To achieve academic excellence by inculcating basic and latest knowledge in which new ideas flourish.
2. To undertake collaborative training which offers opportunities for long-term interaction with academia and industry.

Program Outcomes (POs) for an engineering graduate:

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)

PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences



PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

Program Specific Outcomes

1. **Professional Skills-**The ability to understand, analyze, and develop computer programs in the areas related to algorithms, system software, multimedia, web design, big data analytics, and networking for efficient design of computer-based systems of varying.
2. **Problem-Solving Skills-** The ability to apply standard practices and strategies in software project development using open-ended programming environments to deliver a quality product for business success.
3. **Successful Career and Entrepreneurship** - The ability to employ modern computer languages, environments, and platforms in creating innovative career paths to be an entrepreneur and a zest for higher studies.

Table 1: Abbreviations

Abbreviation	Meaning
CIE	Continuous Internal Evaluation
MSE	Mid Semester Examination
SEE	Semester End Examination
BSC	Basic Science Courses
ESC	Engineering Science Courses
VSEC/VSC	Vocational and Skill Enhancement Courses
VEC	Value Education Courses
AEC	Ability Enhancement Courses
PCC	Program Core Courses
PEC	Program Elective Courses
MDM	Multidisciplinary Minor
OE/OEC	Open Elective - other than a particular program
EEM	Entrepreneurship/Economics/ Management Courses (HSSM)
ELC	Research Methodology
	Computer Engineering Project (CEP)/ Field Project (FP)
	Project
	Internship/ On Job Training (OJT)
IKS	Indian Knowledge System
CC/CCC	Co-Curricular Courses
HOC	Honor Courses
EXT	Exit Courses
DMC	Double Minor Courses
HRC	Honor with Research Courses
SIP	Student Induction Program
L	Lecture
T	Tutorial
P/PR	Practical
TH	Theory
Lab	Laboratory
TW	Term Work
OR	Oral
CE	Civil Engineering
CS	Computer Engineering
ME	Mechanical Engineering
AD	Artificial Intelligence and Data Science Engineering
ET	Electronics and Telecommunication Engineering



GENERAL COURSE STRUCTURE

A. Definition of Credit:

Table 2: Credit Definition

1 Hour Lecture (L) per week	1 Credit
1 Hour Tutorial (T) per week	1 Credit
2 Hours Practical (P) per week	1 Credit

B. Range of Credits: (B.Tech. or Equivalent) in Tech. with Multidisciplinary Minor:

In the light of the fact that a typical NEP Compliant Model Four-year Undergraduate degree program in Technology has about 176 credits, the total number of credits proposed for the four-year B.Tech. in **Computer Engineering** with Multidisciplinary minor degree is kept as **172.**

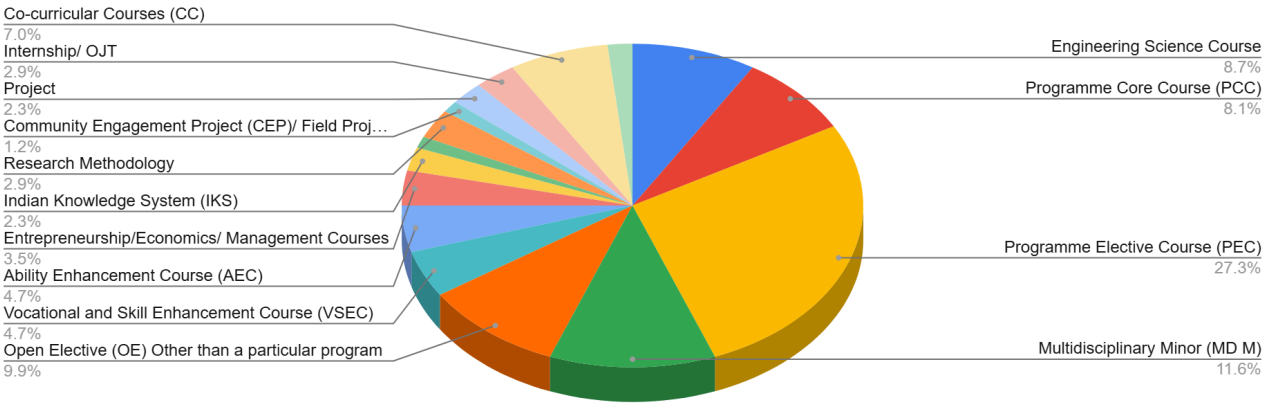
Table 3: Range of Credits

Course Category		Credits As PER NEP Guidelines	Proposed Credits
Basic Science Course	BSC/ESC	14-18	15
Engineering Science Course		16-12	14
Programme Core Course (PCC)	Program Courses	44-56	47
Programme Elective Course (PEC)		20	20
Multidisciplinary Minor (MD M)	Multidisciplinary Courses	14	17
Open Elective (OE) Other than a particular program		8	8
Vocational and Skill Enhancement Course (VSEC)	Skill Courses	8	8
Ability Enhancement Course (AEC)	Humanities Social Science and Management (HSSM)	4	4
Entrepreneurship/Economics/ Management Courses		2	4
Indian Knowledge System (IKS)		2	2
Value Education Course (VEC)		4	7
Research Methodology(RM)	Experiential Learning Courses	4	4
Community Engagement Project (CEP)/ Field Project (FP)		2	2
Project		4	5
Internship/ OJT		12	12
Co-curricular Courses (CC)	Liberal Learning Courses	4	3
Total Credits		**160-176**	**172**

C. Semester wise Credit Distribution Structure for Four Year B. Tech in Computer Engineering with Multidisciplinary Minor:

Table 4: Semester-wise Credit Distribution Structure

Semester		I	II	III	IV	V	VI	VII	VIII	Total Credits
Basic Science Course	BSC/ESC	8	7	-	-	-	-	-	-	15
Engineering Science Course		7	7	-	-	-	-	-	-	14
Programme Core Course (PCC)	Program Courses	-	3	11	8	9	4	9	3	47
Programme Elective Course (PEC)		-	-	-	-	6	5	6	3	20
Multidisciplinary Minor (MD M)	Multidisciplinary Courses	-	-	3	3	3	2	3	3	17
Open Elective (OE) Other than a particular program		-	-	-	3	2	3	-	-	8
Vocational and Skill Enhancement Course (VSEC)	Skill Courses	2	2	-	2	-	2	-	-	8
Ability Enhancement Course (AEC)	Humanities Social Science and Management (HSSM)	1	-	1	2	-	-	-	-	4
Entrepreneurship/Economics/ Management Courses		-	-	2	2	-	-	-	-	4
Indian Knowledge System (IKS)		2	-	-	-	-	-	-	-	2
Value Education Course (VEC)		-	-	3	2	2	-	-	-	7
Research Methodology	Experiential Learning Courses	-	-	-	-	-	4	-	-	4
Community Engagement Project (CEP)/ Field Project (FP)		-	-	2	-	-	-	-	-	2
Project		-	-	-	-	-	2	3	-	5
Internship / OJT		-	-	-	-	-	-	-	12	12
Co-curricular Courses (CC)	Liberal Learning Courses	1	2	-	-	-	-	-	-	3
Total Credits (Major)		**21**	**21**	**22**	**22**	**22**	**22**	**21**	**21**	**172**



Students can opt for any of the following as per the rules and regulations given by the institute:

1. B. Tech with Multidisciplinary Minor = Total 172 Credits
2. B. Tech with Multidisciplinary Minor and Honor = Total 190 Credits
3. B. Tech with Multidisciplinary Minor and Honor by Research = Total 190 Credits
4. B. Tech with Multidisciplinary Minors (Double Minor) = Total 190 Credits

Students will have the flexibility to enter a program in odd semesters and exit a programme after the successful completion of even semesters as per their future career needs. **Students exiting will be awarded provided they secure additional EIGHT credits in skill-based vocational courses.**

The credit structure for different levels under the Four-year Bachelor's Multidisciplinary B. Tech Programme with multiple entries and multiple exit options are as given below**:**

Table 5: Credit Requirements

Level	Qualification Title	Credit Requirements	Semester	Year
4.5	One Year UG Certificate in Tech.	42	2	1
5.0	Two Years UG Diploma in Tech.	86	4	2
5.5	Three Years Bachelor's Degree in Vocation (B. Voc.) or B. Sc. (Tech.)	130	6	3
6.0	4-Years Bachelor's degree (B.Tech. or Equivalent) in Tech. with Multidisciplinary Minor	172	8	4

D. Category-wise Courses

1. MULTIDISCIPLINARY MINOR (MD M)

- List of Multidisciplinary Minor Courses from other departments: Total 17 Credits
- The Minor courses may be from the different disciplines of the Engineering faculty, or they can be from different faculty altogether.
- Students have to choose the MD M in the Second Year and once opted then students can not change it throughout the semesters.

Table 6: Multidisciplinary Minors

Multidisciplinary Minor Baskets (UG)					
MDM Offered by Department	**To be Opted By Department**	**Sr No**	**Course Code**	**Course Name**	**Semester**
Civil Engineering	**Students other than Civil department**	1	24-MDM-CE-2-01	Fundamentals of Green Technology	III
			24-MDM-CE-2-02	Fundamentals of Green Technology Lab	
		2	24-MDM-CE-2-03	Green Building Rating System	IV
		3	24-MDM-CE-3-01	Water and Land Management	V
			24-MDM-CE-3-02	Water and Land Management Lab	
		4	24-MDM-CE-3-03	Socio-economic Management	VI
		5	24-MDM-CE-4-01	Urban Policy Framework	VII
			24-MDM-CE-4-02	Urban Policy Framework Lab	
		6	24-MDM-CE-4-03	Life Cycle Assessment	VIII
Computer Engineering & Artificial Intelligence and Data Science	**Students other than the Computer and AIDS department**	1	24-MDM-CS-2-01	Data Structure	III
			24-MDM-CS-2-02	Data Structure Lab	
		2	24-MDM-CS-2-03	Database Management System	IV
		3	24-MDM-CS-3-01	Object Oriented Programming in Java	V
			24-MDM-CS-3-02	Java Programming Lab	
		4	24-MDM-CS-3-03	Cloud Computing	VI
		5	24-MDM-CS-4-01	Data Science and Machine Learning	VII
			24-MDM-CS-4-02	Data Science and Machine Learning Lab	
		6	24-MDM-CS-4-03	Blockchain Technologies	VIII
Electronics & Telecommunication	**Students other than E&TC department**	1	24-MDM-ET-2-01A	Internet of Things	III
			24-MDM-ET-2-02A	Internet of Things Laboratory	
		2	24-MDM-ET-2-03A	Digital Electronics and Microprocessor	IV

Engineering		3	24-MDM-ET-3-01A	Drone Technology	V
			24-MDM-ET-3-02A	Drone Technology Laboratory	
		4	24-MDM-ET-3-03A	Robotics	VI
		5	24-MDM-ET-4-01A	Mobile Computing	VII
			24-MDM-ET-4-02A	Mobile Computing Laboratory	
		6	24-MDM-ET-4-03A	Wireless Sensor Networks	VIII
Electronics & Telecommunication Engineering	**Students other than E&TC department**	1	24-MDM-ET-2-01B	Lean Systems Fundamentals	III
		2	24-MDM-ET-2-02B	Industry 4.0 Concepts and Technologies	IV
		3	24-MDM-ET-3-01B	Advanced Lean Practices	V
			24-MDM-ET-3-02B	Lean Simulation and Automation Lab	
		4	24-MDM-ET-3-03B	Smart Manufacturing Systems and Digital Transformation	VI
		5	24-MDM-ET-4-01B	Robotics and Automation for Smart Manufacturing	VII
			24-MDM-ET-4-02B	Robotics and AI Implementation Lab	
		6	24-MDM-ET-4-03B	Lean Industry 4.0	VIII
Mechanical Engineering	**Students other than Mechanical department**	1	24-MDM-ME-2-01	Engineering Materials and Safety	III
			24-MDM-ME-2-02	Engineering Materials and Safety Lab	
		2	24-MDM-ME-2-03	Sustainable Energy Technology	IV
		3	24-MDM-ME-3-01	HVAC and Green Building	V
			24-MDM-ME-3-02	HVAC and Green Building Lab	
		4	24-MDM-ME-3-03	Project Planning & Management	VI
		5	24-MDM-ME-4-01	Hydraulic Machinery	VII
			24-MDM-ME-4-02	Hydraulic Machinery Lab	
		6	24-MDM-ME-4-03	Operations Research	VIII

#Note: You can refer syllabus of all SEM V and SEM VI MDM Courses from
https://snjb.org/engineering/syllabi-autonomy



2. OPEN ELECTIVES

- A Student can opt for any one course out of available institute-wide courses defined in the following list as Open Elective – provided he/she has not taken that particular course in his/her Programme core, Programme elective, Multidisciplinary Minor, other Open elective, and Vocational and Skill Enhancement courses, etc. throughout his/her four years of B. Tech Programme.
- The student must opt for a course that is compulsory from another discipline/branch, not from the same Major discipline/branch, and also the course must be not related to his/her major degree/branch courses.
- For Open Electives 8 credits are offered from semester IV to semester VI.
- Two courses of 3 credits and one course of 2 credits.

Table 7: Open Electives

Sr. No	Course Code	Course Name
Open Elective I (SEM-IV)		
1	24-OEC-2-4-01	Precision Agriculture
2	24-OEC-2-4-02	Soil and Water Conservation for Agriculture
3	24-OEC-2-4-03	Business Development, Marketing and Finance
4	24-OEC-2-4-04	Financial Accounting and Management
5	24-OEC-2-4-05	Information Technology Laws and Policies
Open Elective II (SEM-V)		
1	24-OEC-3-5-01	Agronics
2	24-OEC-3-5-02	Digital Marketing
3	24-OEC-3-5-03	Estimating and Costing
4	24-OEC-3-5-04	Sustainable Energy Engineering
5	24-OEC-3-5-05	Occupational Health and Safety
6	24-OEC-3-5-06	Circular Economy
Open Elective III (SEM-VI)		
1	24-OEC-3-6-01	E-Governance in Agriculture
2	24-OEC-3-6-02	Agri Entrepreneurship
3	24-OEC-3-6-03	Innovation Entrepreneur and Venture Development
4	24-OEC-3-6-04	Rural Finance Management and Budgeting
5	24-OEC-3-6-05	Renewable Energy for Agriculture

#Note: You can refer syllabus of sem V and VI Open Elective Courses from
https://snjb.org/engineering/syllabi-autonomy



3. MINORS

- In addition to 172 credits of B. Tech Programmes (Bachelor of Technology) i.e. Major in which the student has taken admission, a student may opt for Specialization Minor in another discipline/branch/emerging areas, not in Major discipline/branch.
- A student is required to earn an additional 18 credits in another discipline/ branch/ emerging areas for Specialization Minor distributed over semesters III to VIII.
- The total number of credits required to complete the Specialization Minor in another discipline/ emerging area is 18 credits, in addition to 172 credits in the Major.
- Minor Courses can be completed through an online platform.

Table 8: Minors

Minor Basket (UG) (2024-28)						
Minor Offered by	**To be Opted By Department**	**Minor Basket Name**	**Sr No**	**Course Code**	**Course Name**	**Semester**
Artificial Intelligence & Data Science Engineering	**Students other than the Computer and AIDS department**	**High Performance Computing**	1	24-DMC-AD-2-01	Computer Networks	**III**
			2	24-DMC-AD-2-02	Cloud Computing	**IV**
			3	24-DMC-AD-3-03	Distributed Computing	**V**
			4	24-DMC-AD-3-04	Blockchain Technology	**VI**
			5	24-DMC-AD-4-05	High Performance Computing	**VII**
			6	24-DMC-AD-4-06	Advance Cloud Computing	**VIII**
Civil Engineering	**Students other than Civil department**	**Infrastructure Engineering**	1	24-DMC-CE-2-01	Infrastructure Planning and Management	**III**
			2	24-DMC-CE-2-02	Infrastructure Economics	**IV**
			3	24-DMC-CE-3-03	Project Formulation and Appraisal	**V**
			4	24-DMC-CE-3-04	Advanced and Sustainable Materials in Infrastructure	**VI**
			5	24-DMC-CE-4-05	Management Information Systems	**VII**
			6	24-DMC-CE-4-06	Computational Methods in Infrastructure Engineering	**VIII**
Computer Engineering	**Students other than Computer and AIDS department**	**Data Science**	1	24-DMC-CS-2-01	Foundation of Data Science	**III**
			2	24-DMC-CS-2-02	Principles of Artificial Intelligence and Machine Learning	**IV**
			3	24-DMC-CS-3-03	Data analytics	**V**



			4	24-DMC-CS-3-04	Business Intelligence & Analytics	**VI**
			5	24-DMC-CS-4-05	Natural Language Processing	**VII**
			6	24-DMC-CS-4-06	Large Language Models	**VIII**
Electronics & Telecommunication Engineering	**Students other than E&TC department**	**Embedded System**	1	24-DMC-ET-2-01	Digital Electronics	**III**
			2	24-DMC-ET-2-02	Microprocessor & Microcontroller	**IV**
			3	24-DMC-ET-3-03	Analog Circuits	**V**
			4	24-DMC-ET-3-04	Mechatronics	**VI**
			5	24-DMC-ET-4-05	Embedded System	**VII**
			6	24-DMC-ET-4-06	Internet of Things	**VIII**
Mechanical Engineering	**Students other than Mechanical department**	**Sustainable Energy Engineering**	1	24-DMC-ME-2-01	Fundamentals of Sustainable Energy Systems	**III**
			2	24-DMC-ME-2-02	Solar PV Design Optimization & Manufacturing	**IV**
			3	24-DMC-ME-3-03	Solar Energy Harnessing Technologies	**V**
			4	24-DMC-ME-3-04	Grid Integration and Smart Grid technologies	**VI**
			5	24-DMC-ME-4-05	Sustainable Engineering Solutions	**VII**
			6	24-DMC-ME-4-06	Sustainability Assessment and Analysis	**VIII**

#Note for NPTEL/SYAYAM: Approved courses and platforms will be enlisted timely by authorities along with rules and regulations

#Note: You can refer syllabus of all SEM V and VI Minor Courses from
https://snjb.org/engineering/syllabi-autonomy



4. HONORS

- In addition to 172 credits of B. Tech Programmes (Bachelor of Technology) i.e. Major in which the student has taken admission, a student may opt for Honors in the same Tech. discipline/branch / Emerging Areas.
- A student is required to earn an additional 18 credits in the same Tech. discipline/ branch / Emerging Areas for Honors distributed over semesters III to VIII.
- The total number of credits required to complete the Honors in the same Tech. discipline/ Emerging Areas is 18 credits, in addition to 172 credits in Major.
- Students will have to compulsorily choose Honors from the same Tech. discipline/branch.
- Honors Degree in the Bachelor of Engineering programme shall be awarded to students earning additional total credits of all six semesters from the second year to final year, i.e., 18 Credits, in addition to 172 credits or 130 credits respectively. The student admitted in the first year must earn 172 credits and 130 credits admitted in lateral entry (admitted after Diploma or B.Sc.) in the second year.
- Minor Courses can be completed through an online platform.

The student has to choose One Honor out of the Two Honor groups provided below

Honors offered by Computer Engineering are as follows:

Table 9: Honors

Sr No	Name of Honors Offered by Department
A.	Blockchain Technologies
B.	Cyber Security

The detailed syllabus structure for the same is as follows:

Table 10A: Honors in Blockchain Technologies

Sr. No	Category	SEM	Course Code	Course Name	Teaching Scheme				
					Hours				Credits
					L	T	P	Total Hours	
01	HOC	III	24-HOC-CS-2-01A	Foundation of Blockchain Technology	3	-	-	3	3
02	HOC	IV	24-HOC-CS-2-02A	Decentralize and Blockchain Technologies	3	-	-	3	3
03	HOC	V	24-HOC-CS-3-03A	Blockchain Architecture Design	3	-	-	3	3
04	HOC	VI	24-HOC-CS-3-04A	Smart Contract and Cryptocurrency	3	-	-	3	3
05	HOC	VII	24-HOC-CS-4-05A	Blockchain Security	3	-	-	3	3
06	HOC	VIII	24-HOC-CS-4-06A	Blockchain Applications	3	-	-	3	3
Total					**18**	-	-	**18**	**18**

Table 10B: Honors in Honors in Cyber Security

Sr. No	Category	SEM	Course Code	Course Name	Teaching Scheme				
					Hours				Credits
					L	T	P	Total Hours	
01	HOC	III	24-HOC-CS-2-01B	Cryptography and Network Security	3	-	-	3	3
02	HOC	IV	24-HOC-CS-2-02B	Information Security	3	-	-	3	3
03	HOC	V	24-HOC-CS-3-03B	Ethical Hacking	3	-	-	3	3
04	HOC	VI	24-HOC-CS-3-04B	Digital Forensic	3	-	-	3	3
05	HOC	VII	24-HOC-CS-4-05B	Network Forensics and Cyber Threat Analysis	3	-	-	3	3
06	HOC	VIII	24-HOC-CS-4-06B	Cyber Physical Systems	3	-	-	3	3
Total					**18**	-	-	**18**	**18**

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#Note: You can refer syllabus of all SEM V and SEM VI Honors Courses from
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5. HONORS WITH RESEARCH AND MULTIDISCIPLINARY MINOR

- The Student will work on a Research Project or Dissertation for 18 Credits in the Fourth Year in the respective discipline.
- The distribution of 18 Credits for Research projects in Sem-VII and Sem-VIII is given below.
- To get a B. Tech in Computer Engineering-Honors with Research and Multidisciplinary Minor degree Students need to earn a total of 190 Credits which consist of 172 credits of regular Multidisciplinary Minor courses, 18 Credits of Honor courses, 18 credits of Research courses.

Table 11: Honors with Research and Multidisciplinary Minor (Sem-VII)

Final Year B. Tech Semester-VII													
Course Code	Course Name	Teaching Scheme				Evaluation Scheme							
		Hours			Credit	Theory Course				Lab Course			Total Marks
		L	T	P	Total	CIE	MSE	SEE	TH Marks	TW	PR	OR	
24-HRC-4-01	Intellectual Property Right (IPR)	2	-	-	2	-	50	50	100	-	-	-	100
24-HRC-4-02	Research Project (Synopsis) Phase-I	-	-	4	2	-	-	-	-	50	-	50	100
24-HRC-4-03	Research Specific Core Course-I (Online NPTEL Course#)	3	-	-	3	-	50	50	100	-	-	-	100
Total		**5**	**-**	**4**	**7**	**-**	**100**	**100**	**200**	**50**	**-**	**50**	**300**

#Note for NPTEL/SYAYAM: Approved courses and platforms will be enlisted timely by authorities along with rules and regulations

Table 12: Honors with Research and Multidisciplinary Minor (Sem-VIII)

Final Year B. Tech Semester-VIII													
Course Code	Course Name	Teaching Scheme				Evaluation Scheme							
		Hours			Credit	Theory Course				Lab Course			Total Marks
		L	T	P	Total	CIE	MSE	SEE	TH Marks	TW	PR	OR	
24-HRC-4-04	Research Project Phase-II	-	-	22	11	-	-	-	-	50	-	50	100
Total		**-**	**-**	**22**	**11**	**-**	**-**	**-**	**-**	**50**	**-**	**50**	**100**

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Curriculum and Evaluation Scheme for Third Year B. Tech. in Computer Engineering with Multidisciplinary Minor

To be implemented for 2024-28 Batch

(With Effect from Academic Year 2026-27)

TEACHING AND EVALUATION SCHEME FOR FIRST YEAR B-TECH

Semester – I

Sr. No	Category	Course Code	Course Name	Teaching Scheme					Evaluation Scheme							
				Hours				Credits	Theory Course				Lab Course			Total Marks
				L	T	P	Total Hours		CIE	MSE	SEE	TH Marks	TW	PR	OR	
1	BSC	24-BSC-1-01	Engineering Physics	3	-	-	3	3	20	20	60	100	-	-	-	100
2	BSC	24-BSC-1-03	Linear Algebra And Differential Calculus	3	1	-	4	4	20	20	60	100	-	-	-	100
3	ESC	24-ESC-1-01	Basic Electrical and Electronics Engineering	3	-	-	3	3	20	20	60	100	-	-	-	100
4	ESC	24-ESC-1-02	Programming and Problem Solving	2	-	-	2	2	20	20	60	100	-	-	-	100
5	BSC	24-BSC-1-05	Engineering Physics Laboratory	-	-	2	2	1	-	-	-	-	25	-	-	25
6	ESC	24-ESC-1-05	Basic Electrical and Electronics Engineering Lab	-	-	2	2	1	-	-	-	-	25	-	-	25
7	ESC	24-ESC-1-07	Programming and Problem Solving Lab	-	-	2	2	1	-	-	-	-	25	-	-	25
8	VSEC	24-VSC-1-01	TechSkill	-	-	4	4	2	-	-	-	-	50	-	-	50
9	CCC		Co-curricular Course -I	-	-	2	2	1	-	-	-	-	50	-	-	50
10	IKS	24-IKS-1-01	Indian Knowledge System	-	2	-	2	2	-	-	-	-	50	-	-	50
11	AEC	24-AEC-1-01	Professional Communication Skills	-	1	-	1	1	-	-	-	-	25	-	-	25
Total				**11**	**4**	**12**	**27**	**21**	**80**	**80**	**240**	**400**	**250**	**-**	**-**	**650**

Course Code	Basket of Co-curricular Course
24-CCC-1-A	Yoga
24-CCC-1-B	Sports
24-CCC-1-C	NSS (National Service Scheme)
24-CCC-1-D	Cultural

Note: Students have to select any one course from the above basket.

Induction Program (Mandatory)	3 Weeks Duration
The induction program (as per AICTE guidelines) is to be completed at the start of the first year.	• SIP Module 1: UHV 1 • SIP Module 2: Physical Health and Related Activities • SIP Module 3: Familiarization of Department/ Branch and Innovation • SIP Module 4: Visit to a Local Area • SIP Module 5: Lectures by Eminent People • SIP Module 6: Proficiency Modules • SIP Module 7: Literature / Literary Activities • SIP Module 8: Creative Practices • SIP Module 9: Extra Curricular Activities



Semester – II

Sr. No	Category	Course Code	Course Name	Teaching Scheme					Evaluation Scheme							
				Hours				Credits	Theory Course				Lab Course			Total Marks
				L	T	P	Total Hours		CIE	MSE	SEE	TH Marks	TW	PR	OR	
1	BSC	24-BSC-1-02	Engineering Chemistry	3	-	-	3	3	20	20	60	100	-	-	-	100
2	BSC	24-BSC-1-04	Statistics and Integral Calculus	3	-	-	3	3	20	20	60	100	-	-	-	100
3	ESC	24-ESC-1-03	Engineering Graphics	3	-	-	3	3	20	20	60	100	-	-	-	100
4	ESC	24-ESC-1-04	Smart Building and Materials	2	-	-	2	2	20	20	60	100	-	-	-	100
5	PCC	24-PCC-CS-1-01	Object Oriented Programming using Java	2	-	-	2	2	20	-	30	50	-	-	-	50
6	BSC	24-BSC-1-06	Engineering Chemistry Laboratory	-	-	2	2	1	-	-	-	-	25	-	-	25
7	ESC	24-ESC-1-08	Engineering Graphics Lab	-	-	2	2	1	-	-	-	-	25	-	-	25
8	ESC	24-ESC-1-09	Smart Building and Materials Lab	-	-	2	2	1	-	-	-	-	25	-	-	25
9	PCC	24-PCC-CS-1-02	Java Programming Lab	-	-	2	2	1	-	-	-	-	25	25	-	50
10	VSEC	24-VSC-1-02	TechShop	-	-	4	4	2	-	-	-	-	50	-	-	50
11	CCC	24-CCC-1-05	Co-curricular Course -II	-	-	4	4	2	-	-	-	-	25	-	-	25
Total				**13**	**-**	**16**	**29**	**21**	**100**	**80**	**270**	**450**	**175**	**25**	**-**	**650**

Level 4.5 Exit Criteria: Mandatory Courses to be completed after the first year to obtain One Year UG Certificate in Computer Engineering

Sr. No	Category	Course Code	Course Name	Teaching Scheme					Evaluation Scheme							
				Hours				Credits	Theory Course				Lab Course			Total Marks
				L	T	P	Total Hours		CIE	MSE	SEE	TH Marks	TW	PR	OR	
1	EXT	24-EXT-1-01	Internship / Fieldwork/OJT	-	-	8	8	4	-	-	-	-	100	-	-	100
2	EXT	24-EXT-1-02	Minor Project	-	-	8	8	4	-	-	-	-	50	-	50	100
Total				**-**	**-**	**16**	**16**	**8**	**-**	**-**	**-**	**-**	**150**	**-**	**50**	**200**

SNJB's Late Sau. K. B. Jain College of Engineering, Chandwad

(An Autonomous Institute Affiliated to Savitribai Phule Pune University, Pune)

Curriculum and Evaluation Scheme for Third Year B. Tech. in Computer Engineering with Multidisciplinary Minor

To be implemented for 2024-28 Batch

(With Effect from Academic Year 2026-27)

TEACHING AND EVALUATION SCHEME FOR SECOND YEAR B-TECH

Semester – III

Sr. No	Category	Course Code	Course Name	Teaching Scheme					Evaluation Scheme							
				Hours				Credits	Theory Course				Lab Course			Total Marks
				L	T	P	Total Hours		CIE	MSE	SEE	TH Marks	TW	PR	OR	
1	PCC	24-PCC-CS-2-01	Discrete Mathematics	3	-	-	3	3	20	20	60	100	-	-	-	100
2	PCC	24-PCC-CS-2-02	Data Structures & Algorithms	3	-	-	3	3	20	20	60	100	-	-	-	100
3	PCC	24-PCC-CS-2-03	Computer Organization and Architecture	3	-	-	3	3	20	20	60	100	-	-	-	100
4	MD M		Multi-Disciplinary Minor-I	2	-	-	2	2	20	-	30	50	-	-	-	50
5	PCC	24-PCC-CS-2-04	Data Structures Lab	-	-	4	4	2	-	-	-	-	50	50	-	100
6	MD M		MultiDisciplinary Minor-I Lab	-	-	2	2	1	-	-	-	-	25	25	-	50
7	EEM	24-EEM-2-01	Engineering Economics	1	-	2	3	2	-	-	-	-	25	-	-	25
8	AEC	24-AEC-2-01	Business Communication Skill	-	-	2	2	1	-	-	-	-	25	-	-	25
9	VEC	24-VEC-2-01	Universal Human Values-II	3	-	-	3	3	-	-	-	-	50	-	-	50
10	ELC (CEP/FP)	24-ELC-CS-2-01	Community Engagement Project (CEP) / Field Project (FP)	-	-	4	4	2	-	-	-	-	25	-	25	50
Total				**15**	**-**	**14**	**29**	**22**	**80**	**60**	**210**	**350**	**200**	**75**	**25**	**650**



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To be implemented for 2024-28 Batch

(With Effect from Academic Year 2026-27)

Semester – IV

Sr. No	Category	Course Code	Course Name	Teaching Scheme					Evaluation Scheme							
				Hours				Credits	Theory Course				Lab Course			Total Marks
				L	T	P	Total Hours		CIE	MSE	SEE	TH Marks	TW	PR	OR	
1	PCC	24-PCC-CS-2-05	Database Management System	3	-	-	3	3	20	20	60	100	-	-	-	100
2	PCC	24-PCC-CS-2-06	Design and Analysis of Algorithm	3	-	-	3	3	20	20	60	100	-	-	-	100
3	MDM		Multi Disciplinary Minor -II	3	-	-	3	3	20	20	60	100	-	-	-	100
4	OE		Open Elective-I	3			3	3	20	20	60	100	-	-	-	100
5	PCC	24-PCC-CS-2-07	Database Management System Lab	-	-	4	4	2	-	-	-	-	50	50	-	100
6	AEC		Modern Language	1	-	2	3	2	-	-	-	-	-	-	25	25
7	EEM	24-EEM-2-02	Entrepreneurship Development	1	-	2	3	2	-	-	-	-	50	-	-	50
8	VSEC	24-VSEC-CS-2-01	Application Development Programming I	-	-	4	4	2	-	-	-	-	25	25	-	50
9	VEC	24-VEC-2-02	Digital and Technological Solutions	1	-	2	3	2	-	-	-	-	25	-	-	25
10	AEC	24-AEC-2-03	Disaster Risk Reduction and Management	1	2	-	3	-	-	-	-	-	-	-	-	-
Total				**16**	**2**	**14**	**32**	**22**	**80**	**80**	**240**	**400**	**150**	**75**	**25**	**650**



AEC- Modern Language Basket		
Course Code	**Course Name**	**Who can Opt**
Indian Languages		
24-AEC-2-02-A	Modern Language- Basic Marathi	Students who have not studied this course from Grade I to Grade X can opt this course in the curriculum
24-AEC-2-02-B	Modern Language- Advance Marathi	Students who have studied this course from Grade I to Grade XIcan opt this course in the curriculum
24-AEC-2-02-C	Modern Language- Basic Hindi	Students who have not studied this course from Grade I to Grade X can opt this course in the curriculum
24-AEC-2-02-D	Modern Language- Advance Hindi	Students who have studied this course from Grade I to Grade X can opt this course in the curriculum
24-AEC-2-02-E	Modern Language- Sanskrit	Students who have not studied this course from Grade I to Grade X can opt this course in the curriculum
Foreign Languages		
24-AEC-2-02-F	Modern Language- Japanese	Students who have not studied this course from Grade I to Grade X can opt this course in the curriculum
24-AEC-2-02-G	Modern Language- German	Students who have not studied this course from Grade I to Grade X can opt this course in the curriculum
24-AEC-2-02-H	Modern Language- French	Students who have not studied this course from Grade I to Grade X can opt this course in the curriculum

#Note: Students have to select any one course from the above basket.



Level 5.0 Exit Criteria

Mandatory Courses to be completed after Second Year for obtaining Two Years UG Diploma in Computer Engineering

Sr. No	Category	Course Code	Course Name	Teaching Scheme					Evaluation Scheme							
				Hours				Credits	Theory Course				Lab Course			Total Marks
				L	T	P	Total Hours		CIE	MSE	SEE	TH Marks	TW	PR	OR	
1	EXT	24-EXT-CS-2-01	Internship / Fieldwork/OJT	-	-	8	8	4	-	-	-	-	100	-	-	100
2	EXT	24-EXT-CS-2-02	Minor Project	-	-	8	8	4	-	-	-	-	50	-	50	100
Total				**-**	**-**	**16**	**16**	**8**	**-**	**-**	**-**	**-**	**150**	**-**	**50**	**200**

TEACHING AND EVALUATION SCHEME FOR THIRD YEAR B-TECH

Semester – V

Sr. No	Category	Course Code	Course Name	Teaching Scheme					Evaluation Scheme							
				Hours				Credits	Theory Course				Lab Course			Total Marks
				L	T	P	Total Hours		CIE	MSE	SEE	TH Marks	TW	PR	OR	
1	PCC	24-PCC-CS-3-01	Automata Theory	3	1	-	4	4	20	20	60	100	-	-	-	100
2	PCC	24-PCC-CS-3-02	Computer Network	3	-	-	3	3	20	20	60	100	-	-	-	100
3	PEC		Program Elective Course- I	4	-	-	4	4	20	20	60	100	-	-	-	100
4	MD M		MultiDisciplinary Minor-III	2	-	-	2	2	20	-	30	50	-	-	-	50
5	OE		Open Elective - II	2	-	-	2	2	20	-	30	50	-	-	-	50
6	PCC	24-PCC-CS-3-03	Computer Network Lab	-	-	4	4	2	-	-	-	-	25		25	50
7	PEC		Program Elective Course- I Lab	-	-	4	4	2	-	-	-	-	50	50		100
8	MD M		MultiDisciplinary Minor-III Lab	-	-	2	2	1	-	-	-	-	25	25		50
9	VEC	24-VEC-3-01	Environmental Studies	2	-	-	2	2	20	-	30	50	-	-	-	50
Total				**16**	**1**	**10**	**27**	**22**	**120**	**60**	**270**	**450**	**100**	**75**	**25**	**650**

Program Elective Course – I				
	Course Code-TH	Name of the Course- TH	Course Code-PR	Name of the Course(PR/OR)
A	24-PEC-CS-3-01A	Computer Graphics	24-PEC-CS-3-02A	Computer Graphics Laboratory
B	24-PEC-CS-3-01B	Cyber Security & Digital Forensic	24-PEC-CS-3-02B	Cyber Security & Digital Forensic Lab
C	24-PEC-CS-3-01C	Artificial Intelligence	24-PEC-CS-3-02C	Artificial Intelligence Lab
D	24-PEC-CS-3-01D	Pervasive Computing	24-PEC-CS-3-02D	Pervasive Computing Lab
E	24-PEC-CS-3-01E	Web Technology	24-PEC-CS-3-02E	Web Technology Lab



Semester – VI

Sr. No	Category	Course Code	Course Name	Teaching Scheme					Evaluation Scheme							
				Hours				Credits	Theory Course				Lab Course			Total Marks
				L	T	P	Total Hours		CIE	MSE	SEE	TH Marks	TW	PR	OR	
1	PCC	24-PCC-CS-3-04	Software Engineering	3	-	-	3	3	20	20	60	100	-	-	-	100
2	PEC		Program Elective Course- II	3	-	-	3	3	20	20	60	100	-	-	-	100
3	MD M		Multi Disciplinary Minor-IV	2	-	-	2	2	20		30	50	-	-	-	50
4	OE		Open Elective -III	3	-	-	3	3	20	20	60	100	-	-	-	100
5	PCC	24-PCC-CS-3-05	Python Programming	-	-	2	2	1	-	-	-	-	-	25	-	25
6	PEC		Program Elective Course- II Lab	-	-	4	4	2	-	-	-	-	25		25	50
7	VSEC	24-VSEC-CS-3-01	Application Development Programming II	-	-	4	4	2	-	-	-	-	25	25	-	50
8	ELC (RM)	24-ELC-3-01	Research Methodology	4	-	-	4	4	20	20	60	100	-	-	-	100
9	ELC (PR)	24-ELC-CS-3-02	Project-stage-I	-	-	4	4	2	-	-	-	-	25		50	75
Total				**15**	**0**	**14**	**29**	**22**	**100**	**80**	**270**	**450**	**75**	**50**	**75**	**650**

Program Elective Courses For SEM VI				
	Course Code-TH	Name of the Course- TH	Course Code-PR	Name of the Course(PR/OR)
A	24-PEC-CS-3-03A	Game Development & Animation	24-PEC-CS-3-04A	Game Development & Animation Lab
B	24-PEC-CS-3-03B	Network Security	24-PEC-CS-3-04B	Network Security Lab
C	24-PEC-CS-3-03C	Data Science	24-PEC-CS-3-04C	Data Science Lab
D	24-PEC-CS-3-03D	High Performance Computing	24-PEC-CS-3-04D	High Performance Computing Lab
E	24-PEC-CS-3-03E	Advanced Web Technology	24-PEC-CS-3-04E	Advanced Web Technology Lab



Level 5.5 Exit Criteria
Mandatory Courses to be completed after Third Year for obtaining Three Year Bachelor's Degree in Vocation (B. Voc.) in Computer Engineering

Sr. No	Category	Course Code	Course Name	Teaching Scheme					Evaluation Scheme							
				Hours				Credits	Theory Course				Lab Course			Total Marks
				L	T	P	Total Hours		CIE	MSE	SEE	TH Marks	TW	PR	OR	
1	EXT	24-EXT-CS-3-01	Internship / Fieldwork/OJT	-	-	8	8	4	-	-	-	-	100	-	-	100
2	EXT	24-EXT-CS-3-02	Minor Project	-	-	8	8	4	-	-	-	-	50	-	50	100
Total				**-**	**-**	**16**	**16**	**8**	**-**	**-**	**-**	**-**	**150**	**-**	**50**	**200**

TEACHING AND EVALUATION SCHEME FOR FINAL YEAR B-TECH

Semester – VII

Sr. No	Category	Course Code	Course Name	Teaching Scheme					Evaluation Scheme							
				Hours				Credits	Theory Course				Lab Course			Total Marks
				L	T	P	Total Hours		CIE	MSE	SEE	TH Marks	TW	PR	OR	
1	PCC	24-PCC-CS-401	Machine Learning	3			3	3	20	20	60	100				100
2	PCC	24-PCC-CS-402	System Programming and Operating System	3			3	3	20	20	60	100				100
3	PEC		Program Elective -III	4			4	4	20	20	60	100				100
4	MD M		MultiDisciplinary Minor-V	2			2	2	20		30	50				50
5	PCC	24-PCC-CS-403	Machine Learning Lab			4	4	2					25	25		50
6	PCC	24-PCC-CS-404	System Programming and Operating System Lab			2	2	1					25		25	50
7	PEC		Program Elective -III Lab			4	4	2					25	25		50
8	MD M		MultiDisciplinary Minor-V Lab			2	2	1					25		25	50
9	ELC (PR)	24-ELC-CS-401	Project-stage-II			6	6	3					50		50	100
Total				**12**	**0**	**18**	**30**	**21**	**80**	**60**	**210**	**350**	**150**	**50**	**100**	**650**

Program Elective Course For SEM VII				
	Course Code-TH	**Name of the Course- TH**	**Course Code-PR**	**Name of the Course(PR/OR)**
A	24-PEC-CS-4-05A	Virtual Reality & Augmented Reality	24-PEC-CS-4-06A	Virtual Reality & Augmented Reality Lab
B	24-PEC-CS-4-05B	Blockchain	24-PEC-CS-4-06B	Blockchain Lab
C	24-PEC-CS-4-05C	Natural Language Processing	24-PEC-CS-4-06C	Natural Language Processing Lab
D	24-PEC-CS-4-05D	Fog and Edge Computing	24-PEC-CS-4-06D	Fog and Edge Computing Lab
E	24-PEC-CS-4-05E	UI/ UX Design	24-PEC-CS-4-06E	UI/UX Design Lab



Semester – VIII

Sr. No	Category	Course Code	Course Name	Teaching Scheme					Evaluation Scheme							
				Hours				Credits	Theory Course				Lab Course			Total Marks
				L	T	P	Total Hours		CIE	MSE	SEE	TH Marks	TW	PR	OR	
1	PCC	24-PCC-CS-4-05	Cloud Computing	3			3	3	40		60	100				100
2	PEC		Program Elective Course-IV	3			3	3	40		60	100				100
3	MD M		Multi Disciplinary Minor VI	3			3	3	40		60	100				100
4	ELC	24-ELC-CS-4-02	Internship			24	24	12					200		150	350
Total				**9**	**0**	**24**	**33**	**21**	**120**	**0**	**180**	**300**	**200**	**0**	**150**	**650**

Note: The above courses from Sr. No. 1 to 3 of SEM-VIII will be conducted in online mode or may be mapped with suitable NPTEL/SWAYAM courses.

Program Elective Course For SEM VIII		
	Course Code-TH	**Name of the Course- TH**
A	24-PEC-CS-4-07A	Compiler
B	24-PEC-CS-4-07B	DevOps
C	24-PEC-CS-4-07C	Deep Learning
D	24-PEC-CS-4-07D	Quantum Computing
E	24-PEC-CS-4-07E	Software Testing

SEM V



24-PCC-CS-3-01: Automata Theory

Teaching Scheme: Theory: 03 Hours/Week Tutorial: 01 Hour/Week	**Credit:** 04	**Examination Scheme:** **CIE** : 20 Marks **MSE** : 20 Marks **SEE** : 60 Marks

Prerequisites Courses: Discrete Mathematics (24-PCC-CS-2-01)

Companion Course: - –

Course Objectives:

- To introduce the students to basics of Theory of Computation
- To study abstract computing models to provide a formal connection between algorithmic
- problem solving and the theory of languages
- To understand Grammar, Pushdown. Automata and Turing Machine for language processing and algorithm design
- To learn about the theory of computability and complexity for algorithm design

Course Outcomes:

After completion of the course, learners should be able to

CO No	CO	BL
CO1	Apply formal language and automata concepts to design FA without output	3
CO2	Build and compare Moore and Mealy machines for output-based automation.	4
CO3	Construct regular expressions to present regular language	3
CO4	Design Context Free Grammars and learn to simplify the grammar.	3
CO5	Generate Pushdown Automaton model for the Context Free Language	3
CO6	Analyze language acceptability using Turing Machines and their variants.	4

Course Contents

Unit I	Formal Language Theory and Finite Automata (FA)	09 Hours

Introduction: Alphabets, strings, and languages, notion of formal languages, introduction to finite automata, simplified notation, state transition diagram, transition table, language accepted by a finite automaton (FA), types of FAs.
FA without output: Deterministic Finite Automata (DFA), Non-Deterministic Finite Automata (NFA) , ε-NFA- formal definitions, examples, equivalence and their inter-conversion, minimization of DFAs.

#Exemplar/Case Studies: Design a finite automata model of a vending machine that handles user interactions including item selection, payment processing, item dispensing etc.

***Mapping of Course Outcomes**	CO1	
Unit II	**Finite Automata with output**	**06 Hours**
FA with output: Moore and Mealy machines – formal definition, examples, equivalence and their inter-conversion, Regular Language and Pumping lemma for regular languages.		
#Exemplar/Case Studies: Design and Verification of an Automatic Toll Booth System		
***Mapping of Course Outcomes**	CO2	
Unit III	**Regular Expressions (RE)**	**07 Hours**
Introduction: Regular expressions (RE) – formal definition, operators of RE, precedence of operators, algebraic laws for RE, converting languages to regular expressions, equivalence of two REs, **Conversions:** RE to NFA, RE to DFA, DFA to RE using loop elimination and Arden's theorem, closure properties of REs.		
#Exemplar/Case Studies: Validation of user email formats using Regular Expressions		
***Mapping of Course Outcomes**	CO3	
Unit IV	**Context free Grammar (CFG)**	**09 Hours**
Introduction: Context-Free Grammar (CFG) - formal definition, derivation, parse tree, examples, ambiguity in grammar, converting ambiguous CFGs to unambiguous CFGs. **Simplification of CFG:** Elimination of ε-productions, unit productions, useless symbols. **Normal Forms:** Chomsky Normal Form (CNF), Greibach Normal Form (GNF), Chomsky hierarchy, regular grammar, left-linear and right-linear grammar, interconversion between DFA and linear grammar, closure properties of context-free languages.		
#Exemplar/Case Studies: Extraction of structured information from social media posts using grammar		
***Mapping of Course Outcomes**	CO4	
Unit V	**Pushdown Automata (PDA)**	**07 Hours**
Introduction: Pushdown Automata (PDA) - formal definition, equivalence of acceptance by final state and by empty stack, types of PDA – Deterministic PDA (DPDA) and Non-Deterministic PDA (NPDA), examples, PDA and context-free languages (CFLs), equivalence of PDA and CFG, interconversion between PDA and CFG.		
#Exemplar/Case Studies: Parsing and PDA: Top-Down Parsing, Bottom-up Parsing simulation showing use of PDA		
***Mapping of Course Outcomes**	CO5	
Unit VI	**Turing Machines (TM)**	**07 Hours**
Introduction: Turing Machines (TM) - formal definition, language acceptability by TM, variants of Turing Machines, universal Turing Machine, halting problem of TM, recursively enumerable languages, recursive languages.		
#Exemplar/Case Studies: Modeling an Intelligent Chatbot Using the Turing Machine Framework		
***Mapping of Course Outcomes**	CO6	

Learning Resources
Text Books :
T1. John E. Hopcroft, Rajeev Motwani, Jeffrey D. Ullman, "Introduction to Automata Theory, Languages and Computation", Addison-Wesley, ISBN 0-201-44124-1. **T2.** Daniel Cohen, "Introduction to Computer Theory", Wiley & Sons, ISBN 97881265133454.
Reference Books :
R1. Sanjeev Arora, Boaz Barak, "Computational Complexity: A Modern Approach", Cambridge University Press, ISBN 0521424267, 97805214242643. **R2.** John Martin, "Introduction to Languages and The Theory of Computation", 2nd Edition, McGraw Hill Education, ISBN 1-25-900558-5, 978-1-25-900558-9. **R3.** J. Carroll, D. Long, "Theory of Finite Automata", Prentice Hall, ISBN 0-13-913708-45. **R4.** Kavi Mahesh, "Theory of Computation: A Problem Solving Approach", Wiley India, ISBN 1081265331106. **R5.** Michael Sipser, "Introduction to the Theory of Computation", Cengage Learning, ISBN 97811331878137. **R6.** Vivek Kulkarni, "Theory of Computation", Oxford University Press, ISBN 0-19-808458.
Additional Resources: (Books, e-Resources) ● https://omereingold.wordpress.com/cs-154-introduction-to-automata-and-complexity-theory/ ● Resources | Automata, Computability, and Complexity | Electrical Engineering and Computer Science | MIT OpenCourseWare ● https://www.udemy.com/course/deterministic-finite-automata-theory-of-computaion
MOOC Courses links : ● https://onlinecourses.nptel.ac.in/noc21_cs83/preview ● https://nptel.ac.in/courses/106105196

24-PCC-CS-3-02: Computer Network		
Teaching Scheme: **Theory: 3 Hours/Week**	**Credit: 3**	**Examination Scheme:** **CIE : 20 Marks** **MSE :20 Marks** **SEE : 60 Marks**

Companion Course: 24-PCC-CS-2-01 Discrete Mathematics

Course Objectives:

- To understand the fundamental concepts of networking standards, protocol , topologies and network devices
- To learn different techniques for framing, error control, flow control and routing and introducing IEEE standards
- To learn different protocols of Network layer and routing algorithms
- To learn different process to process communication in transport layers with socket programming
- To learn application layer protocols for internet communication and services
- To learn modern concepts and technologies in advanced computer network

Course Outcomes:

After completion of the course, learners should be able to

CONo	CO	BL
CO1	Understanding fundamental concepts of Computer Networks, architectures, protocols and technologies	2
CO2	Illustrate the working and functions of data link layer	3
CO3	Analyze the working of different routing protocols and mechanisms	3
CO4	Implement client-server applications using sockets	3
CO5	Illustrate role of application layer with its protocols, client-server architectures	3
CO6	Analyze, and apply advanced networking principles to modern communication systems.	3

Course Contents

Unit I	**Introduction To Computer Networks**	**6 Hours**

Definition, **Types of Networks**: Local area networks (LAN), Metropolitan area networks (MAN), Personal Area Network (PAN) ,Wide area networks (WAN), **Network Models**: The OSI Reference Model, TCP/IP Model, **Network Topologies** : Bus topology ,ring topology star topology , Tree topology , Mesh topology , Hybrid Topology, **Types of Transmission Media** : Twisted-Pair Cable, Coaxial Cable ,Fiber-Optic Cable, **Network Devices**: Hub, Bridge, Switch, Router, Gateway, Access Point, **Line Coding Schemes**: Manchester and Differential Manchester Encodings.

#Exemplar/Case Studies : Study of Campus wide networking.		
***Mapping of Course Outcomes**	**CO1**	
Unit II	**Data Link Layer**	**8 Hours**
Introduction , Function of Data link layer , Framing , types of framing , framing methods ,**ARQ strategies**: Error Detection and correction, Parity Bits, 7 bit Hamming Codes and CRC,**Flow Control Protocols**: Unrestricted Simplex, Stop and Wait, Sliding Window Protocol, **MAC Sub laye**r: Multiple Access Protocols: Pure and Slotted ALOHA, CSMA, WDMA, CSMA/CD, CSMA/CA.		
#Exemplar/Case Studies : Demonstration of DLL protocols on Simulator		
***Mapping of Course Outcomes**	**CO2**	
Unit III	**Network Layer**	**7 Hours**
Introduction: Functions of Network layer. Switching Techniques: Circuit switching, Message Switching, Packet Switching. ,**IP Protocol:** Classes of IP (Network addressing), Subnetting , CIDR , IPv4 , IPv6 , NAT, **Network layer Protocols:** ARP, RARP, ICMP, IGMP., **Network Routing and Algorithms**: Static Routing, Dynamic Routing, Distance Vector Routing, Link State Routing, Path Vector., **Routing Protocols**: RIP, OSPF, BGP, MPLS .		
#Exemplar/Case Studies: Demonstration of Routing Protocols on simulator.		
***Mapping of Course Outcomes**	**CO3**	
Unit IV	**Transport Layer**	**7 Hours**
Process to Process Delivery - TCP & UDP Services, Socket Programming, **Elements of Transport Layer Protocols**: Addressing, Connection establishment, Connection release, Flow control and buffering, Multiplexing, Congestion Control., **Transport Layer Protocols**: TCP and UDP, Congestion control and Quality of Service (QoS) , Concepts of QUIC.		
#Exemplar/Case Studies	Demonstration of transport layer protocols on Simulator.	
***Mapping of Course Outcomes**	**CO4**	
Unit V	**Application layer**	**7 Hours**
Introduction of Application layer , Web and HTTP, Web Caching, DNS, Email: SMTP, MIME, POP3, FTP, TELNET, DHCP, SNMP, HTTPS**.**		
#Exemplar/Case Studies: Study of Application Layer protocols using network protocol analyzer. e.g. Wireshark		
***Mapping of Course Outcomes**	**CO5**	
Unit VI	**Introduction to Advance Computer Network**	**8 Hours**
Next-Generation Networking Technologies: Software-Defined Networking (SDN): Architecture, control and data plane separation, OpenFlow basics, and advantages of network programmability Network Function Virtualization (NFV): Concept, benefits, and integration with SDN , Data Center Networks (DCN): Role in cloud infrastructure, topologies (Fat-tree, Spine-Leaf), and importance in large-scale data		



handling.	
#Exemplar/Case Studies: Study Real-world examples: Google B4 SDN, 5G slicing, IoT in smart cities	
***Mapping of Course Outcomes**	**CO6**
Learning Resources	
Text Books	
T1. Fourauzan B.,"Data Communications and Networking",5thEdition,TataMcGrawHill,Publications, ISBN:0–07 – 058408 – 7 2. **T2.** Andrew S. Tanenbaum, "Computer Networks", 5th Edition, Pearson India, 2012 **T3** .Andrew S. Tanenbaum," Foundations of Modern Networking: SDN, NFV, QoE, IoT, and Cloud,Pearson India, 2012	
Reference Books :	
R1. Kurose, Ross, "Computer Networking a Top Down Approach Featuring the Internet", Pearson, ISBN-10: 0132856204 **R2.** L. Peterson and B. Davie, "Computer Networks: A Systems Approach", 5th Edition, Morgan-Kaufmann, 2012. **R3.** Douglas E. Comer & M.S Narayanan, "Computer Network & Internet", Pearson Education	
Additional Resources: (Books, e-Resources) • https://people.cs.clemson.edu/~jmarty/courses/kurose/KuroseCh1-2.pdf • http://eti2506.elimu.net/Introduction/Books/Data Communications and Networking By Behrouz A.Forouzan.pdf • http://intronetworks.cs.luc.edu/current/ComputerNetworks.pdf • https://www.tutorialspoint.com/data_communication_computer_network/data_communication_computer_network_tutorial.pdf	

24-PEC-CS-3-01A : Computer Graphics

Teaching Scheme: TH: 04 Hours/Week	**Credit:** 04	**Examination Scheme:** **CIE** : 20 Marks **MSE** : 20 Marks **SEE** : 60 Marks

Prerequisites Courses: Basic Mathematics

Companion Course: (24-PEC-CS-3-02A) Computer Graphics Laboratory

Course Objectives:

1. To introduce the fundamental concepts of computer graphics including coordinate systems, graphics modes, drawing basic shapes, colors, patterns, and file formats.
2. To understand and implement fundamental scan conversion techniques and line/circle drawing algorithms for generating basic graphic primitives.
3. To explore polygon representations, filling methods, clipping algorithms, and windowing techniques for accurate rendering in computer graphics.
4. To understand and apply 2D and 3D geometric transformations and projection techniques for accurate representation of graphical objects.
5. To understand the concepts of curves, fractals, and hidden surface detection techniques for advanced computer graphics applications.
6. To understand color models, segment handling, and animation principles for designing dynamic and interactive graphics.

Course Outcomes:

After completion of the course, learners should be able to

CO No	CO	BL
CO1	Demonstrate basic concepts of computer graphics.	3
CO2	Implement mathematical and logical methods to create graphic primitives.	3
CO3	Examine polygon filling, clipping techniques, and viewing transformations to determine efficient rendering methods.	3
CO4	Apply geometric transformations and understand parallel and perspective projections to model and visualize graphical objects effectively.	4
CO5	Construct curve, fractal, and hidden-surface methods for scene creation.	3
CO6	Utilize color models, segment handling, and animation techniques to design motion sequences.	3

Course Contents

Unit I	**Basics of Computer Graphics**	**(07 hours)**



Introduction to Computer Graphics: Coordinate systems and pixels, resolution, aspect ratio, frame buffer, applications of computer graphics.
Graphics initialization and modes: text mode, graphic mode, drawing basic graphics primitives (line, circle, rectangle, ellipse), graphics standards, colors and fill styles, pattern handling
Graphics file formats – File structure, headers, advantages, and disadvantages of BMP, GIF, JPEG, TIFF, PCX, PNG

#Exemplar/Case Studies : Differential Study of Graphics File Formats and Creation of a Sample Graphical File Using Image Editing Tools

***Mapping of Course Outcomes**	**CO1**

Unit II	**Scan Conversion Algorithms**	**(08 Hours)**

Scan Conversion: Random scan display, Raster scan display, Line and line segments, Introduction to aliasing and anti-aliasing, qualities of good line drawing algorithms
Line drawing algorithms: Digital Differential Analyzer (DDA), Bresenham
Circle drawing algorithms: Digital Differential Analyzer (DDA), Bresenham, and Midpoint

#Exemplar/Case Studies: Real-Time Visualization of the Solar System Using Computer Graphics using Line and Circle Drawing Algorithms

***Mapping of Course Outcomes**	**CO2**

Unit III	**Polygons and Clipping**	**(08 Hours)**

Polygons: Representation of polygon and its types: convex, concave and complex, Inside Test
Polygon filling methods: Seed Fill – Flood fill and Boundary Fill, Scan-line Fill algorithms
Windowing: Concept of window and viewport, viewing transformations
Clipping: Point clipping, Cohen Sutherland line clipping algorithm,Sutherland Hodgeman polygon clipping algorithm

#Exemplar/Case Studies: Study Guard clipping Techniques and its use in various rendering softwares.

***Mapping of Course Outcomes**	**CO3**

Unit IV	**Geometric Transformations**	**(08 Hours)**

2D Transformations: introduction, matrices, Translation, Rotation, Scaling, Mirroring and Shearing, Homogeneous coordinate system, matrix representation, Composite transformations, Rotation about an arbitrary point.
3D Transformation: Translation, Scaling, Rotation about X, Y, Z & arbitrary axis reflection about XY, YZ,XZ plane.
Projection: Parallel (Oblique: Cavalier, Cabinet and orthographic: isometric, diametric, trimetric) and Perspective (Vanishing Points – 1 point, 2 point and 3 point)

#Exemplar/Case Studies : Study the use of Affine Transformations
Vlab (Vlab link: https://cse19- iiith.vlabs.ac.in/exp/affinetransformation/theory.html)

***Mapping of Course Outcomes**	**CO4**

Unit V	**Curves, Fractals, Hidden Surfaces**	**(07 Hours)**

Curves: Introduction, Interpolation and Approximation, Blending function, B-Spline curve, Bezier curve
Fractals: Introduction, Classification, Applications, **Fractal generation:** Triadic koch curve, snowflake, Hilbert curve



Hidden Surfaces: Introduction, Back face detection and removal, Algorithms: Depth buffer (z), Depth sorts (Painter), Area subdivision (Warnock)		
#Exemplar/Case Studies: Using the Fractal generation method design mountains, clouds, and trees		
***Mapping of Course Outcomes**	**CO5**	
Unit VI	**Color Model, Segmentation and Animation**	**(07 Hours)**
Colour models: Properties of Light, CIE chromaticity Diagram, RGB, HSV, CMY, YIQ **Segment:** Introduction, Segment table, Segment creation, closing, deleting and renaming, Visibility. **Animation:** Introduction, Design of animation sequences, Animation languages, Key- frame, Morphing, Motion specification.		
#Exemplar/Case Studies: Develop short 2D character animation (walk cycle / Smiling Face Animation etc.)		
***Mapping of Course Outcomes**	**CO6**	
Learning Resources		
Text Books		
T1. S. Harrington, "Computer Graphics", McGraw-Hill Publications, 1987, 2nd Edition, ISBN 0-07-100472-6. **T2.** D. Rogers, "Procedural Elements for Computer Graphics", Tata McGraw-Hill, New Delhi, India, 2001, 2nd Edition, ISBN 0-07-047371-4. **T3.** D. D. Hearn and M. P. Baker, "Computer Graphics: C Version", Pearson Education India, 2009, 2nd Edition, ISBN 817758765X, 9788177587654.		
Reference Books :		
R1. J. Foley, V. Dam, S. Feiner, and J. Hughes, "Computer Graphics Principles and Practice", Pearson Education, 2003, 2nd Edition, ISBN 81-7808-038-9. **R2.** D. Rogers and J. Adams, "Mathematical Elements for Computer Graphics", Tata McGraw-Hill Publication, 2002, 2nd Edition, ISBN 0-07-048677-8.		
Additional Resources: (Books, e-Resources) Photoshop: https://www.adobe.com/africa/learn/photoshop Blender: https://studio.blender.org/training/animation-fundamentals/ Unity: https://docs.unity3d.com/6000.0/Documentation/Manual/UnityManual.html		
MOOC Courses links : • https://nptel.ac.in/courses/106102065 • https://onlinecourses.nptel.ac.in/noc20_cs90/preview		



24-PEC-CS-3-01B Cyber Security & Digital Forensic		
Teaching Scheme: Theory: 3 Hours/Week	**Credit:** 4	**Examination Scheme:** **CIE**: 20 Marks **MSE:** 20 Marks **SEE**: 60 Marks

Prerequisites Courses: Computer Network (24-PCC-CS-3-02)

Companion Course: Cyber Security & Digital Forensic Lab (24-PEC-CS-3-02B)

Course Objectives:

- Introduce the fundamentals of cyber security, and digital forensic.
- Develop understanding of digital evidence acquisition, preservation, and analysis processes.
- Demonstrate forensic techniques for operating systems and file systems.
- Explore methodologies for network forensic, log analysis, and data recovery.
- Examine password cracking, cyber-attack analysis, and forensic investigation tools.
- Provide knowledge of email and mobile forensic, along with forensic report writing and expert testimony.

Course Outcomes:

After completion of the course, learners should be able to

CONo	CO	BL
CO1	Understand fundamental concepts of cyber security, cyber threats, cyber laws, and data protection regulations.	2
CO2	Identify digital evidence and explain standard digital forensic investigation processes.	2
CO3	Apply operating system and file system forensic principles to interpret FAT, NTFS, and EXT file systems in Windows and Linux environments.	3
CO4	Explain network forensic concepts including packet sniffing, traffic analysis, log investigation, data carving, hash auditing, and SIEM log sources.	2
CO5	Explain concepts of password cracking, wireless and web-based attacks, vulnerability scanning, and attack analysis.	2
CO6	Explain email and mobile forensic techniques and outline procedures for preparing professional digital forensic reports.	2

Course Contents		
Unit I	**Fundamentals of Cyber Security and Cyber Laws**	**8 Hours**

Introduction: Introduction and overview of cyber security incidents, Nature and scope of cyber security incidents, Types of Cyber Security Threats: crime against an individual, crime against property, Cyber extortion, Advanced Persistent Cyber Threats (APTs), Cyber threat prevention methods, Indian IT Act 2000 & 2008, The Digital Personal Data Protection Act 2023.		
#Exemplar/Case Studies: Case Study: Kalyani digital-arrest judgment 2025 (conviction in "digital-arrest" cyber-fraud case)		
***Mapping of Course Outcomes**	**CO1**	
Unit II	**Digital Evidence and Investigation Processes**	**7 Hours**
Foundations of Digital Forensics, Language of Computer Crime Investigation, Types and Characteristics of Digital Evidence, Admissibility of Digital Evidence in Court, Conductive Digital Investigation, Handling Digital Crime Scene: Principles, Preservation, Modus Operandi, Motive and Technology, Roles and Responsibilities of Digital Forensic Investigators, Forensic Readiness and Incident Response, Steps in a Digital Forensic Investigation, Techniques in Digital Forensics Investigations.		
#Exemplar/Case Studies: Case Study: Belagavi Fake Call-Centre Case – Transferred to CID for Deep Digital Forensic Analysis (2025)		
***Mapping of Course Outcomes**	**CO2**	
Unit III	**Operating System and File System Forensic**	**8 Hours**
File System Fundamentals: FAT, NTFS, EXT, Windows Forensic: Volatile and Non-Volatile Data, Linux Forensic, Recovering Deleted Files and Partitions, Forensic Tools: FTK Imager, Autopsy, EnCase, RAM Dump Analysis using Volatility.		
#Exemplar/Case Studies: Case Study: Automated Forensic Recovery of Video Evidence from Hikvision & Dahua DVR/NVR Systems (2025).		
***Mapping of Course Outcomes**	**CO3**	
Unit IV	**Network Forensic and Log Analysis**	**7 Hours**
Introduction to Network Forensic, Packet Sniffing and Traffic Analysis Tools, Website and Network Penetration Techniques: Log Analysis and Data Carving using Bulk Extractor and Foremost, Hash Comparison and Audit, SIEM log sources.		
#Exemplar/Case Studies: Case Study: 2024 LockBit Ransomware Attack on Caesars & MGM (Network Breach + Exfiltration)		
***Mapping of Course Outcomes**	**CO4**	
Unit V	**Password Cracking and Cyber Attack Analysis**	**8 Hours**
Introduction to Password Cracking: John the Ripper, Rainbow Tables, Wireless Network Attacks: WiFi Packet Capture		

and Cracking, Web Attacks: SQL Injection, Web Vulnerability Scanning, Image Metadata Extraction, Remote Imaging and PDF File Analysis for forensic analysis and defense.		
#Exemplar/Case Studies: Case Study: RockYou2024 – Massive Plaintext Password Dump.		
***Mapping of Course Outcomes**	**CO5**	
Unit VI	**Email, Mobile Forensics, and Reporting**	**7 Hours**
Email Forensic Investigations, Mobile Forensic Techniques, Preparation for Digital Forensic Investigation, Report Writing and Expert Witness Reports, Audio Video CCTV Forensics, Cloud Forensic.		
#Exemplar/Case Studies: Case Study : Supreme Court of South Korea – Phone Forensic Examines Entire Phone Declared Illegal (2025)		
***Mapping of Course Outcomes**	**CO6**	
Learning Resources		
Text Books		
T1. "Digital Forensics and Cyber Crime", Sanjay Goel, Ersin Uzun, Mengjun Xie, Sumantra Sarkar T3. "Information Security, Privacy and Digital Forensics", Sabyasachi Pramanik, Mangesh M. Ghonge, Ramchandra S. Chincholkar.		
Reference Books :		
R1. "Digital Evidence and Computer Crime", Eoghan Casey. R2. "Incident Response & Computer Forensics" – Jason Luttgens, Matthew Pepe, Kevin Mandia.		
Additional Resources: (Books, e-Resources)		
MOOC Courses links : ● Information Security and Cyber Forensics - https://onlinecourses.swayam2.ac.in/cec26_ge02/preview ● Internet Crimes and Cyber Security - https://onlinecourses.nptel.ac.in/noc26_lw01/preview ● Digital Forensics - https://onlinecourses.swayam2.ac.in/nou25_cs19/preview ● Network Forensics - https://onlinecourses.swayam2.ac.in/ntr26_ed113/preview		

24-PEC-CS-3-01C : Artificial Intelligence

Teaching Scheme:	**Credit:** 04	**Examination Scheme:**
Theory:4 Hours/Week		**CIE**: 20 Marks **MSE:** 20 Marks **SEE**: 60 Marks

Prerequisites Courses: 24-PCC-CS-2-01: Discrete Mathematics

Companion Course: 24-PEC-CS-3-02C : Artificial Intelligence Lab

Course Objectives:

- To understand the concept of Artificial Intelligence (AI) in the form of various Intellectual tasks
- To understand Problem Solving using various peculiar search strategies for AI
- To understand multi-agent environment in competitive environment
- To acquaint with the fundamentals of knowledge and reasoning
- To devise plan of action to achieve goals as a critical part of AI
- To develop a mind to solve real world problems unconventionally with optimality

Course Outcomes:

After completion of the course, learners should be able to

CO No	CO	BL
CO1	**Apply** suitable intelligent agents to solve various real-world AI applications.	3
CO2	**Analyze** informed, uninformed, and heuristic search strategies for solving AI problems.	4
CO3	**Illustarte** ideas underlying modern logical inference systems	2
CO4	**Apply** Generative AI and Large Language Model concepts to demonstrate real-world applications such as text generation and creative systems.	3
CO5	**Design** single-agent and multi-agent systems for real-world problem solving.	3

Course Contents

Unit I	**Introduction to Artificial Intelligence**	**7 Hours**

Foundations of Artificial Intelligence, State of the Art, Risks and Benefits of AI, Intelligent Agent: Sensors, Actuators, Structure of Agents, Agents and Environments, Nature and Types of Environments, Application of AI.

#Exemplar/Case Studies- Intelligent Traffic Signal Management System (ITSMS).

***Mapping of Course Outcomes**	**CO1**

Unit II	**Search-Based Problem Solving**	**8 Hours**

Solving Problems by Searching: problem-solving agent, Blind Search Strategies: Depth Limited Search, Iterative Deepening Depth

First Search , Uniform Cost Search, Bidirectional Search (completeness, optimality, complexity); Informed Search Strategies: Greedy Best-first Search, A* Search (completeness, optimality, complexity)		
#Exemplar/Case Studies- AI-Based Hospital Emergency Ambulance Routing System		
***Mapping of Course Outcomes**	**CO2**	
Unit III	**Adversarial Search and Games**	**7 Hours**
Game Theory, Optimal Decisions in Games, Heuristic Alpha–Beta Tree Search, Limitations of Game Search Algorithms, Constraint Satisfaction Problems (CSP), Constraint Propagation: Inference in CSPs, Backtracking Search for CSPs.		
#Exemplar/Case Studies- AI-Based Smart Examination Timetable Scheduling System		
***Mapping of Course Outcomes**	**CO2**	
Unit IV	**Knowledge Representation and Reasoning Systems**	**8 Hours**
The Wumpus World, Logic, Knowledge-Based Agents; Propositional Logic Knowledge Representation Language, Syntax and Semantics; Logical Reasoning: Entailment and Inference (soundness, completeness); Inference in First-Order Logic, Propositional vs. First-Order Inference, Unification, Forward Chaining, Backward Chaining, Resolution,		
#Exemplar/Case Studies- Knowledge-Based Rescue Robot in a Hazardous Building (Wumpus World Model)		
***Mapping of Course Outcomes**	**CO3**	
Unit V	**Generative AI**	**7 Hours**
Overview of Generative AI and Large Language Models. Basics of attention mechanisms and Transformer architecture. Pre-training techniques and transfer learning strategies, Generative AI Applications: Art and Creativity, Image and Video Generation, Text Generation, Music Composition, Healthcare.		
#Exemplar/Case Studies: AI-Driven Smart Disaster Relief Planning System		
***Mapping of Course Outcomes**	**CO4**	
Unit VI	**Agentic AI**	**8 Hours**
Agentic AI Overview, Agent vs Traditional AI, Types of Agents , Building AI Agents , AI Agent Frameworks, Multi-Agent Systems Programming Foundations for Agentic AI: LangChain, LangGraph, RAG.		
#Exemplar/Case Studies: Agentic AI–based Smart Placement Assistant		
***Mapping of Course Outcomes**	**CO5**	
Learning Resources		
Text Books:		
T1. Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach", Third edition, Pearson, 2003, ISBN :10: 0136042597 **T2.** Deepak Khemani, "A First Course in Artificial Intelligence", McGraw Hill Education(India), 2013, ISBN : 978-1-25-902998-1		

T3. Elaine Rich, Kevin Knight and Nair, "Artificial Intelligence", TMH, ISBN-978-0-07-008770- 5
T4. Generative AI for Everyone: Understanding the Essentials and Applications of This Breakthrough Technology". Altaf Rehmani .

Reference Books :

R1. Nilsson Nils J , "Artificial Intelligence: A new Synthesis", Morgan Kaufmann Publishers Inc. San Francisco, CA, ISBN: 978-1-55-860467-4
R2. Patrick Henry Winston, "Artificial Intelligence", Addison-Wesley Publishing Company, ISBN: 0-201-53377-4
R3. Andries P. Engelbrecht-Computational Intelligence: An Introduction, 2nd Edition-Wiley India- ISBN: 978-0-470-51250-0
R4. "Generative AI in Software Development: Beyond the Limitations of Traditional Coding" Jesse Sprinter, 2024.

Additional Resources: (Books, e-Resources)

- https://cs.calvin.edu/courses/cs/344/kvlinden/resources/AIMA-3rd-edition.pdf
  https://www.cin.ufpe.br/~tfl2/artificial-intelligence-modernapproach.9780131038059.25368.pdf
  http://aima.cs.berkeley.edu/

MOOC Courses links :

- https://nptel.ac.in/courses/106/102/106102220/
- https://nptel.ac.in/courses/106/105/106105077/
- https://nptel.ac.in/courses/106/105/106105078/
- https://nptel.ac.in/courses/106/105/106105079/

24-PEC-CS-3-01D: Pervasive Computing

Teaching Scheme:	**Credit:** 4	**Examination Scheme:**
Theory: 4 Hours/Week		**CIE :** 20 Marks **MSE :** 20 Marks **SEE :** 60 Marks

Prerequisites Courses: 24-PCC-CS-2-03 : Computer Organization and Architecture

Companion Course: 24-PEC-CS-3-02D : Pervasive Computing Lab

Course Objectives:

- To describe ubiquitous computing, its properties, applications and architectural design.
- To explain various smart devices and services used in ubiquitous computing.
- To teach the role of sensors and actuators in designing real time applications using Ubicomp.
- To explore the concept of human computer interaction in the context of Ubicomp.
- To explain Ubicomp privacy and challenges to privacy.
- To describe Ubicomp network with design issues and Ubicomp management.

Course Outcomes:

After completion of the course, learners should be able to

CONo	**CO**	**BL**
CO1	Describe characteristics & architectures of pervasive systems	2
CO2	Analyze the strengths, problems and limitations of the current tools, devices and communications for pervasive computing systems.	4
CO3	Recognize human–system interactions in ubiquitous environment	3
CO4	List and exemplify the key technologies involved in the development Ubicomp systems	3
CO5	Analyze and identify security and privacy issues in pervasive computing systems.	4
CO6	Explore the trends and problems of current pervasive computing systems using examples.	4

Course Contents

Unit I	**Introduction to Pervasive Computing**	**7 Hours**

Concept of Distributed Computing, Mobile Computing, Pervasive Computing, Wearable Computing, Modeling the Key Ubiquitous/Pervasive Computing Properties, Mobile Adaptive Computing , Mobility Management and Caching.

Case Study: Smart Healthcare System using Wearable Devices

***Mapping of Course Outcomes**	**CO1**



Unit II	**Pervasive Computing Devices**	**7 Hours**
Smart Environment: CPI and CCI Smart Devices: Application and Requirements, Device Technology and Connectivity, Human Computer Interaction.		
Case Studies: Philips Hue: Smart Home Lighting at Internet scale		
***Mapping of Course Outcomes**	**CO2**	
Unit III	**Human-Computer Interaction**	**6 Hours**
Explicit HCI, Implicit HCI, User Interface and Interaction for four hand-held widely used devices, Hidden UI via basic smart devices, Hidden UI via wearable and Implanted devices, Human centered design, user models.		
Case Studies: Hostel & Mess Finder Chatbot		
***Mapping of Course Outcomes**	**CO3**	
Unit IV	**Middleware for Pervasive**	**7 Hours**
Adaptive middleware, Context aware middleware, Mobile middleware, Service Discovery, Mobile Agents		
Case Studies: Android Middleware Framework for Context-Aware Mobile Applications		
***Mapping of Course Outcomes**	**CO4**	
Unit V	**Security in Pervasive Computing**	**6 Hours**
Security and Privacy in Pervasive Networks, Experimental Comparison of Collaborative Defense Strategies for Network Security.		
Case Studies: Security in Mobile Healthcare Applications		
***Mapping of Course Outcomes**	**CO5**	
Unit VI	**Challenges and Outlook**	**6 Hours**
Overview of challenges, smart devices, Smart Interaction, Smart physical environment device interaction, Smart human-device interaction, Human Intelligence versus machine intelligence, social issues.		
Case Studies: Analyze Challenges in any Wearable Computing devices		
***Mapping of Course Outcomes**	**CO6**	
Learning Resources		
Text Books		
Textbooks : **T1.** Stefan Poslad, Ubiquitous Computing, Wiley, Student Edition, ISBN:9788126527335 John Krumm, Ubiquitous Computing Fundamentals. **T2.** Frank Adelstein, Sandeep Gupta, Golden Richard III, Loren Schwiebert, "Fundamentals of Mobile and Pervasive Computing," Tata McGraw Hills.		



Reference Books :

R1. Jochen Burkhardt, Horst Henn, Stefan Hepper, Klaus Rindtor, Thomas Schaeck, Pervasive Computing, Pearson, Eighteenth Impression, 2014.

Additional Resources: (Books, e-Resources)

- Adam Greenfield – Everyware: The Dawning Age of Ubiquitous Computing
- Mark Weiser – The Computer for the 21st Century (Xerox PARC Paper)
- Roy Want – Enabling Technologies for Ubiquitous Computing, IEEE
- Daniel Solove – Understanding Privacy
- ACM UbiComp Conference Papers
- IEEE Pervasive Computing Magazine
- MIT Media Lab Projects (Wearables, Smart Environments)

MOOC Courses links :

- https://www.coursera.org/learn/human-computer-interaction
- https://www.coursera.org/specializations/internet-of-things
- https://www.futurelearn.com/courses/smart-cities

24-PEC-CS-3-01E : Web Technology

Teaching Scheme: Theory: 4 Hours/Week	**Credit:** 4	**Examination Scheme:** **CIE** : 20 Marks **MSE** : 20 Marks **SEE** : 60 Marks

Prerequisites Courses: 24-VSEC-CS-2-01 : Application Development Programming I, 24-PCC-CS-2-05: Database Management System, 24-PCC-CS-3-02 : Computer Network

Companion Course: 24-PEC-CS-3-02E : Web Technology Lab

Course Objectives:

- Learn basics of web development.
- To use the Client side and Server side technologies in web development
- To Explore emerging web trends.

Course Outcomes:

After completion of the course, learners should be able to

CONo	CO	BL
CO1	**Apply** advanced CSS and responsive design principles for interactive user interfaces.	3
CO2	**Develop** dynamic web pages using modern JavaScript and DOM manipulation.	3
CO3	**Implement** asynchronous data exchange and API integration	3
CO4	**Implement** server-side applications	3
CO5	**Explore** emerging web technologies	2

Course Contents

Unit I	Fundamentals of Web Technology and CSS	8 Hours

Introduction to web technology: Web clients, web servers, HTTP request and response messages, Design issues of websites **HTML Forms:** Attributes, Elements, Input Types, Input Attributes, and Input Form Attributes. **CSS:** Introduction to CSS, Syntax and Selectors, The Box Model, Positioning Elements, Flexbox and Grid, Typography, Colors and Backgrounds, CSS Variables, Transitions and Animations, and Responsive Design. Introduction to Tailwind CSS

#Exemplar/Case Studies: Study of any e-commerce website

*Mapping of Course Outcomes	CO1

Unit II	JavaScript and DOM Manipulation	8 Hours

JavaScript Basics: Syntax, Variables, Data Types, Operators, Control Structures, Functions & Scopes, Arrays, and Objects. - Events, Form Handling & Client-Side Validation, Error Handling. **ES6 Features:** let/const, arrow functions, template literals, destructuring, modules. **DOM:** Document tree, DOM objects, properties and methods, DOM manipulation, event handling



#Exemplar/Case Studies: Implementation of a To-do-list like web app		
***Mapping of Course Outcomes**	**CO2**	
Unit III	**Data Interchange and Asynchronous Communication**	**8 Hours**
XML: Introduction, structure, elements, attributes, DTD, namespaces. **JSON:** Introduction, syntax, comparison with XML, JSON data Types, Parsing. **AJAX:** Introduction, working, XMLHttpRequest, AJAX Applications, Fetch API, and async communication. **REST API basics:** request methods (GET, POST, PUT, DELETE).		
#Exemplar/Case Studies: Development of a Weather Forecast application using a public API (e.g., OpenWeatherMap)		
***Mapping of Course Outcomes**	**CO3**	
Unit IV	**Server-Side Scripting with Servlet and JSP**	**8 Hours**
Servlet: Architecture overview, lifecycle, generating dynamic content: Hello world, Login Example. **JSP:** Introduction, advantages over Servlet, elements of JSP pages (directives, comments, scripting, actions, templates), MVC architecture and database integration using JDBC.		
#Exemplar/Case Studies: Creation of a Student Management System to demonstrate CRUD operations and database connectivity using JSP and JDBC.		
***Mapping of Course Outcomes**	**CO4**	
Unit V	**Server-Side Scripting with PHP**	**8 Hours**
Introduction to PHP: syntax, variables, data types, operators, arrays, and functions. **Control structures:** conditional statements and loops. **Form handling:** GET/POST methods, form validation. Cookies, sessions, and user authentication. **Database connectivity** using MySQL (CRUD operations, prepared statements). **Security considerations:** SQL Injection, XSS prevention.		
#Exemplar/Case Studies: Use of PHP in developing a blogging application		
***Mapping of Course Outcomes**	**CO4**	
Unit VI	**Emerging Trends**	**8 Hours**
SPA (Single Page Application): Fundamentals and benefits of SPA. **Progressive Web Apps (PWA):** Fundamentals and benefits of PWAs. **Web 3.0:** Overview of blockchain-based web and decentralized applications (DApps). **Cloud Hosting:** Introduction to Firebase Hosting, Vercel, Netlify, and their advanced features. **Serverless Computing:** Basic awareness of serverless architecture and applications. **AI in Web:** Overview of Integration of AI features such as chatbots, personalization, and recommendation engines.		
#Exemplar/Case Studies: Study of Twitter Lite – A Progressive Web App (PWA) Success Story		
***Mapping of Course Outcomes**	**CO5**	
Learning Resources		
Text Books		
T1. Kogent Learning Solutions Inc. Web Technologies: HTML, JavaScript, PHP, Java, JSP, ASP.NET, XML and AJAX, Black Book.		

Reprint ed., Dreamtech Press, 2009.
T2. Kumar, Santosh K., and Kogent Solutions Inc. JDBC, Servlets, and JSP Black Book. New ed., Reprint, Dreamtech Press, 2008.
T5. Gilmore, W. Jason. Beginning PHP & MySQL: From Novice to Professional. 3rd ed., Apress Media LLC, 2008.

Reference Books :

R1. Lemay, Laura, Rafe Colburn, and Jennifer Kyrnin. HTML, CSS & JavaScript Web Publishing in One Hour a Day: Sams Teach Yourself. Sams Publishing, 2015.

Additional Resources: (Books, e-Resources)

- HTML, The Complete Reference http://www.htmlref.com/
- http://w3schools.org/
- http://php.net/
- https://developer.mozilla.org/en-US/docs/AJAX
- http://www.tutorialspoint.com/css/

MOOC Courses links :

- https://freevideolectures.com/course/2308/internet-technology/25
- https://www.digimat.in/nptel/courses/video/106105191/L01.html



24-PCC-CS-3-04 : Computer Network Lab

Teaching Scheme:	**Credit:** 2	**Examination Scheme:**
Practical: 4 Hours/Week		Termwork **(TW)** : 25 Marks Oral **(OR)** : 25 Marks

Prerequisites Courses: 24-PCC-CS-2-01: Discrete Mathematics

Companion Course: 24-PCC-CS-3-02 : Computer Network

Course Objectives:

- To introduce the fundamental concepts of computer networks, layered architecture, and communication protocols.
- To develop the ability to configure, simulate, and analyze networks using modern tools
- To provide knowledge of transport, application, and security mechanisms, enabling students to design and secure real-world network systems.

Course Outcomes:

After completion of the course, learners should be able to

CONo	CO	BL
CO1	**Demonstrate** the use of layered network architectures by selecting suitable devices, protocols, and services.	3
CO2	**Apply** routing, switching, addressing, and packet analysis concepts to configure and troubleshoot networks using simulation tools.	3
CO3	**Analyze** security mechanisms such as ACLs, firewalls, NAT, and wireless encryption to ensure secure communication and advanced controlled environments like Software-Defined Networking (SDN).	4

Guidelines for Instructor's Manual

The instructor‘s manual is to be developed as a reference and hands-on resource. It should include a prologue (about University/program/ institute/ department/foreword/ preface), curriculum of the course, conduction and Assessment guidelines, topics under consideration, concept, objectives, outcomes, set of typical applications/assignments/ guidelines, and references.

Guidelines for Student's Laboratory Journal

Each student must prepare and submit a Laboratory Journal containing all the laboratory assignments performed during the term. The journal should include handwritten write-ups, output for each experiment, which must contain the title of the experiment, date of completion, objectives, problem statement, software and hardware requirements, assessment grade or marks with the assessor's signature, theory or concept in brief, and sample output obtained using simulator tools. All experiments performed using tools such as Cisco Packet Tracer or Wireshark should be documented and submitted as a softcopy. As a conscious effort towards Green IT and environmental awareness, students should avoid attaching printed papers such as program listings or screenshots to the journal. Instead, a softcopy containing all modules, designs, and related materials should be maintained by the Laboratory In-charge for record and verification purposes.

Guidelines for Laboratory /Term Work Assessment

Continuous assessment of laboratory work should be based on the overall performance of Laboratory assignments by a student.



Each Laboratory assignment assessment will assign grades/marks based on parameters, such as timely completion, performance, innovation, efficient codes, and punctuality.

Guidelines for Practical Examination

Problem statements must be decided jointly by the internal examiner and external examiner. During the practical assessment, maximum weightage should be given to satisfactory implementation of the problem statement. Relevant questions may be asked at the time of evaluation to test the student's understanding of the fundamentals and effective and efficient implementation. This will encourage transparent evaluation and a fair approach, and hence will not create any uncertainty or doubt in the minds of the students. So, adhering to these principles will consummate our team efforts to the promising start of student's academics.

Guidelines for Laboratory Conduction

The laboratory is structured into three groups: A, B, and C. Group A contains 6 assignments, Group B contains 3 assignments Group and C contains 1 assignment. For a total 10 Assignments. Students must complete 8 assignments, Mandatory 5 from group A , 2 from group B and 1 from group c.

Software Requirement: Any open Source operating System,Python or C++.

Tools / Simulator: Cisco Packet Tracer / GNS3 / Wireshark / NS2 / POX Controller.

Virtual Laboratory: (If Any):

- Cisco Packet Tracer (Cisco Networking Academy) https://www.netacad.com/

Suggested List of Laboratory Experiments/Assignments

Group A: Assignments (Mandatory Assignment)

Sr No	Assignment Title	*Mapping of Course Outcomes
1.	Design and simulate a LAN with multiple devices and subnets. Assign static and dynamic IP addresses and Verify using ping, ipconfig, netstat, traceroute, nslookup command	**CO1**
2.	Configure static routing protocols between multiple routers. Verify using the Networking command.	**CO2**
3.	Configure inter-VLAN routing to enable communication between different Virtual Local Area Network (VLANs).	**CO2**
4.	Capture traffic in Wireshark of client-server communication using Transmission Control Protocol (TCP) and User Datagram Protocol (UDP).	**CO1,CO2**
5.	Capture real or simulated network traffic. Analyze headers, flags, and sequence numbers of ICMP, TCP, UDP, ARP, and DNS packets.	**CO1,CO2**
6.	Development of a Basic Software-Defined Networking (SDN) Controller Using Python for Traffic Management in a Simulated Network.	**CO3**

Group B: Assignments (Out of List perform any 2)

Sr No	Assignment Title	*Mapping of Course Outcomes
7.	Implement a TCP socket program where a client sends a message (e.g., "**Hello Server**"),	**CO1,CO2**

	processes it at the server (convert to uppercase), and returns it to the client.	
8.	Configure Packet filtering using Access Control List (ACL) on routers. Apply firewall rules to allow/deny specific traffic and verify security policies.	**CO3**
9.	Configure a wireless LAN with security protocols (WEP/WPA2). Capture and analyze wireless packets to study authentication and encryption.	**CO3**
Group C: Mini Project		
Sr No	**Assignment Title**	***Mapping of Course Outcomes**
10.	Design a small office/home network with 3–4 PCs connected via a switch add DHCP, DNS, and firewall rules.	**CO2, CO3**

Learning Resources (If applicable)

Text Books

T1. Andrew S. Tanenbaum, David J. Wetherall, Computer Networks, 6th Edition, Pearson Education, 2017, ISBN: 9789332585495.
T2. Behrouz A. Forouzan, Data Communications and Networking, 5th Edition, McGraw Hill Education, 2017, ISBN: 9780073376226.

Reference Books :

R1. William Stallings, Network Security Essentials: Applications and Standards, 6th Edition, Pearson Education, 2017, ISBN: 9789352866606.
R2. Ajay Singh, Introduction to Cybersecurity: Concepts, Principles, Technologies and Practices, 1st Edition, Universities Press, 2021, ISBN: 9789393330314**.**

Additional Resources: (Books, e-Resources)

- Virtual Labs – https://vlab.co.in
- Cisco Networking Academy – https://www.netacad.com
- Wireshark Official Documentation & Labs – https://www.wireshark.org/docs/

MOOC Courses links :

- Computer Networks and Internet Protocol: IIT Kharagpur Link: https://onlinecourses.nptel.ac.in/noc22_cs19/preview
- Cryptography and Network Security : IIT Kharagpur Link : https://onlinecourses.nptel.ac.in/noc22_cs90/preview



24-PEC-CS-3-02A : Computer Graphics Laboratory

Teaching Scheme:	**Credit:** 02	**Examination Scheme:**
Practical: 04 Hours/Week		**Termwork (TW) :** 50 Marks **Practical (PR) :** 50 Marks

Prerequisites Courses: Basic Mathematics

Companion Course: 24-PEC-CS-3-01A : Computer Graphics

Course Objectives:

- To introduce the fundamental concepts of computer graphics including coordinate systems, graphics modes, drawing basic shapes, colors, patterns and scan conversion techniques.
- To explore polygon representations, filling methods, clipping algorithms and 2D and 3D geometric transformations.
- To understand the concepts of curves, fractals, animation.

Course Outcomes:

After completion of the course, learners should be able to

CO No	CO	BL
CO1	Apply fundamentals of computer graphics to draw and fill basic shapes and mathematical logic to draw graphics primitives using scan conversion algorithms.	3
CO2	Apply polygon filling methods, clipping algorithms to render graphical objects effectively.	3
CO3	Apply mathematical techniques to perform 2D and 3D transformations, including translation, rotation, scaling, reflection.	3
CO4	Design and implement interactive visual elements using curves, fractal patterns.	3
CO5	Implement animations/games using graphics functions, animation techniques and tools.	3

Guidelines for Instructor's Manual

The instructor's manual is to be developed as a reference and hands-on resource. It should include a prologue (about University/program/ institute/ department/foreword/ preface), curriculum of the course, conduction and Assessment guidelines, topics under consideration, concept, objectives, outcomes, set of typical applications/assignments/ guidelines, and references.

Guidelines for Student's Laboratory Journal

The laboratory assignments are to be submitted by students as a journal. The journal consists of a Certificate, table of contents, and handwritten write-up of each assignment (Title, Date of Completion, Objectives, Problem Statement, Software and Hardware requirements, Assessment grade/marks and assessor's sign, Theory- Concept in brief, algorithm, flowchart, test cases, Test Data Set(if applicable), mathematical model (if applicable), conclusion/analysis. Program codes with sample output of all performed assignments will be submitted as softcopy. As a conscious effort and little contribution towards Green IT and environmental awareness, attaching printed papers as part of write-ups and program listing to journals must be avoided. Softcopy containing student programs maintained by the Laboratory in charge is highly encouraged. For reference one or two journals may be maintained with program prints in the Laboratory.



Guidelines for Laboratory /Term Work Assessment

Continuous assessment of laboratory work should be based on the overall performance of Laboratory assignments by a student. Each Laboratory assignment assessment will assign grades/marks based on parameters, such as timely completion, performance, innovation, efficient codes, and punctuality.

Guidelines for Practical Examination

Problem statements must be decided by the internal examiner in consultation with the external examiner. During practical assessment, maximum weightage should be given to satisfactory implementation of the problem statement. Relevant questions may be asked at the time of evaluation to test the student's understanding of the fundamentals, effective and efficient implementation. The questions asked will in no way be the deciding factor for passing the students. This will encourage, transparent evaluation and fair approach, and hence will not create any uncertainty or doubt in the minds of the students. So adhering to these principles will consummate our team efforts to the promising start of student's academics.

Guidelines for Laboratory Conduction

The instructor is expected to frame the assignments by understanding the prerequisites, technological aspects, utility and recent trends related to the topic. The assignment framing policy needs to address the average students and inclusive of an element to attract and promote the intelligent students. Use of open source software is encouraged. Based on the concepts learned. Instructors may also set one assignment or mini-project that is suitable to each branch beyond the scope of the syllabus. Operating System recommended :- 64-bit Open source Linux or its derivative Programming tools recommended: - Open Source Programming tool like G++/GCC. , Programming Language C, C++, Java, Python, OpenGL

Virtual Laboratory: (If Any):
https://cse18-iiith.vlabs.ac.in/

Suggested List of Laboratory Experiments/Assignments

Group A: Assignments (Mandatory Assignment)

Sr No	Assignment Title	*Mapping of Course Outcomes
1.	Write a program to draw a man walking in the rain with an umbrella.	CO1
2.	Write a program to draw lines using DDA and Bresenham's algorithm with patterns such as solid, dotted, dashed, dash dot and thick.	CO1
3.	Write a program to draw a concave polygon and fill it with desired color using any polygon filling algorithm.	CO2
4.	Write a program to implement Cohen-Sutherland line clipping algorithm for a given window.	CO2
5.	Write a program to draw 2-D objects and perform the following basic transformations, translation,rotation, scaling.	CO3
6.	Write a program to generate fractal patterns by using Koch curves.	CO4

Group B: Assignments (Out of List perform any 2) (Optional)

Sr No	Assignment Title	*Mapping of Course Outcomes
1.	Write a program to draw Circle using DDA and Bresenham's and midpoint	CO1

	algorithm	
2.	Write a program to implement Sutherland-Hodgeman polygon clipping algorithm for a given window.	CO2
3.	Write a program to draw 2-D objects and perform the following transformations rotation about an arbitrary point, reflection and shearing for square (0,0,0,100,100,100,100,0)	CO3
4.	Write a program to generate snowflake fractal patterns by using Koch curves or implement Hilbert curves.	CO4
5.	Write a program to implement a bouncing ball using sine wave form	CO1, CO5
6.	Write a program to implement the game of 8 puzzles.	CO1, CO5
Group C: Assignments (Any One) Mini Project		
Sr No	**Assignment Title**	***Mapping of Course Outcomes**
1.	Design and implement Game/ Animation Clip/ Graphics Editor/ Mobile App etc. using Blender/Maya/ OpenGL/Unity Software etc.	CO1, CO5
2.	Write an animated program to design a social theme–based graphical flyer using basic graphics primitives (Like Woman Empowerment…)	CO1, CO5



24-PEC-CS-3-02B : Cyber Security & Digital Forensic Lab		
Teaching Scheme: Practical: 4 Hours/Week	**Credit:** 2	**Examination Scheme:** Termwork **(TW)**: 50 Marks Practical **(PR)**: 50 Marks

Prerequisites Courses: Computer Network Lab (24-PCC-CS-3-03)

Companion Course: Cyber Security & Digital Forensics (24-PEC-CS-3-01B)

Course Objectives:

CO1: Develop hands-on skills in acquiring, preserving, and analysing digital evidence from Windows and Linux OS.
CO2: Gain practical expertise in using forensic tools such as FTK Imager, Autopsy, EnCase, and Volatility.
CO3: Perform network traffic capture, analysis, and web reconnaissance to investigate cyber incidents.
CO4: Apply password-cracking, Wi-Fi-attack, and web-vulnerability-assessment techniques in a controlled environment.
CO5: Extract and analyse metadata from digital media and files to gather forensic evidence.
CO6: Prepare professional forensic reports and document investigation findings accurately.

Course Outcomes:

After completion of the course, learners should be able to

CONo	CO	BL
CO1	Acquire, preserve, and analyse static and live data from Windows and Linux OS	3
CO2	Analyse volatile memory and recover deleted files, partitions, and artefacts	3
CO3	Capture and analyse network traffic, perform website reconnaissance, and identify vulnerabilities.	3
CO4	Apply password cracking and wireless network attacks to assess and understand security weaknesses.	3
CO5	Extract metadata and perform forensic analysis on images, files, and mobile devices.	3
CO6	Prepare comprehensive forensic reports	3

Guidelines for the Instructor's Manual

The instructor's manual is to be developed as a reference and hands-on resource. It should include a prologue (about the university, program/ institute/ department/foreword/ preface), the curriculum of the course, conduct and Assessment guidelines, topics under consideration, concepts, objectives, outcomes, a set of typical applications/assignments/ guidelines, and references.

Guidelines for Students' Laboratory Journal

The laboratory assignments are to be submitted by students as a journal. The journal consists of a Certificate, table of contents, and handwritten write-up of each assignment (Title, Date of Completion, Objectives, Problem Statement, Software and Hardware requirements, Assessment grade/marks and assessor's sign, **Theory** - Concept in brief, algorithm, flowchart, test cases, Test Data Set(if applicable), mathematical model (if applicable), conclusion/analysis. Program codes, along with sample output of all performed assignments, will be submitted as a soft copy. As a conscious effort and small contribution towards Green IT and environmental awareness, attaching printed papers as part of write-ups and program listings to journals should be avoided. For reference, one or two journals may be maintained with program prints in the Laboratory.

Guidelines for Laboratory /Term Work Assessment

Continuous assessment of laboratory work should be based on the overall performance of Laboratory assignments by a student. Each Laboratory assignment assessment will assign grades/marks based on parameters, such as timely completion, performance, innovation, efficient coding, and punctuality.

Guidelines for Practical Examination

The internal examiner and the external examiner must jointly decide upon problem statements. During the practical assessment, maximum weightage should be given to satisfactory implementation of the problem statement. Relevant questions may be asked at the time of evaluation to test the student's understanding of the fundamentals and effective and efficient implementation. This will encourage transparent evaluation and a fair approach, thereby eliminating any uncertainty or doubt in the minds of the students. So, adhering to these principles will consummate our team efforts to the promising start of students' academics.

Guidelines for Laboratory Conduction

Virtual Laboratory: (If Any):

Suggested List of Laboratory Experiments/Assignments

Group A: Assignments (Mandatory Assignment)

Sr No	Assignment Title	*Mapping of Course Outcomes
1.	Acquire Static Data (Static disk images) from Windows OS using FTK Imager	CO1
2.	Live Data (Volatile memory, running processes, network connections) Acquisition from Windows and Linux OS, using tools like Velociraptor, WinPmem/LinPMem, and Rekall.	CO1
3.	RAM Dump Analysis using Volatility Framework (Volatility 3)	CO2
4.	Recovering Deleted Files and Partitions from Disk Images using forensic tools - TestDisk, Photorec, Foremost, The Sleuth Kit, DigiLAR	CO2
5.	Forensic Analysis of disk images using Autopsy on	CO2



	Windows/Linux	
6.	Web Vulnerability Assessment: SQL Injection, Nikto, HTTRACK, and Netcraft	CO4
Group B: Assignments (Out of List, perform any 2)		
Sr No	**Assignment Title**	***Mapping of Course Outcomes**
7.	Password Cracking using John the Ripper and Rainbow Tables	CO4
8.	WiFi Packet Capture and Password Cracking using Aircrack-ng	CO4
9.	Image Metadata Extraction using Imago and File Analysis	CO5
10.	Introductory mobile forensics using tools - OpenMF, MobSF (Mobile Security Framework), ALEAPP.	CO5, CO6
Group C: Mini Project		
Sr No	**Assignment Title**	***Mapping of Course Outcomes**
11.	Mini Project - Design a small tool for any type of forensic case study automation	CO1 to CO6
Learning Resources (If applicable)		
Text Books		
T1. "Digital Forensics and Cyber Crime", Sanjay Goel, Ersin Uzun, Mengjun Xie, Sumantra Sarkar **T2.** "Information Security, Privacy and Digital Forensics", Sabyasachi Pramanik, Mangesh M. Ghonge, Ramchandra S. Chincholkar.		
Reference Books :		
R1. "Digital Evidence and Computer Crime", Eoghan Casey **R2. "Incident Response & Computer Forensics",** Jason Luttgens, Matthew Pepe, Kevin Mandia		



24-PEC-CS-3-02C: Artificial Intelligence Lab

Teaching Scheme: Practical: 4 Hours/Week	**Credit:** 2	**Examination Scheme:** **Termwork (TW)** : 50 Marks **Practical (PR)** : 50 Marks

Prerequisites Courses: 24-PCC-CS-2-01: Discrete Mathematics

Companion Course: 24-PEC-CS-3-01C : Artificial Intelligence

Course Objectives:

- To learn and apply various search strategies, suitable algorithms for solving AI problems.
- To Formalize and Implement constraints in search problems
- To learn and apply various Generative AI Tools.

Course Outcomes:

After completion of the course, learners should be able to

CONo	CO	BL
CO1	**Develop** suitable algorithms for solving diverse AI problem domains.	3
CO2	**Design** logical inference mechanisms for intelligent reasoning.	3
CO3	**Apply** Generative AI tools to design and deploy intelligent solutions for real-world problem scenarios.	3
CO4	**Design** an agent to search and collect information on a given topic.	3

Guidelines for Instructor's Manual

The instructor‘s manual is to be developed as a reference and hands-on resource. It should include a prologue (about University/program/ institute/ department/foreword/ preface), curriculum of the course, conduction and Assessment guidelines, topics under consideration, concept, objectives, outcomes, set of typical applications/assignments/ guidelines, and references.

Guidelines for Student's Laboratory Journal

The laboratory assignments are to be submitted by students as a journal. The journal consists of a Certificate, table of contents, and handwritten write-up of each assignment (Title, Date of Completion, Objectives, Problem Statement, Assessment grade/marks and assessor's sign, a brief explanation of the underlying AI theory and concepts, pseudocode or algorithm, evaluation parameters used for model performance), answers to the questions provided at the end of the lab manual, and the conclusion or analysis).. Program codes with sample output of all performed assignments will be submitted as softcopy. As a conscious effort and little contribution towards Green IT and environmental awareness, attaching printed papers as part of write-ups and program listing to journals must be avoided. Softcopy containing student programs maintained by the Laboratory in charge is highly encouraged. For reference one or two journals may be maintained with program prints in the Laboratory.

Guidelines for Laboratory /Term Work Assessment

Continuous assessment of laboratory work should be based on the overall performance of Laboratory assignments by a student. Each Laboratory assignment assessment will assign grades/marks based on parameters, such as timely completion, performance, innovation, efficient codes, and punctuality.



Guidelines for Practical Examination

Problem statements must be decided jointly by the internal examiner and external examiner. During the practical assessment, maximum weightage should be given to satisfactory implementation of the problem statement. Relevant questions may be asked at the time of evaluation to test the student's understanding of the fundamentals and effective and efficient implementation. This will encourage transparent evaluation and a fair approach, and hence will not create any uncertainty or doubt in the minds of the students. So, adhering to these principles will consummate our team efforts to the promising start of student's academics.

Guidelines for Laboratory Conduction

Students must complete any 8 assignments out of 10 assignments.
Hardware Requirement: Desktop or laptop with minimum 8 GB RAM, multi-core processor, and stable internet connectivity.
Software Requirement: Open-source operating system, VS Code and required Python libraries.

Virtual Laboratory: (If Any):

- https://ds1-iiith.vlabs.ac.in/exp/depth-first-search/index.html
- https://ds1-iiith.vlabs.ac.in/exp/breadth-first-search/bfs/index.html

Suggested List of Laboratory Experiments/Assignments

Group A: Assignments (Perform any 8 Assignment)

Sr No	Assignment Title	*Mapping of Course Outcomes
1.	Implement A star Algorithm for any game search problem.	CO1
2.	Implement Minimax algorithm for game playing (Alpha-Beta pruning)	CO1
3.	Implement a solution for a Constraint Satisfaction Problem using Backtracking for n-queens problem.	CO1
4	Implement basic search strategies Cryptarithmetic.	CO1
5.	Design and Implementation of Forward Chaining Reasoner	CO2
6.	Design and Implementation of Backward Chaining Reasoner	CO2
7.	Develop a simple AI assistant that answers user queries (education, career, general help).	CO3
8.	Design intelligent solutions for real-world problem scenarios using Generative AI tools.	CO3
9.	Autonomous Research Agent: Build a system that takes a topic, searches the web, verifies sources, and compiles a comprehensive report.	CO4
10.	Personalized Tutor Agent: Design an AI Agent that Adapts Learning Material Based on Quiz Results	CO4



24-PEC-CS-3-02D : Pervasive Computing Lab

Teaching Scheme: Practical: 4 Hours/Week	**Credit:** 2	**Examination Scheme:** **Termwork (TW)** : 50 Marks **Practical (PR)** : 50 Marks

Prerequisites Courses: 24-PCC-CS-2-03 : Computer Organization and Architecture

Companion Course: 24-PEC-CS-3-01D: Pervasive Computing

Course Objectives:

- To develop problem solving abilities for smart devices.
- To develop problem solving abilities for gamification.
- To develop problem solving abilities of pervasiveness & embedded security.

Course Outcomes:

After completion of the course, learners should be able to

CONo	CO	BL
CO1	**Develop** basic Android applications and implement context-aware features.	3
CO2	**Apply** intelligent problem-solving techniques such as MiniMax and clustering for mobile applications.	3
CO3	**Design** smart applications using user profiling, recommendation systems, and scheduling models.	4
CO4	**Implement** database operations and analyze user data for smart pervasive applications.	4

Guidelines for Instructor's Manual

The instructor's manual is to be developed as a reference and hands-on resource. It should include a prologue (about University/program/ institute/ department/foreword/ preface), curriculum of the course, conduction and Assessment guidelines, topics under consideration, concept, objectives, outcomes, set of typical applications/assignments/ guidelines, and references.

Guidelines for Student's Laboratory Journal

The laboratory assignments are to be submitted by students as a journal. The journal consists of a Certificate, table of contents, and handwritten write-up of each assignment (Title, Date of Completion, Objectives, Problem Statement, Assessment grade/marks and assessor's sign, a brief explanation of the underlying Pervasive Computing theory and concepts, pseudocode or algorithm, evaluation parameters used for model performance), answers to the questions provided at the end of the lab manual, and the conclusion or analysis).. Program codes with sample output of all performed assignments will be submitted as softcopy. As a conscious effort and little contribution towards Green IT and environmental awareness, attaching printed papers as part of write-ups and program listing to journals must be avoided. Softcopy containing student programs maintained by the Laboratory in charge is highly encouraged. For reference one or two journals may be maintained with program prints in the Laboratory.

Guidelines for Laboratory /Term Work Assessment

Continuous assessment of laboratory work should be based on the overall performance of Laboratory assignments by a student. Each Laboratory assignment assessment will assign grades/marks based on parameters, such as timely completion, performance,



innovation, efficient codes, and punctuality.

Guidelines for Practical Examination

Problem statements must be decided jointly by the internal examiner and external examiner. During the practical assessment, maximum weightage should be given to satisfactory implementation of the problem statement. Relevant questions may be asked at the time of evaluation to test the student's understanding of the fundamentals and effective and efficient implementation. This will encourage transparent evaluation and a fair approach, and hence will not create any uncertainty or doubt in the minds of the students. So, adhering to these principles will consummate our team efforts to the promising start of student's academics.

Guidelines for Laboratory Conduction

The laboratory is structured into a total 9 Assignments. Students must complete all 9 assignments.
Software Requirement: Android OS with 64-bit Intel-i5/i7 or latest higher processor computers.
Hardware Requirement: Intel Internet of Things (IoT) Developer Kit or Intel Galileo board or BBB or Open Source equivalent, VxWorks R, the real-time operating system (RTOS) for IoT, NS3, Scala, Python.

Suggested List of Laboratory Experiments/Assignments

Assignments (Mandatory Assignment)

Sr No	Assignment Title	*Mapping of Course Outcomes
1.	Setting-up the environment: Android development environment. Installing and setting up the environment. Hello world application. Running the emulator.	CO1
2.	UI Design: Design a User Interface using pre-built UI components such as structured layout objects, UI controls and special interfaces such as dialogs, notifications, and menus. Also make this UI attractive using Android graphics platform OpenGL.	CO1
3..	Write a program to build a smart mobile app for context management. The objective of this assignment is to build a smart phone mobile app which can sense some parameters of the user and convert these parameters into some contextual information in order to do the context management.	CO1
4.	Write a program to build a smart mobile app for user profiling. The objective of this assignment is to develop a smart mobile app which can create user profiles based on their preferences so that smart recommendations can be provided at run time.	CO3
5.	Implementation of MiniMax approach for TIC-TAC-TOE using Java/ Scala/ Python-Eclipse Use GUI Player X and Player O are using their mobiles for the play. Refresh the screen after the move for both the players. /8 PUZZLE	CO2
6.	Developing a book recommender (a book that the reader should read and is new) Expert system or (any other).	CO3
7.	Develop a POP for scheduling your higher studies exam. Assume suitable data like college submission schedule, college exams, constraints that a paper publication must to appear before the exam, a family function at home and so on.	CO3



8	Database Connectivity: Create a SQLite Database for an Android Application and perform CRUD (Create, Read, Update and Delete) database operations.	CO4
9.	Implement k-means for clustering data of children belonging to different age groups to perform some specific activities. Formulate the Feature vector for following parameters: i. height ii. weight iii. age iv. IQ Formulate the data for 40 children to form 3 clusters.	CO4

Learning Resources

Additional Resources: (Books, e-Resources)

- Android Studio Setup & First App – Get started with Android Studio
- UI Design & Graphics (Layouts, Menus, OpenGL) – Build UI, OpenGL ES
- Context Management (Sensors) – Sensors Overview
- User Profiling & Smart Recommendations – Google ML, Recommender Systems – Coursera
- MiniMax (Tic-Tac-Toe GUI) – JavaFX, Python Tkinter
- Book Recommender / Expert System – Intro to Recommender Systems
- Scheduling App (POP / Tasks) – WorkManager
- SQLite Database CRUD – SQLite, Room DB
- K-Means Clustering – Scikit-learn: K-Means

MOOC Courses links :

- https://www.coursera.org/learn/introduction-to-android-mobile-application-development
- https://www.coursera.org/specializations/android-app-development

24-PEC-CS-3-02E: Web Technology Lab

Teaching Scheme: Practical: 4 Hours/Week	**Credit:** 2	**Examination Scheme:** **Termwork (TW)** : 50 Marks **Practical (PR)** : 50 Marks

Prerequisites Courses: 24-VSEC-CS-2-01: Application Development Programming I, 24-PCC-CS-2-07: Database Management System Lab

Companion Course: 24-PEC-CS-3-01E: Web Technology

Course Objectives:

- To learn the web based development environment
- To use client side and server side web technologies
- To design and develop web applications using front end technologies and backend databases

Course Outcomes:

After completion of the course, learners should be able to

CONo	CO	BL
CO1	Create responsive and structured web pages.	3
CO2	Develop web applications using client side technologies.	3
CO3	Apply the server side technologies for web application development	3
CO4	Make use of emerging trends in web application development.	3

Guidelines for Instructor's Manual

The instructor's manual is to be developed as a reference and hands-on resource. It should include a prologue (about University/program/ institute/ department/foreword/ preface), curriculum of the course, conduction and assessment guidelines, topics under consideration, concept, objectives, outcomes, set of typical applications/assignments/ guidelines, and references.

Guidelines for Student's Laboratory Journal

The laboratory assignments are to be submitted by students as a journal. The journal consists of a Certificate, table of contents, and handwritten write-up of each assignment (Title, Date of Completion, Objectives, Problem Statement, Software and Hardware requirements, Assessment grade/marks and assessor's sign, Theory- Concept in brief, conclusion/analysis. Program codes with sample output of all performed assignments will be submitted as softcopy. As a conscious effort and little contribution towards Green IT and environmental awareness, attaching printed papers as part of write-ups and program listing to journals must be avoided. For reference one or two journals may be maintained with program prints in the Laboratory.

Guidelines for Laboratory /Term Work Assessment

Continuous assessment of laboratory work should be based on the overall performance of Laboratory assignments by a student. Each Laboratory assignment assessment will assign grades/marks based on parameters provided in the rubrics.



Guidelines for Practical Examination

Problem statements must be decided jointly by the internal examiner and external examiner. During the practical assessment, maximum weightage should be given to satisfactory implementation of the problem statement. Relevant questions may be asked at the time of evaluation to test the student's understanding of the fundamentals and effective and efficient implementation. This will encourage transparent evaluation and a fair approach, and hence will not create any uncertainty or doubt in the minds of the students. So, adhering to these principles will consummate our team efforts to the promising start of student's academics.

Guidelines for Laboratory Conduction

The instructor is expected to frame the assignments by understanding the prerequisites, technological aspects, utility and recent trends related to the topic. The assignment framing policy needs to address the average students and inclusive of an element to attract and promote the intelligent students. Use of open source software is encouraged. Based on the concepts learned. Miniproject should be implemented by the students in a group of 2-3 students.

Virtual Laboratory: (If Any):

- https://code.org/en-US/tools/web-lab

Suggested List of Laboratory Experiments/Assignments

Group A: Assignments (Mandatory Assignment)

Sr No	Assignment Title	*Mapping of Course Outcomes
1.	Create a multipanel layout (header, navigation, main content, footer) for a college or e-commerce homepage using HTML and CSS.	CO1, CO2
2.	To implement a functional registration form with client-side validation using JavaScript and ES6.	CO2
3.	Create a To-Do List app where users can add, edit, and delete tasks dynamically using DOM	CO2
4.	Design the XML document to store the information of the employees of any business organization, display the content using CSS and demonstrate the use of DTD.	CO2
5.	To fetch and display real-time weather data using a public API. ● Use Fetch API to retrieve data from any free weather API. ● Display temperature, humidity, and city name dynamically.	CO3
6.	Build a Student Management module: Add, View, Update, Delete student records using JSP and MySQL.	CO3
7.	Create login and registration pages using PHP and MySQL.	CO3
8.	Build a simple blogging module using PHP and MySQL.: Add, edit, delete blog posts with user authentication.	CO3
9.	Build a chatbot or recommendation feature.	CO2, CO4



Group B: Assignments		
Sr No	**Assignment Title**	***Mapping of Course Outcomes**
1.	Develop a fully functional web application that demonstrates responsive design, accessibility compliance, interactivity, and secure server-side integration.	CO1, CO2, CO3, CO4

24-VEC-3-01: Environmental Studies

Teaching Scheme: Theory: 2 Hours/Week	**Credit:** -2	**Examination Scheme:** **CIE** : 20 Marks **SEE** : 30 Marks

Prerequisite Course: 24-BSC-1-02: Engineering Chemistry

Companion Course: - --

Course Objectives:

- To define the environment and its types, raise awareness, and classify resources for sustainable development.
- To understand the sources and impacts of pollution, and to explore prevention and mitigation strategies.
- To examine climate change, ozone depletion, acid rain, deforestation, desertification, and international monitoring efforts (e.g., the GLOBE Program).
- To explore key environmental protection laws and understand the EIA process.

Course Outcomes:

After completion of the course, learners will be able to

CO No	CO	BL
1	**Understand** environmental principles, which in turn help in sustainable development.	2
2	**Apply** pollution control measures and environmental standards.	3
3	**Understand** global issues and global efforts & their impact on human life	2
4	**Understand** key environmental protection laws and **apply** EIA guidelines to ensure sustainable development.	3

Course Contents

Unit I	Introduction & Natural Resources	7 Hours

Definition and Concept of Environment, Types of Environment, Components of the environment (atmosphere, hydrosphere, lithosphere, biosphere), Raising Environmental Awareness in India. Classification of Resources: Living and Non-Living resources, Land resources: Forest resources, Energy resources: Growing energy needs. Ecosystems: Structure and function, energy flow (food chains, food webs, ecological succession), Biodiversity: levels, hotspots in India, conservation (in-situ, ex-situ).

#Exemplar/Case Studies: Ganga rejuvenation plan (Namami Gange), InterLinking of Rivers Project, Use of alternate energy sources.(Any One)

*Mapping of Course Outcomes	CO1

Unit II	**Environmental Pollution and Its Impact**	**8 Hours**
Environmental Pollution: Air Pollution: Primary & secondary pollutants, causes, effects, & control of pollution, Automobile & Industrial pollution, Water pollution: Sources & types of pollution, effects, & control of pollution, Protection of the Source of water. Drinking water quality standards & Disposal of Sewage standards. Land Pollution, Types, causes, & effects, Noise Pollution. Causes & effects. Introduction to Marine, thermal, and nuclear pollution. Solid waste, biomedical, and E-waste management.		
#Exemplar/Case Studies/Self Study: Bhopal Gas Tragedy, Nanar oil refinery project, Ganga Pollution case. (Any One)		
***Mapping of Course Outcomes**	**CO2**	
Unit III	**Global Environmental Issues and Global Efforts:**	**7 Hours**
Climate change and impacts on the human environment. Ozone depletion. Acid rain, Deforestation, and desertification. Introduction to The GLOBE Program and GLOBE Protocols, Carbon footprint and environmental ethics, Human population, health, urbanization, women & child welfare. Disaster management: floods, earthquakes, cyclones, landslides.		
#Exemplar/Case Studies: Chornobyl nuclear accident case, Montreal Protocol, 2023 Delhi Air Pollution Crisis (Anyone)		
***Mapping of Course Outcomes**	**CO3**	
Unit IV	**Environmental Policy and Legislation**	**8 Hours**
Introduction to Environmental Protection Act, Air and Water Acts, Wildlife Protection Act, Forest Conservation Act, National Green Tribunal Act -2010, Introduction to Environmental Impact Assessment (EIA). Introduction to Sustainable Development Goals (SDGs), Environmental movements: Chipko, Silent Valley, Bishnois of Rajasthan, Role of religions and cultures in conservation, Environmental Agreements: Kyoto Protocol, Cartegena Protocol, Ramsar Convention, Vienna Convention, Basel Convention.		
#Exemplar/Case Studies: Impact of Pesticides and Fertilizers on Soil in Punjab, India		
***Mapping of Course Outcomes**	**CO4**	
Learning Resources		
Text Books:		
T1. R. Rajagopalan, "Environmental Studies from Crisis to Cure", Oxford University Press. **T2.** Erach Bharucha, "Textbook of Environmental Studies for Undergraduate Courses", University Grants Commission. **T3.** K.Raghavan Nambiar, "Textbook Of Environmental Studies", Scitech Publications (India) Pvt Limited.		
Reference Books :		
R1. P.C.Thakur, "Environmental Studies", A.B.D. Publishers **R2.** Basu Mahua Xavier S, "Fundamentals Of Environmental Studies", Cambridge University Press.		
Additional Resources: (Books, e-Resources: https://bgc.ac.in/pdf/env-book-2018.pdf		
MOOC Courses links: https://onlinecourses.nptel.ac.in/noc23_hs155/preview https://onlinecourses.nptel.ac.in/noc25_hs195/preview		

SEM VI



24-PCC-CS-3-04: Software Engineering

Teaching Scheme:	**Credit:** 3	**Examination Scheme:**
Theory: 3 Hours/Week		**CIE** : 20 Marks **MSE :** 20 Marks **SEE** : 60 Marks

Prerequisites Courses: 24-ESC-1-02 Programming and Problem Solving

Companion Course: NA

Course Objectives:

- To introduce fundamental concepts of software engineering, SDLC, and key process models.
- To develop skills in software project planning, cost estimation, and scheduling.
- To explain software design principles and their application in system design.
- Apply Agile principles, Scrum framework, and JIRA tools to effectively execute software development projects.
- To provide knowledge of software testing strategies, quality assurance, and evaluation techniques.
- To enable the use of AI techniques and tools to improve software development workflows.

Course Outcomes:

After completion of the course, learners should be able to

CONo	CO	BL
1	**Understand** software engineering fundamentals, SDLC, and major process models.	2
2	**Apply** project planning, estimation, and scheduling techniques in software projects.	3
3	**Apply** software design concepts in system design.	3
4	**Understand** Agile concepts, frameworks, practices, and their application in modern software development.	3
5	**Understand** software testing, quality concepts, and evaluation techniques.	2
6	**Apply** AI to enhance software development workflows.	3

Course Contents

Unit I	**Introduction to Software Engineering**	**7 Hours**

Software Engineering Fundamentals: Introduction to software engineering, The Nature of Software, Defining Software, Software Engineering Practice. **Software Process**: The Waterfall Model, Incremental Process Models, Evolutionary Process Models, Concurrent Models **Requirements Engineering**: Eliciting Requirements, Collaborative Requirements Gathering, Usage Scenarios, Elements of the Requirements Model, Negotiating Requirements, Validating Requirements.

#Exemplar/Case Studies: Studying a Banking App's lifecycle by analyzing how the Waterfall Model secures core transactions while the Evolutionary Model allows for constant UI updates, demonstrating modern Software Engineering Practice.

***Mapping of Course Outcomes**	**CO1**	
Unit II	**Estimation and Scheduling**	**7 Hours**
Estimation for Software Projects: The Project Planning Process, Defining Software Scope and Checking Feasibility, Resources management, Reusable Software Resources, Environmental Resources, Software Project Estimation, Decomposition Techniques, Software Sizing, Problem-Based Estimation, LOC-Based Estimation, FP-Based Estimation, Object Point (OP)-based estimation, ProcessBased Estimation. **Project Scheduling:** Project Scheduling, Defining a Task for the Software Project, Scheduling tools (Gantt Project, Agantty, Project Libre.		
#Exemplar/Case Studies : Estimating and scheduling a Smart Health App by calculating complexity (Function Points), reusing existing login modules, and mapping the 6-month timeline on a Gantt Chart to ensure feasibility.		
***Mapping of Course Outcomes**	**CO2**	
Unit III	**Design Engineering**	**7 Hours**
Design Concepts: Design within the Context of Software Engineering, The Design Process, Software Quality Guidelines and Attributes, Design Concepts - Abstraction, Architecture, Design Patterns, Separation of Concerns, Modularity, Information Hiding, Functional Independence, Refinement, Aspects, Refactoring,Design Classes, The Design Model , Data Design Elements, Architectural Design Elements, Interface Design Elements, Component-Level Design Elements, Component Level Design for Web Apps, Content Design at the Component Level, Functional Design at the Component Level, Deployment-Level Design Elements.		
#Exemplar/Case Studies : Analyzing a Food Delivery System by using Abstraction to simplify order flows, Modularity to separate payments from menus, and Refactoring the code to improve Software Quality without changing its core behavior.		
***Mapping of Course Outcomes**	**CO3**	
Unit IV	**Agile Software Development Methodology**	**8 Hours**
Introduction to Agile: Need for Agile, Agile vs Traditional, Development, Overview of Agile Methodologies, **Agile Manifesto & Principles:** Agile Manifesto,Four Core Values,Twelve Agile Principles, Agile Mindset **Agile Process & Practices:** Iterative and Incremental Development, Time-boxed Iterations, Continuous Integration **Scrum Framework:** Scrum Overview, Scrum Roles ,Scrum Artifacts, Scrum Events **Other Agile Models:** Kanban, Lean Software Development **Benefits, Challenges & Applications:** Advantages of Agile, Challenges in Agile Adoption, Agile in Large-Scale Projects, Real-world Applications **Agile project management using JIRA**: issue tracking, user stories, backlogs, sprint planning, boards, and reporting.		
#Exemplar/Case Studies: Agile using JIRA in Mobile App Development (Scenario: A startup develops a mobile fitness app with evolving user needs.)		
***Mapping of Course Outcomes**	**CO4**	
Unit V	**Software Testing**	**7 Hours**
Role of testing in software projects and research prototypes , Levels of testing: unit, integration, system, acceptance Types of testing: Black-box, white-box (basis path, decision, etc. only concept-level) Regression, performance, usability (overview) Test case design: equivalence class partitioning, boundary value analysis Software quality concepts: Quality attributes (reliability, usability, maintainability, etc.), Quality assurance vs quality control Software metrics (intro): Product metrics, process metrics, project metrics (concept-level examples) Evaluation of research software: performance analysis, comparison with existing systems.		



#Exemplar/Case Studies: Selenium Testing with any online application		
***Mapping of Course Outcomes**	**CO5**	
Unit VI	**AI in Software Engineering**	**7 Hours**
Introduction to AI in Software Engineering: AI-assisted code generation and auto-completion (GitHub Copilot), Intelligent bug detection and code quality analysis, Automated testing and test case generation, AI-based code review and pull request analysis, Issue tracking and requirement management using AI, Version control, collaboration, and DevOps automation with AI.		
#Exemplar/Case Studies: Study the application of AI-based testing and defect prediction tools for a Student Management System and discuss their impact on software quality		
***Mapping of Course Outcomes**	**CO6**	
Learning Resources		
Text Books		
T1. Roger Pressman, "Software Engineering: A Practitioner's Approach", McGraw Hill, ISBN 0–07–337597–7 **T2.** Ian Sommerville, "Software Engineering", Addison and Wesley, ISBN 0-13-703515-2		
Reference Books :		
R1. Carlo Ghezzi, "Fundamentals of Software Engineering", PHI, ISBN-10: 0133056996 **R2.** Rajib Mall, "Fundamentals of Software Engineering", PHI, ISBN-13: 978-8120348981 **R3.** S K Chang, "Handbook of Software Engineering and Knowledge Engineering", World Scientific, Vol I, II, ISBN: 978-981-02-4973-1 **R4.** Tom Halt, "Handbook of Software Engineering", Clanye International, ISBN- 10: 1632402939		
Additional Resources: (Books, e-Resources) • https://ebookpdf.com/roger-s-pressman-software-engineering		
MOOC Courses links : • https://swayam.gov.in/nd1_noc19_cs69/preview • https://swayam.gov.in/nd2_cec20_cs07/preview		

24-PEC-CS-3-03A: Game Development & Animation

Teaching Scheme:	**Credit:** 3	**Examination Scheme:**
Theory: 3 Hours/Week		**CIE** : 20 Marks **MSE :** 20 Marks **SEE** : 60 Marks

Prerequisites Courses: (24-PEC-CS-3-01A)Computer Graphics, (24-PEC-CS-3-02A) Computer Graphics Laboratory

Companion Course: (24-PEC-CS-3-04A) Game Development & Animation Lab

Course Objectives:

- **To provide** students with a strong foundation in the fundamentals of game development, including the roles of designers, programmers, artists, and testers, as well as the evolution and types of games.
- **To develop** the ability to design, document, and implement game systems, including storytelling, game mechanics, level design, animation workflows, object management, scripting, and audio integration.
- **To equip** students with skills to analyze, test, and optimize games, while evaluating monetization strategies, performance metrics, and emerging trends in the gaming industry.

Course Outcomes:

After completion of the course, learners should be able to

CONo	CO	BL
CO1	Understand the fundamentals of games and participants in game development.	2
CO2	Develop and document game design concepts, including storytelling, mechanics, and level design strategies.	2
CO3	Apply principles of animation to create and manage animation workflows for games	3
CO4	Understand game objects, components, inputs, and physics concepts in a game development environment	2
CO5	Develop game logic, manage objects, integrate audio, scripting and programming concepts to functional game systems..	3
CO6	Interpret game testing methods, performance metrics, optimization techniques, monetization models, and emerging trends in gaming.	2

Course Contents

Unit I	**Introduction to Game Development**	**6 Hours**

Game Definition, **History and Evolution of Video Games,** Types of Games: 2D, 3D, **Game development process:** Pre-production, Production, Testing, Post-production, Introduction to Game Engines, **Game development roles:** designer, programmer, artist, tester

#Exemplar/Case Studies: Study of coordinated roles of designers, programmers, artists, and testers in the development of the game Angry Birds or any other game		
***Mapping of Course Outcomes**	**CO1**	
Unit II	**Game Design Strategies**	**8 Hours**
Game storytelling:Need of Stories in Games, The Storytelling Engine, Linear Stories, Nonlinear Stories, **Game mechanics**: Space, Objects, Attributes, and States, Actions, Rules, Skill, Chance, **Level design and environment setup:** Level Design, Key Design Principles, Layouts, Expanding on the Principles of Level Design, The Level Design Process, Pitfalls of Level Design, **Game Design Document (GDD) preparation**: Introduction to Game Design Document, The Myth of the Game Design Document ,The Purpose of Documents, Types of documents		
#Exemplar/Case Studies: Prepare game design documentation for the game Candy Crush Saga, covering the purpose of design documents and the structure/layout of the document.		
***Mapping of Course Outcomes**	**CO2**	
Unit III	**Introduction to Animation**	**7 Hours**
Principles of Animation**,** Conventional and computer based animation:2D and 3D, **Animation techniques:** Keyframing, tweening, 3D model- Texturing-skeleton-skinning, rigging, and motion paths, **Animation workflow and rendering basics:** Pre-production:scripting, storyboarding, Production: modeling, animating, and Post-production:rendering, editing, sound design		
#Exemplar/Case Studies: Study on Pixar's Toy Story focusing on 3D animation principles, modeling, rigging, texturing, keyframing, and complete animation workflow from pre-production to post-production.		
***Mapping of Course Outcomes**	**CO3**	
Unit IV	**Game Development Environment**	**8 Hours**
Game Objects & Components, Sprites (2D Assets), **Basic Transform Operations:** Simple Movement, Simple Rotation, Object Scaling, **Input Handling:** Easy Input Handling, Player Controls (Keyboard / Mouse / Touch), **2D Physics Concepts:** Rigidbody2D Components:,Colliders (Box, Circle, Polygon), Physics Materials, Collision Detection & Triggers, **Primitive Data & Math Fundamentals:** Data Types and Variables, Mathematical Operations, Variable Scope and Access, **Organizing Game Objects:** Parent–Child Hierarchy, Sorting Layers,Tagging Game Objects,Collision Layers		
#Exemplar/Case Studies: Design a simple 2D game by selecting an appropriate game development platform such as Unity, Godot, or Unreal.		
***Mapping of Course Outcomes**	**CO4**	
Unit V	**Game Logic & Object Management**	**8 Hours**
Introduction to Scripting: C# Coding Fundamentals, Game Loops & Functions, Defining and Using Functions, **Object-Oriented Concepts:** Defining Classes, Creating and Using Classes, Accessing Game Objects, Constructor and Property Functions, **Managing Game Objects:** Prefabs (Creation and Usage), Instantiating (Creating) Objects, Destroying Objects, Activating and Deactivating Game Objects, Controlling Object Lifespans with Invoke,. **Sound Effects:** Importing Audio Files,Adding Sounds to Game Objects, Playing Audio Through Scripts, **3D Physics Concepts:** Rigidbody (3D), Colliders (Box, Sphere, Capsule), Gravity, Mass, Drag, Basic 3D Collision & Trigger Events		



#Exemplar/Case Studies: Design a simple 3D game by selecting an appropriate game development platform such as Unity, Godot, or Unreal.		
***Mapping of Course Outcomes**	**CO5**	
Unit VI	**Testing, Deployment and Trends in Gaming**	**7 Hours**
Game TestingTypes: Functional, Usability, Performance, Optimization and Debugging, **GameMonetization and Publishing Platforms** :Game monetization- In Play Store, Advertising, Subscriptions, Publishing platforms-Distribution, Marketing and promotion,Backend services Steam, **Emerging Trends:** AR/VR, AI in games,Metaverse, **Ethics and Career Scope in Game Industry**		
#Exemplar/Case Studies: State of freemium mobile game monetisation in 2025		
***Mapping of Course Outcomes**	**CO6**	
Learning Resources		
Text Books:		
T1. Erik Bethke, " Game Development and Production", Wordware Publishing,Inc., 2003, ISBN 1-55622-951-8 **T2.** Jesse Schell, "The Art of Game Design", Morgan Kaufmann, Elsevier, ISBN: 978-0-12-369496-6		
Reference Books :		
R1. Jeremy Gibson Bond, "*Introduction to Game Design, Prototyping, and Development*",Addison Wisely		
Additional Resources: (Books, e-Resources) Unity (https://docs.unity3d.com/6000.0/Documentation/Manual/UnityManual.html) Unreal Engine Documentation (https://docs.unrealengine.com) Case study for unit2 :https://www.tmv.edu.in/pdf/BCA-Game%20Development%20Syllabus.pdf Unit 3 case study: https://ieeexplore.ieee.org/abstract/document/501812 Unit 6 case study: https://www.utupub.fi/bitstream/handle/10024/182531/Kamari_Elmeri_opinnayte.pdf?sequence=1		
MOOC Courses links : • https://www.freecodecamp.org/news/game-development-for-beginners-unity-course/ • https://www.coursera.org/specializations/game-design-and-development • https://books.google.co.in/books?id=XhrBBeA0-LMC&pg=PR11&source=gbs_selected_pages&cad=1#v=onepage&q&f=false • https://littleflowercollege.edu.in/upload/e_contents/files/ca10931070103bc44feafb883cfbb02e.pdf • https://www.geeksforgeeks.org/computer-networks/difference-between-2-d-and-3-d-animation/		



24-PEC-CS-3-03B : Network Security

Teaching Scheme:	**Credit:** 3	**Examination Scheme:**
Theory: 4 Hours/Week		**CIE** : 20 Marks **MSE** : 20 Marks **SEE** : 60 Marks

Prerequisites Courses: 24-PCC-CS-3-02 Computer Network

Companion Course: 24-PEC-CS-3-04B : Network Security Lab

Course Objectives:

- Understand the fundamentals of network security, security services, attacks, and mechanisms.
- Apply symmetric and asymmetric cryptographic algorithms to ensure data confidentiality and integrity.
- Analyze authentication mechanisms, hashing techniques, and digital signatures for secure communication.
- Evaluate network and web security protocols to design secure and reliable communication systems.

Course Outcomes:

After completion of the course, learners should be able to

CONo	CO	BL
CO1	Understand the principles of network security services, attacks, and mechanisms.	2
CO2	Apply symmetric key encryption algorithms to ensure confidentiality in data transmission.	3
CO3	Implement asymmetric encryption techniques for secure key distribution and confidentiality.	3
CO4	Demonstrate hashing and message authentication mechanisms for integrity verification.	3
CO5	Evaluate digital signature and user authentication techniques for secure system access.	3
CO6	Analyze network and web security protocols to protect communication systems.	3

Course Contents

Unit I	**Introduction to Security Concepts**	**7 Hours**

Need for Security, Computer Security Concepts, Security Attacks, Security Services, Security Mechanisms, Classical Encryption Techniques, A Model for Network Security, Substitution Techniques, Transposition Techniques .

#Exemplar/Case Studies: WhatsApp allegedly exposed 3.5 billion phone numbers via the "Click to Chat" feature.

***Mapping of Course Outcomes**	**CO1**

Unit II	**Symmetric Key Cryptography**	**7 Hours**

Block Cipher Principles, DES and Weaknesses of DES, Block Cipher Design Principles, Stream Ciphers, RC4 and Security issues

with RC4, AES Structure, Transformation Functions, Key Expansion, Block Cipher Modes of Operation, Key Management and Distribution for Symmetric Encryption		
#Exemplar/Case Studies Malicious attachments (Trojan Horse) like "Invoice.pdf" or "Security_Update.exe" and its impact.		
***Mapping of Course Outcomes**	**CO2**	
Unit III	**Asymmetric Key Cryptography**	**7 Hours**
Principles of Public-Key Cryptosystems, RSA Algorithm, Computational Aspects and Security, Diffie–Hellman Key Exchange, Man-in-the-Middle Attack, Key Management and Distribution for Asymmetric Encryption.		
#Exemplar/Case Studies: Social media encryption—such as end-to-end and server-side encryption—to ensure secure communication and protect user privacy.		
***Mapping of Course Outcomes**	**CO3**	
Unit IV	**Hashing and Message Authentication**	**8 Hours**
Authentication Requirements, Message Authentication Functions, Cryptographic Hash Functions, Secure Hash Algorithms (SHA-256 and SHA-512) , Message Authentication Codes (MACs) , Requirements and Security of MACs , HMAC overview.		
#Exemplar/Case Studies SHA-256/SHA-512 and MACs secure online banking, digital payments.		
***Mapping of Course Outcomes**	**CO4**	
Unit V	**Digital Signatures and User Authentication**	**7 Hours**
Digital Signature: concepts, properties, requirements, and security; various digital signature schemes such as ElGamal and Schnorr; remote user authentication principles; and symmetric and asymmetric authentication techniques.		
#Exemplar/Case Studies DigiLocker's digital signature service, e-Sign, to digitally sign documents by linking to Aadhaar number and mobile number.		
***Mapping of Course Outcomes**	**CO5**	
Unit VI	**Network and Web Security**	**6 Hours**
Need for Web Security; Common Web Security Threats (Phishing, SQL Injection, Session Hijacking), Web Security Approaches: CAPTCHA Mechanisms and Bot Prevention; Emerging Trends in Web Security: Zero Trust Architecture,AI-based Threat Detection.		
#Exemplar/Case Studies Examining the effectiveness of CAPTCHA mechanisms in preventing automated bot activities and ensuring secure user interactions on web applications.		
***Mapping of Course Outcomes**	**CO5,CO6**	
Learning Resources		
Text Books		
T1. William Stallings, "Cryptography and Network Security, Principles and Practice", Pearson Education, India		

T2. N Harini, T R Padmanabhan, C K Shyamala, "Cryptography and Security", Wiley-India

Reference Books :

R1. Mark Stamp, "Information Security Principles and Practice", Willy India Edition
R2. Atul Kahate, "Cryptography and Network Security", Tata Mc Grawhill, India
R3. Forouzan, "Cryptography & Network Security", PHI

Additional Resources: (Books, e-Resources)
Vlabs, "Cryptography Lab" - http://cse29-iiith.vlabs.ac.in/
Cryptool - https://www.cryptool.org/en/
Wireshark - https://www.wireshark.org/download.html

MOOC Courses links :

- NPTEL Videos and PDF - "Cryptography and Network Security by Debdeep Mukhopadhyay, IIT Kharagpur"- https://nptel.ac.in/courses/106105031/
- "Network and Computer Security by Prof Ronald Rivest, MIT" - https://ocw.mit.edu/courses/6-857-network-and-computer-security-spring-2014/



24-PEC-CS-3-03C: Data Science

Teaching Scheme: Theory: 3 Hours/Week	**Credit:** 3	**Examination Scheme:** **CIE** : 20 Marks **MSE :** 20 Marks **SEE** : 60 Marks

Prerequisites Courses: 24-PCC-CS-2-05: Database Management System, 24-PEC-CS-3-01C: Artificial Intelligence

Companion Course: 24-PEC-CS-3-04C Data Science Lab

Course Objectives:

- To introduce fundamental concepts and applications of Data Science across diverse domains.
- To develop skills in data preprocessing, transformation, and quality improvement.
- To provide understanding of multimodal data types and large-scale data systems.
- To enable analysis and visualization of data for effective insight communication and decision-making.

Course Outcomes:

After completion of the course, learners should be able to

CONo	CO	BL
1	**Understand** fundamental concepts of data science along with the different application domains.	2
2	**Apply** data preprocessing and transformation techniques to improve data quality for analysis.	3
3	**Apply** concepts of multimodal data to interpret image, audio, and video data characteristics.	3
4	**Analyze** datasets using descriptive, diagnostic, predictive, and prescriptive analytics techniques.	4
5	**Analyze** large-scale data systems and **communicate** insights using Big Data and visualization concepts.	4

Course Contents

Unit I	**Foundations of Data Science**	**7 Hours**

Introduction and Applications, **The Relationship between Data Science and various fields** : Statistics, Computer Science, Business Analytics, Information Science, Issues of Ethics, Bias and Privacy, **Data** : Definition, Classification(structured, semi-structured, and unstructured data) ; **Types of data attributes(Variables):** qualitative and quantitative; D**ata Collection sources** : Open Data, Social Media Data, Multimodal Data, Data Storage, and Data Presentation; **Data Science Lifecycle**

#Exemplar/Case Studies : Understand the Data Science lifecycle to support smart decisions for insurance customer churn.

***Mapping of Course Outcomes**	**CO1**

Unit II	**Data Preprocessing**	**8 Hours**

Data Preprocessing: Types of Missing Data, Identification and Handling of incomplete, inconsistency and noisy data; **Data Transformation:** Data Smoothing, Data Aggregation, Generalization, Discretization, Normalization (Min-Max normalization,

z-score normalization, Decimal scaling); **Detection and Handling Outlier:** Detection using Min-Max normalization, z-score normalization, and InterQuartile Range(IQR), Removal of Outlier, and Different Strategies to Replace Outlier.		
#Exemplar/Case Studies : Identification of fraudulent credit card transactions using anomaly detection techniques		
***Mapping of Course Outcomes**	**CO1, CO2**	
Unit III	**Modalities of Data**	**7 Hours**
Introduction and Importance Data Modalities; **Applications**; **Types of Multimodal Data**: image, audio, and video data, **Image Data Fundamentals:** digital image structure, pixels and resolution, common formats; **Audio Data Fundamentals:** basics of sampling and amplitude, common formats, data representation; **Video Data Fundamentals**: structure, frames and frame rate, common formats, relationship between image and video data; **Multimodal Data Challenges, Future trends in multimodal data systems**		
#Exemplar/Case Studies: ImageBind framework for multimodal data binding and unified representation		
***Mapping of Course Outcomes**	**CO3**	
Unit IV	**Data Analysis and Analytics**	**7 Hours**
Introduction and relation between Data Analysis and Analytics; Descriptive Analysis: Frequency distribution and measures of centrality and dispersion; **Diagnostic Analytics:** Correlation analysis and variance analysis; **Predictive Analytics:** Trend analysis and simple time-based extrapolation; **Prescriptive Analytics:** Rule-based decision analysis and scenario analysis; **Exploratory Data Analysis:** Summary statistics and visual inspection for data summarization, pattern discovery, trend identification, and anomaly detection.		
#Exemplar/Case Studies: Data Analytics to Derive Business Intelligence		
***Mapping of Course Outcomes**	**CO3, CO4**	
Unit V	**Big Data Analytics and Data Management**	**8 Hours**
Relation between Big Data and Big Data Analytics; Need and Characteristics of Big Data (5Vs); **Overview of Big Data and Hadoop ecosystem:** HDFS and MapReduce, Apache Spark; **Higher-level tools**: Pig and Hive; **Supporting tools**: Sqoop and Flume; **Big Data architecture flow:** data ingestion, distributed storage, distributed processing, data transformation and analytics using DBT(Data Build Tool) framework, and visualization.		
#Exemplar/Case Studies: Manage large-scale Netflix data to drive decisions and improve customer experience.		
***Mapping of Course Outcomes**	**CO3, CO4**	
Unit VI	**Data Visualization and Visual Analytics**	**7 Hours**
Introduction to Data Visualization; Importance and role of visualization in analytics; Challenges in Data Visualization; Types of Data Visualization; Data Visualization Techniques; Chart Selection Strategies; **Types of Plots**: Categorical, Numerical, and Time-Series Data; **Visual Encoding principles:** color, shape, size, scale, and layout; **Best practices and guidelines** for effective visualization; Visualizing Big Data; Common Data Visualization Tools and Framework		
#Exemplar/Case Studies: Choose suitable chart types to communicate data insights effectively.		



***Mapping of Course Outcomes**	**CO5**

Learning Resources

Text Books

T1. Chirag Shah, A Hands-On Introduction to Data Science, Cambridge University Press, 2020, ISBN 978-1108472449.
T2. David Dietrich, Barry Hiller, "Data Science and Big Data Analytics", EMC education services, Wiley publication, 2012, ISBN0-07-120413-X

Reference Books :

R1. Andy Kirk, Data Visualisation: A Handbook for Data Driven Design, SAGE Publications, 2016, ISBN 978-1473963371.
R2. Stephan Kudyba, Big Data, Mining, and Analytics: Components of Strategic Decision Making, Auerbach Publications (CRC Press), 2014, ISBN 978-1466568709

Additional Resources: (Books, e-Resources)

- https://www.google.co.in/books/edition/_/6ZQ7EQAAQBAJ?hl=en&gbpv=1
- https://www.projectpro.io/article/data-analysis-process/612
- https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud
- Using Data Analytics to Derive Business Intelligence: A Case Study. https://arxiv.org/abs/2305.19021
- Imagebind: https://ai.meta.com/blog/imagebind-six-modalities-binding-ai/ , https://arxiv.org/abs/2305.05665
- Big Data Analytics: Big Data Case Studies: How Four Industry Leaders Overcame Data Challenges | by NAGADHYANESH R | Medium
- https://www.gooddata.com/blog/how-to-choose-the-best-chart-type-to-visualize-your-data/

MOOC Courses links :

- https://onlinecourses.nptel.ac.in/noc25_cs20/preview
- https://onlinecourses.nptel.ac.in/noc25_cs60/preview

24-PEC-CS-3-03D: High Performance Computing

Teaching Scheme:	**Credit:** 3	**Examination Scheme:**
Theory: 3 Hours/Week		**CIE :** 20 Marks **MSE :** 20 Marks **SEE :** 60 Marks

Prerequisites Courses: 24-PCC-CS-2-03 : Computer Organization and Architecture

Companion Course: 24-PEC-CS-3-04D:High Performance Computing Lab

Course Objectives:

- To understand different parallel programming models
- To analyze the performance and modeling of parallel programs
- To illustrate the various techniques to parallelize the algorithm
- To Understand parallel programming using OpenMP and MPI to enhance performance and scalability.
- To Understand GPU-based parallel programming concepts with CUDA for accelerated computing.
- To Understand Scope of Parallel Computing and its search algorithms.

Course Outcomes:

After completion of the course, learners should be able to

CONo	**CO**	**BL**
CO1	**Understand** various Parallel Paradigm	2
CO2	**Design** and Develop an efficient parallel algorithm to solve given problem	3
CO3	**Illustrate** data communication operations on various parallel architecture	3
CO4	**Apply** OpenMP and MPI constructs and communication techniques to develop scalable parallel applications.	3
CO5	**Analyze** CUDA-based GPU programming techniques to evaluate their efficiency in parallel execution.	4
CO6	**Analyze** the performance of HPC applications	4

Course Contents

Unit I	**Introduction to Parallel Computing**	**7 Hours**

Introduction to Parallel Computing: Motivating Parallelism, Modern Processor: Stored- program computer architecture, General-purpose Cache-based Microprocessor architecture. Parallel Programming Platforms: Implicit Parallelism, Dichotomy of Parallel Computing Platforms, Physical Organization of Parallel Platforms, Communication Costs in Parallel Machines. Levels of parallelism, Models: SIMD, MIMD, SIMT, SPMD, Data Flow Models, Demand-driven Computation, Architectures: N-wide superscalar architectures, multi-core, multi-threaded.

#Case Studies: Weather Forecasting using Parallel Computing.



***Mapping of Course Outcomes**	**CO1**	
Unit II	**Parallel Algorithm Design**	**7 Hours**
Principles of Parallel Algorithm Design: Preliminaries, Decomposition Techniques, Characteristics of Tasks and Interactions, Mapping Techniques for Load Balancing, Methods for Containing Interaction Overheads, Parallel Algorithm Models: Data, Task, Work Pool and Master Slave Model, Complexities: Sequential and Parallel Computational Complexity, Anomalies in Parallel Algorithms.		
#Case Studies: Parallel Traffic Signal Optimization System for Smart City.		
***Mapping of Course Outcomes**	**CO2**	
Unit III	**Parallel Communication**	**7 Hours**
Basic Communication: One-to-All Broadcast, All-to-One Reduction, All-to-All Broadcast and Reduction, All-Reduce and Prefix-Sum Operations, Collective Communication using MPI: Scatter, Gather, Broadcast, Blocking and non blocking MPI, All-to-All Personalized Communication, Circular Shift, Improving the speed of some communication operations.		
#Case Studies: Real-Time Stock Market Risk Analytics System.		
***Mapping of Course Outcomes**	**CO3**	
Unit IV	**Parallel programming using OpenMP and MPI**	**6 Hours**
Parallel programming using OpenMP(Open Multi-Processing): OpenMP Components,Data Scope Attribute/Clauses,Work-sharing Constructs,Synchronization Constructs,Library Routines & Environment Variables. MPI(Message Passing Interface): Introduction to MPI,Basic terms in MPI,MPI Implementation,Sample programs,Point to Point Communication,Collective Communication.		
Case Studies: DNA Sequence Analysis for Genomic Research using OpenMP & MPI.		
***Mapping of Course Outcomes**	**CO4**	
Unit V	**Parallel Programming with Co-Processors**	**6 Hours**
Need of Co-processors, Introduction to GPUs, GPU Architecture,Types of GPUs, CUDA Programming Basics.		
Case Studies: Real-Time Medical Image Segmentation using GPUs (CUDA).		
***Mapping of Course Outcomes**	**CO5**	
Unit VI	**High Performance Computing Applications**	**6 Hours**
Scope of Parallel Computing, Parallel Search Algorithms: Depth First Search(DFS), Breadth First Search(BFS), Parallel Sorting: Bubble and Merge, Distributed Computing: Document classification, Parallel Computing for AI/ ML.		
Case Studies: Intelligent Traffic Management & Emergency Vehicle Routing using HPC.		
***Mapping of Course Outcomes**	**CO6**	
Learning Resources		

Text Books
Textbooks : **T1.** Ananth Grama, Anshul Gupta, George Karypis, and Vipin Kumar, "Introduction toParallel Computing", 2nd edition, Addison-Wesley, 2003, ISBN: 0-201-64865-2 **T2.** Seyed H. Roosta, —Parallel Processing and Parallel Algorithms Theory and Computation, Springer-Verlag 2000 ,ISBN 978-1-4612-7048-5 ISBN 978-1-4612-1220-1 **T3.** John Cheng, Max Grossman, and Ty McKercher, —Professional CUDA C Programming‖, John Wiley & Sons, Inc., ISBN: 978-1-118-73932-7
Reference Books :
R1. Kai Hwang, "Scalable Parallel Computing", McGraw Hill
Additional Resources: (Books, e-Resources) CDAC: https://www.cdac.in/index.aspx?id=tap&tapcat=High_Performance_Computing
MOOC Courses links : 1: https://www.coursera.org/learn/introduction-high-performance-computing 2: mooc-list.com/tags/high-performance-computing



24-PEC-CS-3-03E : Advanced Web Technology

Teaching Scheme: Theory: 3 Hours/Week	**Credit:** 3	**Examination Scheme:** **CIE** : 20 Marks **MSE** : 20 Marks **SEE** : 60 Marks

Prerequisites Courses: 24-PEC-CS-3-01E : Web Technology

Companion Course: 24-PEC-CS-3-04E : Advanced Web Technology Lab

Course Objectives:

- Understand modern full-stack web architecture and frameworks.
- Design scalable and secure front-end applications using React.js and Angular.
- Develop backend services with Node.js and Express.js.
- Explore advanced web technologies. .

Course Outcomes:

After completion of the course, learners should be able to

CONo	**CO**	**BL**
CO1	**Understand** the architecture and ecosystem of modern web applications.	2
CO2	**Develop** responsive, component-based frontends using React.js and Angular frameworks	3
CO3	**Build** secure and scalable backend services using Node.js and Express.js.	3
CO4	**Implement** advanced concepts for dynamic web systems.	3
CO5	**Apply** web security and performance optimization techniques for reliable full-stack applications.	3

Course Contents

Unit I	**Modern Web Ecosystem & Architecture**	**7 Hours**

Evolution from monolithic to microservice and serverless models. MERN & MEAN stack overview and architecture. Single Page Applications (SPA) vs Multi Page Application (MPA), Client-Side Rendering (CSR) vs Server-Side Rendering (SSR), vs Static Site Generation (SSG), RESTful APIs vs GraphQL. Development ecosystem: npm, yarn, Vite, Webpack, Babel. Environment management and API versioning.

#Exemplar/Case Studies: Study the evolution of Amazon/Uber Website from Monolithic to Microservices.

***Mapping of Course Outcomes**	**CO1**

Unit II	**React.js: Component-Based Architecture**	**8 Hours**

Introduction to React fundamentals, JSX, and component-based architecture. Working with components, props, and state for

dynamic UI rendering. Understanding lifecycle and React Hooks – useState, useEffect, useMemo. Managing global state using Context API and introduction to Redux Toolkit. Implementing routing with React Router and integrating REST APIs. Handling errors, optimizing performance, and applying memoization. Exploring Next.js for Server-Side Rendering (SSR) and Static Site Generation (SSG).		
#Exemplar/Case Studies: Study of Netflix's improved performance by using React concepts.		
***Mapping of Course Outcomes**	**CO2**	
Unit III	**Angular for Enterprise Web Development**	**8 Hours**
Angular JS : Overview, MVC architecture, directives, expression, controllers, filters, tables, modules, Form handling (template-driven & reactive), includes, views, scopes, services, dependency injection and Observables (RxJS). , custom directives, Internationalization.		
#Exemplar/Case Studies: Study of how Gmail uses Angular for dynamic UI rendering, routing, data binding, and real-time updates across emails, chats, and labels.		
***Mapping of Course Outcomes**	**CO2**	
Unit IV	**Backend Development using Node.js & Express.js**	**8 Hours**
Introduction to Node.js, Browser JS vs Node.js, Node.js REPL (Read–Eval–Print–Loop). Node.js event loop, modules, and asynchronous model. Introduction to Express, Getting started with Express Application, Request and Response Objects Routes and Middlewares, Templates, Template Engines, and Rendering Views. Input validation and error handling. File uploads and environment configuration.		
#Exemplar/Case Studies: Understand the BookMyShow to learn how large-scale ticketing systems handle authentication, events, databases and payments using Node.js and Express.js.		
***Mapping of Course Outcomes**	**CO3**	
Unit V	**Advanced Web Concepts**	**7 Hours**
WebSockets and Socket.io for real-time communication. GraphQL schema design, queries, and mutations. Microservices and API gateway concepts. Serverless architecture (AWS Lambda, Vercel Functions overview). Event-driven systems and message queues (Kafka/RabbitMQ overview). Integration of third-party APIs (maps, payment gateways).		
#Exemplar/Case Studies: Chat Application using WebSockets/Payment Gateway integration in an e-commerce site.		
***Mapping of Course Outcomes**	**CO4**	
Unit VI	**Web Security & Performance Engineering**	**7 Hours**
Authentication & Authorization (JWT-JSON Web Token, OAuth2, RBAC-Role-Based Access Control, bcrypt), OWASP Top 10 (updated 2025 version). CORS(Cross-Origin Resource Sharing), CSRF(Cross-Site Request Forgery), Secure Headers -CSP (Content Security Policy, HTTPS (Hypertext Transfer Protocol Secure), Basic DevSecOps awareness, Zero Trust overview, Performance metrics awareness (Core Web Vitals: LCP-Largest Contentful Paint, FID-First Input Delay, CLS-Cumulative Layout Shift)		
#Exemplar/Case Studies: LinkedIn authentication flow/Amazon performance optimization/Monitoring real-time metrics with Grafana.		



***Mapping of Course Outcomes**	**CO5**
Learning Resources	
Text Books :	
T1. Ethan Brown, Web Development with Node and Express, O'Reilly Media. **T2.** Alex Banks & Eve Porcello, Learning React: Functional Web Development with React and Redux, O'Reilly. **T3.** Shyam Seshadri, Angular Up & Running, O'Reilly.	
Reference Books :	
R1. Vasan Subramanian. Pro MERN Stack: Full Stack Web App Development with Mongo, Express, React, and Node.	
Additional Resources: (Books, e-Resources) • React Official Docs (https://react.dev/) • Angular Official Docs (angular.io) • Node.js Docs (nodejs.org)	
MOOC Courses links : • Coursera: Full-Stack Web Development with React (HKUST) • Udemy: Angular – The Complete Guide	

24-PCC-CS-3-05: Python Programming		
Teaching Scheme: Practical: 2 Hours/Week	**Credit:** 1	**Examination Scheme:** **Practical (PR)** : 25 Marks

Prerequisites Courses: 24-ESC1-07: Programming and Problem Solving Lab

Companion Course: NA

Course Objectives:

- Understand and apply Python basics including loops, conditions, functions, and strings.
- Apply lists, arrays, and dictionaries for efficient data storage, sorting, and searching operations.
- Utilize Python libraries to perform array operations integrated with file handling and GUI-based application development.
- Implement object-oriented programming with classes and objects.

Course Outcomes:

After completion of the course, learners should be able to

CONo	CO	BL
CO1	Apply Python basics including loops, conditions, functions, and string operations to solve real-world problems.	3
CO2	Implement lists, arrays, and dictionaries to store, sort, and search data effectively.	3
CO3	Use Python libraries to perform array operations, apply file handling techniques, and develop GUI-based applications.	3
CO4	Demonstrate object-oriented programming concepts using classes and objects	3

Guidelines for Instructor's Manual

The instructor's manual is to be developed as a reference and hands-on resource. It should include a prologue (about University/program/ institute/ department/foreword/ preface), curriculum of the course, conduction and Assessment guidelines, topics under consideration, concept, objectives, outcomes, set of typical applications/assignments/ guidelines, and references.

Guidelines for Student's Laboratory Journal

The laboratory assignments are to be submitted by students as a journal. The journal consists of a Certificate, table of contents, and handwritten write-up of each assignment (Title, Date of Completion, Objectives, Problem Statement, Software and Hardware requirements, Assessment grade/marks and assessor's sign, Theory- Concept in brief, algorithm, flowchart, test cases, Test Data Set(if applicable), mathematical model (if applicable), conclusion/analysis. Program codes with sample output of all performed assignments will be submitted as softcopy. As a conscious effort and little contribution towards Green IT and environmental awareness, attaching printed papers as part of write-ups and program listing to journals must be avoided. DVD containing student programs maintained by the Laboratory in charge is highly encouraged.

Guidelines for Laboratory /Term Work Assessment

Continuous assessment of laboratory work should be based on the overall performance of Laboratory assignments by a student.



Each Laboratory assignment assessment will assign grades/marks based on parameters, such as timely completion, performance, innovation, efficient codes, and punctuality.

Guidelines for Practical Examination

Problem statements must be decided jointly by the internal examiner and external examiner. During the practical assessment, maximum weightage should be given to satisfactory implementation of the problem statement. Relevant questions may be asked at the time of evaluation to test the student's understanding of the fundamentals and effective and efficient implementation. This will encourage, transparent evaluation and a fair approach, and hence will not create any uncertainty or doubt in the minds of the students. So, adhering to these principles will consummate our team efforts to the promising start of student's academics.

Guidelines for Laboratory Conduction

Virtual Laboratory: (If Any):NA

Suggested List of Laboratory Experiments/Assignments

Assignments

Sr No	Assignment Title	*Mapping of Course Outcomes
1.	Accept marks of N students in a subject and perform analysis using loops and conditions: 1. Calculate average marks 2. Count students above and below average 3. Identify topper(s) **Note: Use of Input Function,for loop, if_else, list**	CO1
2.	Write a Python program to perform string operation 1. Longest word detection 2. Character frequency analysis 3. Palindrome check **Note: Use of Function and string**	CO1
3.	Write a Python program to store roll numbers of student arrays who attended training programs in sorted order using any sort algorithm and search whether a particular student attended training program or not, using any search method.	CO2
4.	Develop a Python program to manage student information (roll no, name, marks) using **lists and dictionaries** with operations: 1.Add 2. Search 3. Update 4. Display	CO2
5.	Write a Python program to validate user inputs such as: 1. Email ID format 2. Mobile number format 3. Password strength	CO2
6.	Write a Python program to create and display matrices, and perform operations like addition, subtraction, multiplication, transpose. **Note:Introduce Numpy library and file handling**	CO3
7.	Design a Python program using **Tkinter GUI** to simulate a traffic signal. The GUI should: 1. Display a circle representing the traffic signal light. 2. Have three buttons: **Red**, **Green**, and **Orange**. 3. Change the color of the circle according to the button clicked.	CO3

8.	Design a Python program to simulate a simple banking system for a single customer using **classes and objects.** The program should allow the customer to deposit money, withdraw money, and display account details.	CO4

Learning Resources

Text Books :
T1: Let Us Python By Kanetkar Yashwant
T2: Core Python Programming By Rao R Nageswara, New Delhi Dreamtech Press; 2018

Reference Books :
R1. Python Programming: Using Problem Solving Approach,By Thereja Reema,New Delhi Oxford University; 2020
R2. Core Python Programming By Chun Wesley J,Phi Learning Private Limited ; 2007

Additional Resources: (Books, e-Resources)
- https://www.w3schools.com/python/

MOOC Courses links :
- https://onlinecourses.swayam2.ac.in/cec22_cs20/preview
- https://onlinecourses.swayam2.ac.in/cec26_cs01/preview

24-PEC-CS-3-04A: Game Development & Animation Lab

Teaching Scheme:	**Credit:** 2	**Examination Scheme:**
Practical:4 Hours/Week		Termwork **(TW)** : 25 Marks **Oral/Practical (OR)** : 25 Marks

Prerequisites Courses: (24-PEC-CS-3-01A) Computer Graphics, (24-PEC-CS-3-02A)Computer Graphics Laboratory

Companion Course: (24-PEC-CS-3-03A) Game Development & Animation

Course Objectives:

- To introduce students to the Editor environment, essential tools, interface windows, and asset management workflows required for building 2D games.
- To enable learners to apply GameObjects, components, physics, and colliders for designing interactive 2D and 3D scenes with functional gameplay mechanics.
- To develop the ability to create, test, and deploy small-scale 2D and 3D games using prefabs, animations, UI systems, and scene management techniques.

Course Outcomes:

After completion of the course, learners should be able to

CONo	CO	BL
CO1	**Demonstrate** understanding of Unity Editor, tools, windows, and asset workflow required for 2D game development.	3
CO2	**Apply** appropriate GameObjects, components, physics, and colliders to design functional 2D scenes and gameplay mechanics.	3
CO3	**Develop** interactive 2D and 3D games with movement, collision, animation, UI, and object management.	3
CO4	**Build, test, and deploy** small-scale functional 2D and 3D games using workflows, prefabs, and scene management techniques.	3

Guidelines for Instructor's Manual

The instructor's manual is to be developed as a reference and hands-on resource. It should include a prologue (about University/program/ institute/ department/foreword/ preface), curriculum of the course, conduction and Assessment guidelines, topics under consideration, concept, objectives, outcomes, set of typical applications/assignments/ guidelines, and references.

Guidelines for Student's Laboratory Journal

The laboratory assignments are to be submitted by students as a journal. The journal consists of a Certificate, table of contents, and handwritten write-up of each assignment (Title, Date of Completion, Objectives, Problem Statement, Software and Hardware requirements, Assessment grade/marks and assessor's sign, Theory- Concept in brief, algorithm, flowchart, test cases, Test Data Set(if applicable), mathematical model (if applicable), conclusion/analysis. Program codes with sample output of all performed assignments will be submitted as softcopy. As a conscious effort and little contribution towards Green IT and environmental



awareness, attaching printed papers as part of write-ups and program listing to journals must be avoided. DVD containing student programs maintained by the Laboratory in charge is highly encouraged. For reference one or two journals may be maintained with program prints in the Laboratory.

Guidelines for Laboratory /Term Work Assessment

Continuous assessment of laboratory work should be based on the overall performance of Laboratory assignments by a student. Each Laboratory assignment assessment will assign grades/marks based on parameters, such as timely completion, performance, innovation, efficient codes, and punctuality.

Guidelines for Practical Examination

Problem statements must be decided jointly by the internal examiner and external examiner. During the practical assessment, maximum weightage should be given to satisfactory implementation of the problem statement. Relevant questions may be asked at the time of evaluation to test the student's understanding of the fundamentals and effective and efficient implementation. This will encourage transparent evaluation and a fair approach, and hence will not create any uncertainty or doubt in the minds of the students. So, adhering to these principles will consummate our team efforts to the promising start of student's academics.

Guidelines for Laboratory Conduction

The laboratory is structured into three groups: A, B. Students must complete all assignments from Group A (mandatory) and any two assignments each from Group B and Group C as per instructor guidance.

Hardware Requirement:

64-bit Windows / Linux machine or its derivative, capable of running Unity; basic input devices (keyboard, mouse), optional graphics tablet, and AR/VR headset (if available).

Software Requirement:

Unity (Free/Personal Edition), C# scripting tools (built-in Unity editor), Blender (open-source), and optional Adobe Photoshop or any open-source image editing tool (GIMP/Krita).

Virtual Laboratory: (If Any):

Suggested List of Laboratory Experiments/Assignments

Group A: Assignments (Mandatory Assignment)

Sr No	Assignment Title	*Mapping of Course Outcomes
1.	**Getting Started With Unity & Interface** Set up the Unity environment by creating a new 2D project, arranging necessary windows (Scene, Game, Hierarchy, Project, Inspector), and customizing the workspace layout. Save the layout as a reusable preset.	CO1
2.	**Asset Organization & Importing Assets** Organize a new Unity project using a clean folder structure. Import image, audio, and script assets into the project and configure appropriate import settings for each type.	CO1
3.	**Working With Primitive GameObjects** Create a scene using built-in Unity primitive GameObjects such as Cube, Sphere,	CO1



	Capsule, and Plane. Modify their transforms and organize them into a structured Hierarchy.	
4.	**Scene Creation for 2D and 3D Gameplay** Create a 2D and 3D platformer scene by designing ground tiles, player objects, background layers, and lighting settings. Configure the orthographic camera to match pixel-perfect settings.	CO2
5.	**Implementing Physics 2D and 3D Using Rigidbody2D** Attach a Rigidbody2D and 3D to a ball object and apply gravity-based motion. Modify Body Type, Mass, Drag, and Gravity Scale values and analyze how each affects motion.	CO2
6.	**Physics Material 2D and 3D** Create Physics Material 2D and 3D assets with different friction and bounciness values. Apply them to objects and analyze collision results through practical experiments.	CO3
7.	**Sprite Editor – Slicing and Animation** Import a sprite sheet and use Unity's Sprite Editor to slice it into individual frames. Create an animation using sliced sprites and apply it to a Player GameObject.	CO3
8.	**Collider 2D Comparison** Add different Collider to 2D components (Box, Circle, Capsule, Polygon, Edge) to various objects and test their collision behavior in a 2D scene. Document use cases for each collider type.	CO3
9.	**Asset Store Integration** Download a free Unity Asset Store package, import it into the project, and integrate selected assets into a simple demo scene. Adjust import settings as needed.	CO3
10.	**Creating and Using Prefabs** Design a reusable Prefab for a Collectible object. Add components such as Sprite Renderer, Collider2D, and custom script. Instantiate multiple instances in the scene and demonstrate prefab overrides.	CO4
Group B: Assignments (Miniproject)		
Sr No	**Assignment Title**	***Mapping of Course Outcomes**
1.	Design a simple Unity 2D/3D Game of your choice showing the score of the Game.	CO1,CO2,CO3,CO4
Learning Resources (If applicable)		
Text Books		
T1. Dave Calabrese, Unity 2D Game Development, Packt Publishing, ISBN 978-1-84969-256-4 **T2.** Michelle Menard and Bryan Wagstaff , Game Development with Unity,2nd Edition, Cengage Learning PTR,,ISBN-13: 978-1-305-11054-0		



Reference Books :
R1. Jared Halpern, Developing 2D Games with Unity: Independent Game Programming with C#, 1st Edition, Apress, ISBN-10 1484237714
Additional Resources: (Books, e-Resources) ● Unity: https://docs.unity3d.com/6000.0/Documentation/Manual/UnityManual.html ● Unity Learn (https://learn.unity.com) ● https://darajat.ly/wp-content/uploads/2024/06/Game-Development-with-Unity-2nd-Edition-PDF-Books-1-Copy.pdf



24-PEC-CS-3-04B : Network Security Lab

Teaching Scheme:	**Credit:** 02	**Examination Scheme:**
Practical: 4 Hours/Week		Termwork **(TW)** : 25 Marks Oral **(OR)** : 25 Marks

Prerequisite Courses: 24-PCC-CS-3-03: Computer Network Lab

Companion Course: 24-PEC-CS-3-03B : Network Security.

Course Objectives:

- Implement classical symmetric and asymmetric encryption and decryption algorithms.
- Apply hashing, message authentication, and digital signature mechanisms on sample data.
- Perform key exchange and user authentication techniques using practical examples.
- Analyze network and web security protocols using tools like Wireshark and Cryptool.

Course Outcomes:

After completion of the course, learners should be able to

CONo	CO	BL
CO1	Implement classical and modern symmetric encryption algorithms for secure communication.	3
CO2	Apply asymmetric encryption algorithms and key exchange techniques.	3
CO3	Demonstrate data integrity and authentication in the data communication.	3
CO4	Examine network and web security protocols.	3

Guidelines for Instructor's Manual

The instructor's manual is to be developed as a reference and hands-on resource. It should include a prologue (about University/program/ institute/ department/foreword/ preface), curriculum of the course, conduction and Assessment guidelines, topics under consideration, concept, objectives, outcomes, set of typical applications/assignments/ guidelines, and references.

Guidelines for Student's Laboratory Journal

The laboratory assignments are to be submitted by students as a journal. The journal consists of a Certificate, table of contents, and handwritten write-up of each assignment (Title, Date of Completion, Objectives, Problem Statement, Software and Hardware requirements, Assessment grade/marks and assessor's sign, Theory- Concept in brief, algorithm, flowchart, test cases, Test Data Set(if applicable), mathematical model (if applicable), conclusion/analysis. Program codes with sample output of all performed assignments will be submitted as softcopy. As a conscious effort and little contribution towards Green IT and environmental awareness, attaching printed papers as part of write-ups and program listing to journals must be avoided. DVD containing student programs maintained by the Laboratory in charge is highly encouraged. For reference one or two journals may be maintained with program prints in the Laboratory.

Guidelines for Laboratory /Term Work Assessment



Continuous assessment of laboratory work should be based on the overall performance of Laboratory assignments by a student. Each Laboratory assignment assessment will assign grades/marks based on parameters, such as timely completion, performance, innovation, efficient codes, and punctuality.

Guidelines for Practical Examination

Problem statements must be decided jointly by the internal examiner and external examiner. During the practical assessment, maximum weightage should be given to satisfactory implementation of the problem statement. Relevant questions may be asked at the time of evaluation to test the student's understanding of the fundamentals and effective and efficient implementation. This will encourage transparent evaluation and a fair approach, and hence will not create any uncertainty or doubt in the minds of the students. So, adhering to these principles will consummate our team efforts to the promising start of student's academics.

Guidelines for Laboratory Conduction

The laboratory is structured into three groups: A, B, and C. Group A and Group B contain 6 assignments each; Group C contains 1 assignment, making a total of 13 assignments.
Students must complete 9 assignments.
Hardware Requirement: Any standard computer with core i3 Processor or equivalent, an open-source or Windows OS.
Software Requirement: CrypTool, Wireshark, OpenSSL, Python with Anaconda Navigator, Spyder, Jupyter, or PyCharm.

Virtual Laboratory:

- Vlabs, "Cryptography Lab" - http://cse29-iiith.vlabs.ac.in/

Suggested List of Laboratory Experiments/Assignments

Assignments (9 out of 11)

Sr No	Assignment Title	*Mapping of Course Outcomes
1.	Develop a program that securely encrypts and decrypts text by shifting characters using the Caesar Cipher technique.	**CO1**
2.	Design a system to perform message encryption and decryption using the Playfair Cipher for secure communication.	**CO1**
3.	Implement matrix-based Hill Cipher and transposition methods (Columnar or Rail Fence) to ensure data confidentiality.	**CO1**
4.	Implement AES-based symmetric key encryption and decryption.	**CO1**
5.	Implement RSA public-key encryption and decryption to achieve secure and authenticated data transmission.	**CO2**
6.	Develop a model to establish a shared secret key between two users using the Diffie–Hellman Key Exchange mechanism.	**CO2**
7.	Develop a program to generate secure message digests using SHA-256 and HMAC for integrity verification with Message Authentication Codes (MACs).	**CO3**

8.	Implement and demonstrate the process of digital signing and verification using ElGamal or Schnorr algorithms for message authenticity.	**CO3**
9.	Capture and analyze SSL/TLS handshake packets using Wireshark to understand how secure connections are established.	**CO4**
10.	Use CrypTool or Wireshark to study the working of HTTPS and SSH protocols and evaluate their cryptographic security mechanisms.	**CO4**
11.	Compare and analyze the performance, speed, and security of symmetric and asymmetric encryption algorithms through report-based evaluation.	**CO2, CO3**

Learning Resources (If applicable)

Text Books

T1. Andrew S. Tanenbaum, David J. Wetherall, Computer Networks, 6th Edition, Pearson Education, 2017, ISBN: 9789332585495.
T2. Behrouz A. Forouzan, Data Communications and Networking, 5th Edition, McGraw Hill Education, 2017, ISBN: 9780073376226.
T3. William Stallings, "Cryptography and Network Security, Principles and Practice", Pearson Education, India

Reference Books :

R1. Mark Stamp, "Information Security Principles and Practice", Willy India Edition
R2. Atul Kahate, "Cryptography and Network Security", Tata Mc Grawhill, India
R3. N Harini, T R Padmanabhan, C K Shyamala, "Cryptography and Security", Wiley-India

Additional Resources: (Books, e-Resources)

- Virtual Labs – https://vlab.co.in
- Cisco Networking Academy – https://www.netacad.com

MOOC Courses links :

- NPTEL Videos and PDF - "Cryptography and Network Security by Debdeep Mukhopadhyay, IIT Kharagpur"- https://nptel.ac.in/courses/106105031/

24-PCC-CS-3-04C: Data Science Lab

Teaching Scheme:	**Credit:** 2	**Examination Scheme:**
Practical: 4 Hours/Week		Termwork **(TW)** : 25 Marks Oral **(OR)** : 25 Marks

Prerequisites Courses: 24-PCC-CS-2-07: Database Management System Lab

Companion Course: 24-PEC-CS-3-03C: Data Science

Course Objectives:

- To develop competencies in data preprocessing, transformation, and quality management techniques.
- To provide understanding of multimodal data analysis techniques and large-scale data systems.
- To enable application of analytical and visualization techniques for effective insight communication and decision-making.

Course Outcomes:

After completion of the course, learners should be able to

CONo	**CO**	**BL**
CO1	**Apply** data preprocessing techniques to handle missing, duplicate, inconsistent, and noisy data.	3
CO2	**Apply** data preprocessing techniques to handle missing, duplicate, inconsistent, and noisy data using normalization and encoding methods.	3
CO3	**Analyze** datasets using exploratory data analysis techniques to compute statistical measures and interpret patterns through appropriate visualizations.	4
CO4	**Analyze** multimodal data (text, image, audio, and video) by extracting basic characteristics, performing modality-specific preprocessing, and interpreting data properties.	4
CO5	**Analyze** large-scale and business datasets using Big Data tools and BI platforms to derive insights through aggregation, filtering, and dashboard-based visualization.	4

Guidelines for Instructor's Manual

The instructor's manual is to be developed as a reference and hands-on resource. It should include a prologue (about University/program/ institute/ department/foreword/ preface), curriculum of the course, conduction and Assessment guidelines, topics under consideration, concept, objectives, outcomes, set of typical applications/assignments/ guidelines, and references.

Guidelines for Student's Laboratory Journal

The laboratory assignments are to be submitted by students as a journal. The journal consists of a Certificate, table of contents, and handwritten write-up of each assignment (Title, Date of Completion, Objectives, Problem Statement, Assessment grade/marks and assessor's sign, a brief explanation of the underlying DS/ML theory and concepts, pseudocode or algorithm, evaluation parameters used for model performance), answers to the questions provided at the end of the lab manual, and the conclusion or analysis).. Program codes with sample output of all performed assignments will be submitted as softcopy. As a



conscious effort and little contribution towards Green IT and environmental awareness, attaching printed papers as part of write-ups and program listing to journals must be avoided. Softcopy containing student programs maintained by the Laboratory in charge is highly encouraged. For reference one or two journals may be maintained with program prints in the Laboratory.

Guidelines for Laboratory /Term Work Assessment

Continuous assessment of laboratory work should be based on the overall performance of Laboratory assignments by a student. Each Laboratory assignment assessment will assign grades/marks based on parameters, such as timely completion, performance, innovation, efficient codes, and punctuality.

Guidelines for Practical Examination

Problem statements must be decided jointly by the internal examiner and external examiner. During the practical assessment, maximum weightage should be given to satisfactory implementation of the problem statement. Relevant questions may be asked at the time of evaluation to test the student's understanding of the fundamentals and effective and efficient implementation. This will encourage transparent evaluation and a fair approach, and hence will not create any uncertainty or doubt in the minds of the students. So, adhering to these principles will consummate our team efforts to the promising start of student's academics.

Guidelines for Laboratory Conduction

The laboratory is structured into three groups: A, B, and C. Group A contains 6 assignments, Group B contains1 assignmentmaking a total of 7 assignments. It is mandatory for students to complete all assignments.

Hardware Requirement: Desktop or laptop with minimum 8 GB RAM, multi-core processor, and stable internet connectivity.

Software Requirement: Open-source operating system, VS Code and required Python libraries.

Suggested List of Laboratory Experiments/Assignments

Group A: Data Science

Sr No	Assignment Title	*Mapping of Course Outcomes
1.	**Extract** data from APIs, explore and organize it into a mini data warehouse, **analyze** data quality, **perform** data slicing, identify noisy (IQR method) and inconsistent data, and prepare the dataset for preprocessing.	CO1, CO2
2	**Apply** data preprocessing techniques to handle missing values through mean and median imputation, remove duplicate records, perform normalization (Min–Max, Z-score, Decimal scaling), and encode categorical variables using Label and One-Hot encoding.	CO2
3	**Perform** exploratory data **analysis** by computing measures of central tendency and dispersion, and **visualizing** data through univariate (histogram, boxplot), bivariate (scatter plot), and multivariate (heatmap) plots using Matplotlib and Seaborn.	CO3
4	**Analyze** text data analysis by applying basic preprocessing techniques, computing word frequency statistics, and visualizing text characteristics (bar charts or word clouds).	CO4
5	**Analyze** image data analysis applying basic image operations, and visualizing image samples and pixel intensity distributions.	CO4
6	**Analyze** audio data by examining signal properties, visualizing waveforms, and summarizing basic audio characteristics.	CO4

7	**Analyze** video data by extracting basic video properties, visualizing selected frames, and Interpret video data characteristics.	CO4
Group B Big Data Analytics		
Sr No	**Assignment Title**	***Mapping of Course Outcomes**
1	Using Apache Spark, **perform big data analysis** by loading a text-based log dataset, examining log structure, applying basic preprocessing, computing log statistics, filtering records by log levels, and performing aggregation using Spark RDD.	CO5
2	Using Power BI, **perform data cleaning and visualization** by importing a dataset, identifying data types, applying transformations using Power Query Editor, creating summary reports, and designing basic and advanced visuals to derive insights.	CO5
Group C Mini Project		
Collect data from suitable sources, perform data cleaning and preprocessing, analyze the dataset using descriptive and exploratory techniques, and present insights through data storytelling using Power BI or Python-based visualizations.		
Learning Resources (If applicable)		
Text Books		
T1. Jake VanderPlas, *Python Data Science Handbook*, O'Reilly Media, 2016, ISBN 978-1491912058. **T2.** Wes McKinney, *Python for Data Analysis*, O'Reilly Media, 2017, ISBN 978-1491957660.		
Additional Resources: (Books, e-Resources) ● data.gov.in (Open Government Data Platform – India): https://data.gov.in/apis ● Zomato Restaurant Dataset: https://www.kaggle.com/code/nitinchoudhary012/zomato-restaurant-analysis-detailed-eda ● IPL Dataset (Data Analysis and Visualization): https://machinelearningknowledge.ai/ipl-data-analysis-and-visualization-project-using-python/ ● India Spam SMS Dataset (Text Data): https://www.kaggle.com/datasets/junioralive/india-spam-sms-classification ● Indian Traffic Signs Dataset (Image Data): https://www.kaggle.com/datasets/aryashah2k/indian-traffic-signs ● Speech Commands Dataset (Audio Data): https://www.kaggle.com/datasets/mohamedhanyyy/speech-command-dataset ● Open Video Dataset (Short Clips): https://www.kaggle.com/datasets/kalpeshkumar259/open-video-dataset ● System Log Dataset (Big Data – Spark): https://www.kaggle.com/datasets/shawon10/web-log-dataset ● Insurance Dataset (Power BI Analysis): https://www.kaggle.com/datasets/mirichoi0218/insurance		
MOOC Courses links : ● https://onlinecourses.nptel.ac.in/noc25_cs20/preview ● https://onlinecourses.nptel.ac.in/noc25_cs60/preview		



24-PEC-CS-3-04D:High Performance Computing Lab

Teaching Scheme: Practical: 4 Hours/Week	**Credit:** 2	**Examination Scheme:** **Termwork (TW)** : 25 Marks **Oral (OR)** : 25 Marks

Prerequisites Courses: 24-PCC-CS-2-04:Data Structures Lab

Companion Course: 24-PEC-CS-3-03D: High Performance Computing

Course Objectives:

- To understand and implement searching and sorting algorithms.
- To learn the fundamentals of GPU Computing in the CUDA environment.
- To Understand Scope of Parallel Computing and its search algorithms.

Course Outcomes:

After completion of the course, learners should be able to

CONo	CO	BL
CO1	Analyze and measure performance of sequential and parallel algorithms.	4
CO2	Design and Implement solutions for multicore/Distributed/parallel environments.	3
CO3	Analyze the performance of HPC applications	4

Guidelines for Instructor's Manual

The instructor‘s manual is to be developed as a reference and hands-on resource. It should include a prologue (about University/program/ institute/ department/foreword/ preface), curriculum of the course, conduction and Assessment guidelines, topics under consideration, concept, objectives, outcomes, set of typical applications/assignments/ guidelines, and references.

Guidelines for Student's Laboratory Journal

The laboratory assignments are to be submitted by students as a journal. The journal consists of a Certificate, table of contents, and handwritten write-up of each assignment (Title, Date of Completion, Objectives, Problem Statement, Software and Hardware requirements, Assessment grade/marks and assessor's sign, Theory- Concept in brief, algorithm, flowchart, test cases, Test Data Set(if applicable), mathematical model (if applicable), conclusion/analysis. Program codes with sample output of all performed assignments will be submitted as softcopy. As a conscious effort and little contribution towards Green IT and environmental awareness, attaching printed papers as part of write-ups and program listing to journals must be avoided. For reference one or two journals may be maintained with program prints in the Laboratory.

Guidelines for Laboratory /Term Work Assessment

Continuous assessment of laboratory work should be based on the overall performance of Laboratory assignments by a student. Each Laboratory assignment assessment will assign grades/marks based on parameters, such as timely completion, performance, innovation, efficient codes, and punctuality.



Guidelines for Practical Examination

Problem statements must be decided jointly by the internal examiner and external examiner. During the practical assessment, maximum weightage should be given to satisfactory implementation of the problem statement. Relevant questions may be asked at the time of evaluation to test the student's understanding of the fundamentals and effective and efficient implementation. This will encourage transparent evaluation and a fair approach, and hence will not create any uncertainty or doubt in the minds of the students. So, adhering to these principles will consummate our team efforts to the promising start of student's academics.

Guidelines for Laboratory Conduction

The instructor is expected to frame the assignments by understanding the prerequisites, technological aspects, utility and recent trends related to the topic. The assignment framing policy needs to address the average students and inclusive of an element to attract and promote the intelligent students. Use of open source software is encouraged. Based on the concepts learned. Mini projects should be implemented by the students in a group of 2-3 students.

Suggested List of Laboratory Experiments/Assignments

Group A: Assignments

Sr No	Assignment Title	*Mapping of Course Outcomes
1.	Design and implement Parallel Breadth First Search and Depth First Search based on existing algorithms using OpenMP. Use a Tree or an undirected graph for BFS and DFS.	CO1
2.	Write a program to implement Parallel Breadth First Search (BFS) and Depth First Search (DFS) based on existing algorithms using MPI. Use a Tree or an Undirected Graph for BFS and DFS.	CO2
3.	Design and implement Parallel Bubble Sort and Merge sort using OpenMP. Use existing algorithms and measure the performance of sequential and parallel algorithms.	CO1
4.	Write a program to implement Parallel Bubble Sort and Merge sort using MPI. Use existing algorithms and measure the performance of sequential and parallel algorithms.	CO2
5.	Implement Min, Max, Sum and Average operations using Parallel Reduction.	CO2
6.	Write a CUDA Program for Addition of two large vectors using CUDA C.	CO3
7.	Implement a CUDA Program for Matrix Multiplication using CUDA C.	CO3

Group B: Mini Project

Sr No	Assignment Title	*Mapping of Course Outcomes
1.	Design and implement a Mini Project based on Openmp/MPI/CUDA C	CO1, CO2, CO3



24-PEC-CS-3-04E: Advanced Web Technology Lab

Teaching Scheme: Practical: 4 Hours/Week	**Credit:** 2	**Examination Scheme:** **Termwork (TW)** : 25 Marks **Oral (OR)** : 25 Marks

Prerequisites Courses: 24-PEC-CS-3-01E: Web Technology Lab

Companion Course: 24-PEC-CS-3-03E: Advanced Web Technology

Course Objectives:

- To learn the advanced web development environment
- To use client side and server side web technologies
- To design and develop web applications using advanced front end technologies and backend databases

Course Outcomes:

After completion of the course, learners should be able to

CONo	CO	BL
CO1	**Implement** project setup and dependency management using npm or yarn.	3
CO2	**Develop** interactive and component-based front-end modules.	3
CO3	**Construct** server-side applications and real-time communication services using Nodejs, Expressjs, WebSockets.	3
CO4	**Demonstrate** secure authentication mechanisms and implement security best practices.	3

Guidelines for Instructor's Manual

The instructor‘s manual is to be developed as a reference and hands-on resource. It should include a prologue (about University/program/ institute/ department/foreword/ preface), curriculum of the course, conduction and Assessment guidelines, topics under consideration, concept, objectives, outcomes, set of typical applications/assignments/ guidelines, and references.

Guidelines for Student's Laboratory Journal

The laboratory assignments are to be submitted by students as a journal. The journal consists of a Certificate, table of contents, and handwritten write-up of each assignment (Title, Date of Completion, Objectives, Problem Statement, Software and Hardware requirements, Assessment grade/marks and assessor's sign, Theory - Concept in brief, conclusion/analysis. Program codes with sample output of all performed assignments will be submitted as softcopy. As a conscious effort and little contribution towards Green IT and environmental awareness, attaching printed papers as part of write-ups and program listing to journals must be avoided. For reference one or two journals may be maintained with program prints in the Laboratory.

Guidelines for Laboratory /Term Work Assessment

Continuous assessment of laboratory work should be based on the overall performance of Laboratory assignments by a student. Each Laboratory assignment assessment will assign grades/marks based on parameters, such as timely completion, performance, innovation, efficient codes, and punctuality.



Guidelines for Practical Examination

During the assessment, maximum weightage should be given to the student's clarity of concepts, depth of understanding, and ability to apply fundamentals effectively. Examiners may ask relevant questions to evaluate conceptual knowledge, analytical thinking, and problem-solving skills.This approach will ensure transparency, fairness, and consistency in evaluation, eliminating uncertainty or doubt among students. By following these guidelines, we can uphold the quality of the academic process and support students in achieving a strong academic foundation.

Guidelines for Laboratory Conduction

The instructor is expected to frame the assignments by understanding the prerequisites, technological aspects, utility and recent trends related to the topic. The assignment framing policy needs to address the average students and inclusive of an element to attract and promote the intelligent students. Use of open source software is encouraged. Based on the concepts learned, mini projects should be implemented by a group of 2-3 students.

Suggested List of Laboratory Experiments/Assignments

Group A: Assignments

Sr No	Assignment Title	*Mapping of Course Outcomes
1.	**Setup a Project Environment and Managing Dependencies with npm/yarn.** Hint: Focuses on the development ecosystem by initializing a Node.js project, using npm and yarn to install, update, and manage project dependencies, and creating scripts for running the application, demonstrating environment management.	CO1
2.	**Creating a Multi-Step Survey Form using Components and State Management.** Hint: Develops a component-based Survey Form to practice creating multiple components, managing form data and component state using the useState Hook, and implementing conditional rendering or routing to handle multiple form steps.	CO2
3.	**Implementing a Simulated Live Score Dashboard using Services and RxJS.** Hint: Creates an Angular module to demonstrate Dependency Injection and Observables (RxJS) by setting up a Service that simulates streaming data (like a score or stock price) and displaying the changing data in a component.	CO2
4.	**Building a Simple CRUD API for Resources (e.g., Tasks or Books).** Hint: Builds a Node.js/Express.js server to define a RESTful API that implements all four CRUD operations (POST/Create, GET/Read, PUT/Update, DELETE/Delete) for a specific resource, utilizing Request and Response Objects and basic Routing.	CO3
5.	**Develop Real-time Chat Application using WebSockets/Socket.io.** Hint: Implements a multi-user chat application to demonstrate real-time communication using WebSockets and Socket.io, covering event handling and broadcasting, as applied in a case study.	CO3
6.	**Implementing Authentication using Auth0 and Applying Secure Headers.** Hint: Integrates a third-party service like Auth0 to secure an Express API using JWT for authentication, demonstrating the use of modern security flows (like OAuth2) and applying Secure Headers.	CO4

Group B: Mini Project		
Sr No	**Assignment Title**	***Mapping of Course Outcomes**
1.	Using the advanced web technologies covered in Group A, develop a full-stack web application with following details: 1. Choose any application relevant to Group A. 2. Develop application considering Frontend and Backend 3. Test and validate applications using Manual/Automation testing. 4. Deploy on any free web hosting platform.	CO1, CO2, CO3, CO4

24-VSEC-CS-3-01 : Application Development Programming II

Teaching Scheme: Practical: 4 Hours/Week	**Credit:** 2	**Examination Scheme:** Termwork **(TW)** : 25 Marks **Practical (PR)** : 25 Marks

Prerequisites Courses: NA

Companion Course: NA

Course Objectives:

- **Understand** the fundamentals of Android/Flutter development and environment setup.
- **Apply** UI components and event handling to develop interactive applications.
- **Implement** sensors, authentication, and advanced mobile app features.
- **Design and develop** small mobile applications using Android/Flutter platforms.

Course Outcomes:
After completion of the course, learners should be able to

CONo	**CO**	**BL**
CO1	**Use** Android/Flutter development tools to create basic apps.	2
CO2	**Build** interactive mobile applications using UI elements and event handling.	3
CO3	**Integrate** sensors, animations, and advanced components into mobile apps.	4
CO4	**Develop** complete mobile applications with authentication and user-friendly interface.	6

Guidelines for Instructor's Manual

The instructor‘s manual is to be developed as a reference and hands-on resource. It should include a prologue (about the University/program/ institute/ department/foreword/ preface), curriculum of the course, conduction and Assessment guidelines, topics under consideration, **concept, objectives, outcomes, set of typical applications/assignments/ guidelines, and references.**

Guidelines for Student's Laboratory Journal

The laboratory assignments are to be submitted by students as a journal. The journal consists of a Certificate, table of contents, and handwritten write-up of each assignment (Title, Date of Completion, Objectives, Problem Statement, Software and Hardware requirements, Assessment grade/marks and assessor's sign, Theory- Concept in brief, algorithm, flowchart, test cases, Test Data Set(if applicable), mathematical model (if applicable), conclusion/analysis. Program codes with sample output of all performed assignments will be submitted as softcopy. As a conscious effort and little contribution towards Green IT and environmental awareness, attaching printed papers as part of write-ups and program listing to journals must be avoided.

Guidelines for Laboratory /Term Work Assessment

Continuous assessment of laboratory work should be based on the overall performance of Laboratory assignments by a student. Each Laboratory assignment assessment will assign grades/marks based on parameters, such as timely completion, performance,

innovation, efficient codes, and punctuality.

Guidelines for Practical Examination

Problem statements must be decided jointly by the internal examiner and external examiner. During the practical assessment, maximum weightage should be given to satisfactory implementation of the problem statement. Relevant questions may be asked at the time of evaluation to test the student's understanding of the fundamentals and effective and efficient implementation. This will encourage transparent evaluation and a fair approach, and hence will not create any uncertainty or doubt in the minds of the students. So, adhering to these principles will consummate our team efforts to the promising start of student's academics.

Guidelines for Laboratory Conduction

The laboratory is structured into three groups: Group A consists of 6 mandatory assignments, Group B consists of 6 assignments out of which students must perform any 2, and Group C consists of a Mini Project; thus, out of the total 13 assignments, each student must complete 9 by selecting 6 from Group A, 2 from Group B, and 1 from Group C.

Hardware Requirement: Any open-source OS, Dual Core+, 8 GB RAM, 250 GB+ storage, Internet.
Software Requirement: Android Studio, Flutter SDK, JDK, Gradle, Emulator.

Virtual Laboratory: (If Any):

- FlutLab.io - Flutter IDE online

Suggested List of Laboratory Experiments/Assignments

Group A: Assignments (Mandatory Assignment)

Sr No	Assignment Title	*Mapping of Course Outcomes
1.	**Install Android Studio** and **create a virtual device (AVD)** for running Android applications.	CO1
2.	**Develop a simple Android app** that displays "Hello World" using **TextView** and **EditText**.	CO2
3.	**Create an application** demonstrating the use of **Button**, **ImageButton**, **ToggleButton**, and **RadioButton** for user interaction.	CO2
4.	Develop an Android application that accesses and displays real-time data from built-in device sensors like the accelerometer, gyroscope, and proximity sensor.	CO3
5.	Develop a CALCULATOR mobile app. Include as many arithmetic operations as possible: Addition,Subtraction,Multiplication,Division,Exponant,Square.	CO4
6.	Develop a Quiz App that displays 10 multiple-choice questions, allows users to select answers, and shows the final score at the end. Make necessary assumptions.	CO4

Group B: Assignments (Out of List perform any 2) (Optional)

Sr No	Assignment Title	*Mapping of Course

		Outcomes
1.	Setting up and configuring the Flutter development environment.	CO1
2.	**Create a Flutter application** using GUI components, custom fonts, and color themes.	CO2
3.	Develop **an animation-based application** to enhance user interaction.	CO4
4.	Deploy **a map-based application** using location and map APIs.	CO4
5.	Implementing a Flutter program to display a progress bar.	CO4
6.	Develop a to-do list application in Flutter to manage daily tasks with add, edit, and delete features.	CO4
Group C: Assignment		
Sr No	**Assignment Title**	***Mapping of Course Outcomes**
1	Develop & Deploy an Android/Flutter Application. Mini Project	CO4
Learning Resources (If applicable)		

Text Books

T1. Android Application Development Black Book by Pradeep Kothari, DreamTech

Reference Books :

R1. Android application development all-in-one for dummies.
R2. Flutter: A complete book for mobile app development – A practical approach with concepts and programming. TechPress.

Additional Resources: (Books, e-Resources)

- **https://developer.android.com/**
- **https://flutter.dev/**

MOOC Courses links :

- **https://developer.android.com/courses/android-basics-compose/course**
- **https://onlinecourses.swayam2.ac.in/nou25_ge63/preview**

24-ELC-3-01: Research Methodology

Teaching Scheme: Theory: 4 Hours/Week	**Credit:** 4	**Examination Scheme:** **CIE** : 20 Marks **MSE :** 20 Marks **SEE** : 60 Marks

Prerequisites Courses: 24-bsc-1-01: Linear Algebra And Differential Calculus, 24-BCS-1-02 Engineering Chemistry, 24-BCS-1-01: Engineering Physics, 24-AEC-1-01: Professional Communication Skills.

Companion Course:

Course Objectives:

- Introduce students to the basic concepts, purpose, and process of research in engineering.
- Develop understanding of literature review and problem formulation techniques.
- Explain the fundamentals of research design and sampling methods.
- Equip students with basic data collection, analysis, and interpretation skills.
- Train students to prepare professional research reports and presentations.
- Create awareness about research ethics, plagiarism, and intellectual property rights.

Course Outcomes:

After completion of the course, learners should be able to

CO No	CO	BL
CO1	Explain the basic concepts, objectives, and types of research in engineering.	2
CO2	Identify research problems through effective literature review.	3
CO3	Select suitable research designs and sampling methods for studies.	3
CO4	Collect and analyze data using basic statistical tools and techniques.	3
CO5	Prepare concise research reports and present results effectively.	3
CO6	Apply research ethics and understand fundamentals of intellectual property rights.	4

Course Contents

Unit I	**Introduction to Research**	**06 Hours**

What is research?-Meaning and purpose of research, Research process and basic steps, Types of research: fundamental, applied, experimental; Research questions and objectives, Concept of variables and hypotheses

#Exemplar/Case Studies

Comparison of Fuel Efficiency in Petrol vs LPG Engines

***Mapping of Course Outcomes**	**CO1**



Unit II	**Literature Review & Problem Formulation**	**06 Hours**
Importance and purpose of literature review, Major sources: books, journals, internet databases (Google Scholar, IEEE Xplore, etc.), How to summarize and identify research gaps, Framing and defining the research/problem statement. Introduction to AI tools.		
#Exemplar/Case Studies Review of Heat Exchanger Materials for Improved Thermal Efficiency		
***Mapping of Course Outcomes**	**CO2**	
Unit III	**Research Design & Sampling Basics**	**06 Hours**
Research design types: descriptive, experimental, Concept of variables (independent, dependent, control), Sampling techniques: random and non-random, Introduction to pilot study and data reliability		
#Exemplar/Case Studies Sampling and Testing of Lubricating Oils for Wear Reduction		
***Mapping of Course Outcomes**	**CO3**	
Unit IV	**Data Collection & Analysis**	**06 Hours**
Methods of data collection: survey, observation, experiment; Data organization and basic statistical tools: mean, median, standard deviation, Introduction to hypothesis testing and correlation, Data presentation: tables, charts, graphs (Excel / basic tools)		
#Exemplar/Case Studies Data Analysis of Stress-Strain Test Results of Mild Steel Samples		
***Mapping of Course Outcomes**	**CO4**	
Unit V	**Report Writing & Presentation**	**06 Hours**
Structure of research/project report, Sections: abstract, introduction, methodology, results, conclusion; Citation and referencing (basic IEEE / APA format), Introduction to Plagiarism tools, Plagiarism awareness and paraphrasing, Effective presentation and project proposal skills		
#Exemplar/Case Studies: Project Report on Design and Fabrication of Pedal-Powered Water Pump		
***Mapping of Course Outcomes**	**CO5**	
Unit VI	**Research Ethics & Intellectual Property**	**06 Hours**
Research ethics: honesty, plagiarism, data integrity; Introduction to patents, copyrights, and innovation; Overview of current and emerging research trends in engineering		
#Exemplar/Case Studies: Intellectual Property Case Study on Innovative Gear Mechanism Design		
***Mapping of Course Outcomes**	**CO6**	



Learning Resources
Text Books
T1. C.R. Kothari & Gaurav Garg, Research Methodology: Methods and Techniques, New Age International Publishers. **T2.** Ranjit Kumar, Research Methodology: A Step-by-Step Guide for Beginners, Sage Publications. **T3.** Deepak Chawla & Neena Sondhi, Research Methodology: Concepts and Cases, Vikas Publishing. **T4.** Dr. S. Rajasekar, Research Methodology for Engineers, CRC Press / (Available as Open Source Reference).
Reference Books :
R1. Wayne C. Booth, Gregory G. Colomb, Joseph M. Williams, The Craft of Research, University of Chicago Press. **R2**. K.N. Krishnaswamy, Appa Iyer Sivakumar, M. Mathirajan, Management Research Methodology: Integration of Principles, Methods and Techniques, Pearson Education. **R3.** Donald R. Cooper & Pamela S. Schindler, Business Research Methods, McGraw Hill. **R4.** John W. Creswell, Research Design: Qualitative, Quantitative, and Mixed Methods Approaches, Sage Publications. **R5.** T. Ramachandran, Research Methodology for Engineers, Notion Press. **R6.** IEEE / Elsevier / Springer Digital Library Resources, for literature review and citation practice.
Additional Resources: (Books, e-Resources)
MOOC Courses links : https://onlinecourses.nptel.ac.in/noc26_ge43/preview **(Research Methodology)**

24-ELC-CS-3-02: Project-stage-I

Teaching Scheme:	**Credit:** 02	**Examination Scheme:**
Practical: 4 Hours/Week		Termwork **(TW)**: 25 Marks Oral/Practical **(PR)**: 50 Marks

Prerequisites Courses: 24-ELC-CS-2-01: Community Engagement Project, 24-AEC-2-01: Business Communication Skill, 24-VEC-2-03: Digital and Technological Solutions

Companion Course: 24-ELC-3-01: Research Methodology; 24-PCC-CS-3-04: Software Engineering

Course Objectives:

- To develop research-oriented, analytical, and systematic problem-solving abilities in students.
- To enable students to identify and select real-world engineering, technological, or societal problems for project work.
- To train students in conducting structured literature surveys and critically evaluating existing research and solutions.
- To guide students in formulating precise problem statements, project objectives, and preliminary system design for successful project execution.

Course Outcomes:

After completion of the course, learners should be able to

CONo	CO	BL
CO1	Identify and justify a real-world engineering or societal problem using foundational and discipline-specific knowledge.	02
CO2	Analyze existing solutions and research gaps through a structured review of journals, patents, standards, and technical reports.	03
CO3	Formulate a precise problem statement with defined objectives, constraints, and sustainability/ethical considerations.	04
CO4	Develop a preliminary system design or methodology that covers the project, including tool selection and planning, as well as pilot execution.	05

Preamble

Project Stage 1 aims to prepare students for independent, research-oriented project work by introducing them to the fundamentals of problem identification, literature review, research gap analysis, problem statement formulation, and objective setting. This stage builds the foundation for innovative, sustainable, and socially relevant project execution in subsequent semesters. It fosters critical thinking, creativity, teamwork, and alignment with global standards of project-based learning.

Group Structure

Working in supervisor- or mentor-monitored groups, students plan, manage, and complete a task/project/activity that addresses the stated problem.

1. A project group may consist of **2–4 students**.
2. Interdisciplinary group formation is encouraged.
3. Faculty supervisors and mentors will guide each group, providing periodic feedback.

4. The same supervisor may continue guiding students through the final year project for the next two years.

Modules of Project Stage-1

1. **Problem Identification –** Understanding real-world challenges in engineering, technology, and society.
2. **Literature Survey –** Reviewing past and current research, benchmarking existing solutions, and recognizing limitations.
3. **Research Gap Identification –** Analyzing gaps and unmet needs in the chosen domain.
4. **Problem Statement Framing –** Formulating precise, clear, and researchable project statements.
5. **Objective Definition –** Preparing specific, measurable, achievable, relevant, and time-bound (SMART) objectives.
6. **Methodology & System Design–** Outlining possible approaches, tools, and techniques to address the problem, along with pilot work completion.
7. **Documentation & Presentation Skills –** Preparing structured reports and professional presentations.

Assessment

Continuous Internal Assessment (CIA) will be conducted through project reviews, presentations, and mentor evaluations. Weightage will be given to **problem originality, research depth, clarity of objectives, teamwork, and communication skills**. Ethical and societal considerations in problem selection are mandatory.
Students are expected to maintain a culture of collaboration, self-motivation, peer learning, and responsibility. The institution will support students through guidance programs and resources. Both supervisors/mentors and students must actively participate in the assessment process. Assessment Rubrics: View

Project Stage-1 Report Guidelines

- The report should follow a structured format:
    1) Title Page
    2) Abstract
    3) Introduction (Background, Problem Context)
    4) Literature Review (with references)
    5) Identified Research Gap
    6) Problem Statement
    7) Objectives
    8) Proposed Methodology/System Design
    9) Partial Execution (50% work)
    10) References (APA/IEEE format)
- Recommend length: 3,000 to a maximum of 6,000 words.
- Reports should demonstrate originality, critical thinking, and academic integrity

Guidelines for Students

- Students must select a problem that is feasible to implement within available resources and a timeframe.
- Students should apply **Science, Technology, Engineering and Mathematics (STEM) principles and adopt an interdisciplinary approach wherever possible.**
- Students must **conduct a structured literature survey** to identify research gaps and **frame a clear problem statement.**
- Students should **align their project with at least one UN Sustainable Development Goal (SDG) and the action of the UN Pact for the Future.**
- Students must ensure **originality, ethical research practices**, and focus on **sustainability** with measurable **societal or environmental impact.**

- Students should maintain a logbook, prepare structured reports, and participate actively in all review assessments.
- Refer to the Student's detailed Project Stage-1 guidelines for more details, including the Checklist.

Schedule of Project Stage 1

Project Stage-1 Plan (4 Months Overview)

- **Month 1:** Foundation & Problem Finalization.
- **Month 2:** Literature Survey & Gap Analysis.
- **Month 3:** Methodology & Design
- **Month 4:** Report Writing

Learning Resources

Reference Books:

R1. Literature Review Simplified: The Checklist Edition: A Checklist Guide to Literature Review, Rafiq Muhammad, ISBN: 978-9198900798, 2023
R2. Research Methodology - Methods and Techniques, C.R. Kothari, Gaurav Garg, New Age International Publishers, ISBN: 978-9386649225, 2020
R3. How to Be Good at Science, Technology & Engineering, Dorling Kindersley (DK), ISBN: 978-0241227862, 2018

Online Resources:

1. Research Projects Broad Domains and Priority Areas described by the Indian Council of Social Science Research, Ministry of Education, Government of India: https://icssr.org/research-projectsmajor-and-minor
2. IoT Open Edu co-funded by the European Union: https://iot-open.eu/
3. United Nations Sustainable Development Goals (SDGs): https://sdgs.un.org/goals
4. UN SDGs Student Resources: https://www.un.org/sustainabledevelopment/student-resources/

Additional Resources: (Books, e-Resources)

- SDGs: A book of action for educators and young learners | UNDP https://sdgknowledgehub.undp.org.in/wp-content/uploads/2023/10/SDG-Hand-book-English.pdf
- SDG Guidebook for youth in action | UNDP + SDG MOVE Thailand (https://www.undp.org/sites/g/files/zskgke326/files/2022-09/EN-%20SDG%20Guidebook%20for%20youth%20EN%206%203%2022%20without%20mark.pdf)
- Resources for Researchers (https://www.un.org/en/academic-impact/page/resources-researchers)
- The European Competence Framework for Researchers (https://opusproject.eu/openscience-news/the-european-competence-framework-for-researchers/)

MOOC Courses links :

- NASA Open Science 101 Training | https://stemgateway.nasa.gov/s/course-offering/a0BSJ0000029hSn/open-science-101
- Open Science MOOC | https://opensciencemooc.eu/
- Academic Writing | Swayam https://onlinecourses.swayam2.ac.in/cec21_ge32/preview

	SNJB's Late Sau KBJ College of Engineering, Chandwad (An Autonomous Institute affiliated to Savitribai Phule Pune University, Pune)		
	Examination: Mid Semester Examination **(MSE)**		**Programme:** UG
	Class:	**Branch:**	
	Academic Year:	**Semester:**	**Pattern:**
Course Name:		**Course Code:**	
Time: 1Hour		**Maximum Marks:** 20	

Instructions to the candidates:

1. *Solve Q.1 OR Q.2; Q.3 OR Q.4; Q.5 OR Q.6*
2. *Bold-faced figures to the right indicate full marks.*
3. *Assume the suitable data if necessary*
4. *Any other instruction required for particular course may be added by subject/course chairman*

QN	Question	Mark
1a)	Q 1 can be bifurcated to maximum two sub questions (Unit I)	7
1b)		
	OR	
2	Q 2 can be bifurcated to maximum two sub questions (Unit I)	7
3	Q 3 can be bifurcated to maximum two sub questions (Unit II)	7
	OR	
4	Q 4 can be bifurcated to maximum two sub questions (Unit II)	7
5	Q 5 can be bifurcated to maximum two sub questions (Unit III)	6
	OR	
6	Q 6 can be bifurcated to maximum two sub questions (Unit III)	6

Shree Neminath Jain Brahmacharyashram's
Late Sau. Kantabai Bhavarlalji Jain College of Engineering
Neminagar, Chandwad -423 101 Dist. Nashik.
An Autonomous Institute, Affiliated to Savitribai Phule Pune University, Pune

Semester End Examination (Regular) : Month Year

Programme: UG (Program Name) Class:

Course and Code: Semester: (Odd/Even)

Academic Year: Pattern:

Time: 2Hr 30 Min Examination: SEE (Month Year) Max. Marks: 60

Instructions to the candidates:

1. *Solve Q.1 OR Q.2, Q.3 OR Q.4, Q.5 OR Q.6, Q.7 OR Q.8*
2. *Bold-faced figures to the right indicate full marks.*
3. *Assume the suitable data if necessary, but Justify it.*
4. *Draw the neat labelled diagrams, wherever necessary.*

QN	Question	Marks
1 a)	Unit I	6
1 b)	Unit II	7
1 c)	Unit III	7
	OR	
2 a)	Unit I	6
2 b)	Unit II	7
2 c)	Unit III	7
3 a)	Unit IV	7
3 b)	Unit IV	7
	OR	
4 a)	Unit IV	7
4 b)	Unit IV	7
5 a)	Unit V	7
5 b)	Unit V	6
	OR	
6 a)	Unit V	7
6 b)	Unit V	6
7 a)	Unit VI	7
7 b)	Unit VI	6
	OR	
8 a)	Unit VI	7
8 b)	Unit VI	6

Supporting Document

Sr. No.	Syllabus Contains	Short Answer	Yes / No	Page No. (In Syllabus)
1	अभ्यासक्रम	Enclosed in Syllabus	Yes	1
2	पात्रता	(As per the Rules and Regulations mentioned in MoM)	Yes	119
3	अभ्यासक्रमाची उद्दिष्टे	Enclosed in Syllabus	Yes	31
4	विषयाचे नाव	Enclosed in Syllabus	Yes	24
5	घटकांचा तपशील	Enclosed in Syllabus	Yes	24
6	तासिका	Enclosed in Syllabus	Yes	24
7	श्रेयांक पद्धत	Enclosed in Syllabus	Yes	24
8	संदर्भ साहित्य	Enclosed in Syllabus	Yes	33
9	संदर्भ ग्रंथ	Enclosed in Syllabus	Yes	33
10	प्रश्नपत्रिकेचे स्वरूप	Enclosed in Syllabus	Yes	117
11	अंतर्गत मूल्यमापनाचे स्वरूप	Enclosed in Syllabus	Yes	24
12	सत्र परीक्षेचे स्वरूप	Enclosed in Syllabus	Yes	118
13	गुणांकन	Enclosed in Syllabus	Yes	24